

水泥烧成系统电除尘器的协同优化控制

摘 要

水泥烧成系统电除尘器需在颗粒物排放达标与运行电耗之间协同取舍。本文以七天分钟级运行数据为基础,先对附件逐列作可识别性审计,把“可由数据验证的”与“只能靠机理推演的”分开,再分别建立功率实证模型与排放灰箱模型,最后在历史运行范围内求解约束优化问题,依次回答四问。

针对问题一,本文建立了由可识别性审计、物理形式功率模型与 Deutsch-Anderson 灰箱构成的分层响应模型。首先逐列审计:出口浓度 10030 条有效记录全部落在 48.74~50.00 mg/Nm³,其中 5491 条恰为 50,与全部 12 个变量的相关系数绝对值均小于 0.017,封顶分类 AUC=0.5098、未封顶子样本回归 $R^2 = -0.019$,判定该列为量程截顶、不可识别,排放层只能按机理推演并逐处标注其性质。其次按电场功率与电压平方成正比、振打功率与振打频率成正比的物理形式,用最小二乘拟合总功率,得 $P = 73.1 + 0.0850 \sum_i U_i^2 + 5.39 \times 10^4 \sum_i T_i^{-1}$,在未参与拟合的测试日上 $R^2 = 0.993$ 、RMSE=5.98 kW、平均偏差 -0.02 kW,据此算出振打环节占总功率的 38.8%,而入口温度、浓度、流量对总功率无额外解释力。再以相关分析提取历史操作规律:入口浓度升高时四个电场的振打周期一律缩短($r = -0.21 \sim -0.28$,四个时段方向一致)。最后由每周期积灰量 $\propto L \cdot T$ 的质量守恒推导振打峰值标度,不必假定指数即得:单次峰值随周期近似线性放大,而其对小时均值的贡献 $\propto L \cdot T \cdot (1/T) = L$,与周期无关。

针对问题二,本文建立了 K-means 工况聚类模型与支持域约束下的非线性规划模型。先以入口温度、入口浓度、烟气流量三个外生变量作 K-means 聚类,由轮廓系数($k = 4$ 时取最大值 0.4084)并结合温度两档 $\times$ 流量两档的 2 $\times$ 2 物理结构,确定划分为 4 类典型工况。再以上述功率模型为目标函数,以记录值 50 mg/Nm³ 为锚点、不作任何缩放而建立的 Deutsch-Anderson 灰箱排放式为约束,以各工况历史运行范围与工程排序关系为边界,构成 8 变量非线性规划,用多起点 SLSQP (每组 53 个起点)确定性求解,并以 4×10^5 点随机搜索独立对照。求得四类工况达到 10 mg/Nm³ 的最低功率为 1851.1~2104.3 kW,加权 1973.0 kW,比历史实际的 1783.2 kW 高 10.6% (日电耗 42.8→47.4 MWh)。收紧排放必然增加电耗,同一排放水平下不存在额外的节能空间。

针对问题三,本文建立了单位电功率去除收益的解析判据模型。把去除指数 K 与电场功率 P_{field} 同时对 U_i^2 求导,得判据 $\alpha_i / (b_i U_i^{ref2})$ 。第三电场参考电压最低,该收益在四类工况中均居首;原“前级优先”的结论实由排序约束 $U_1 \geq U_3 + 3$ 导出,而非权重 α_i 本身。振打侧同理按 c_i / T_i^2 排序,应优先延长周期较短的前级。据此给出低负荷工况四场同步升压、高负荷工况择优升压的差异化策略,并整理为七条控制优先级决策规则。

针对问题四,本文建立了四场电压同比缩放假设下的解析估计模型,并以八变量数值重解校验。由 $K \propto U^p$ 解得 $\Delta P = P_{field} [(1 + \Delta K / K_{10})^{2/p} - 1]$,即 $\Delta P = P_{field} \ln[(10 - B)/(5 - B)] / K_{10}$,给出 123.1~142.4 kW,与数值最优解相差不足 1%;加权功率由 1973.0 升至 2105.5 kW,增幅 6.72%,日电耗多 3.2 MWh。增幅与 $2/p$ 成正比,就功率增量与目标可达性而言, p 是主导性的待标定参数,一次电压阶跃试验即可确定;其余灰箱参数仍须由后续现场数据一并标定。更关键的约束不是电耗而是电源容量:10 mg/Nm³ 仍在历史电压范围内,5 mg/Nm³ 则需各场电压超出历史最大值最多 5%。若要求控制组合严格落在历史经验支持的范围内,八组解中七组仍可行,代价为多耗 1.5%~6.1% 电力。

最后,本文以 81 组结构情景扫描、电场权重消融与幂次参考表检验结论的稳健性,并明确排放层各系数均属工程先验、须经现场标定,所给控制参数只能作为现场试验起点。

关键词: 电除尘器 可识别性审计 Deutsch-Anderson 灰箱 K-means 聚类 多起点 SLSQP

1 问题重述与分析

1.1 问题背景

电除尘器通过高压电场使粉尘荷电并向收尘极迁移。提高电场电压通常能够增强荷电与迁移,但会提高运行功率并受到火花、绝缘和联锁条件限制;振打过于频繁会增加再飞扬次数,周期过长又可能导致积灰加重和单次脱落量增大。水泥工业颗粒物排放受到国家标准约束 [1], 题目进一步给出 10 mg/Nm^3 和 5 mg/Nm^3 两个控制目标, 要求同时考虑排放、总功率和振打峰值。

在建立模型之前, 须先明确本题的变量在设备上各处于什么位置, 否则后文的“可调量”与“结果量”容易混淆。图 1 给出研究对象的结构与变量定义: 含尘烟气依次流经四个电场, 每个电场由独立高压电源施加电压 U_i , 并由独立振打装置按周期 T_i 清灰。

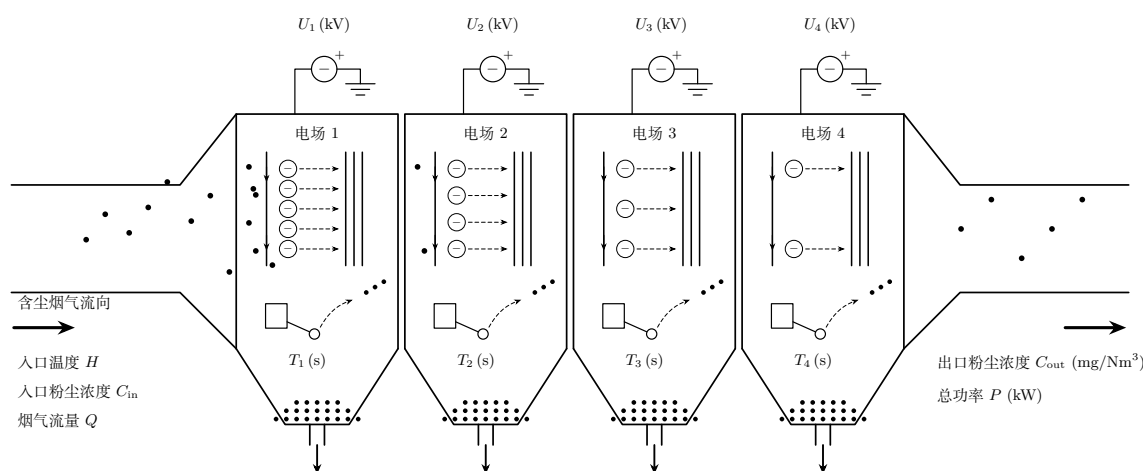


图 1 四电场电除尘器结构与变量定义

由图 1 可读出本题的变量结构: 入口侧的入口温度 H 、入口粉尘浓度 C_{in} 与烟气流量 Q 由上游工艺决定, 属**不可调的工况量**; 四电场电压 U_i 与振打周期 T_i 共 8 个量属**可调的控制量**; 出口侧的出口粉尘浓度 C_{out} 与总功率 P 则是需要同时兼顾的**两个结果量**。四问都可归结为: 在工况量给定的条件下如何配置这 8 个控制量, 使一个结果量达标而另一个尽可能小。本文不涉及设备本体的机械结构设计。

题目包含四项相互依赖的任务: 第一, 解释入口温度、入口浓度、烟气流量、四电场电压和振打周期与出口浓度之间的关系, 并分析振打峰值; 第二, 划分典型工况, 在 10 mg/Nm^3 约束下寻找最低功率控制参数; 第三, 选取差异显著的工况解释控制优先级; 第四, 将约束收紧至 5 mg/Nm^3 并估计功率代价。

1.2 总体思路

附件中有两列结果变量, 信息量却相差悬殊: 总功率几乎完全由控制量决定, 而出口浓度被压缩在量程上限附近, 几乎没有变化。若不加区分地将二者同时作为建模目标, 模型在出口浓度上学到的只会是传感器的截顶读数。本文因此采取先分层、后优化的路线:

1. 逐列检查哪些关系能被数据辨认出来, 把“可由附件验证的”与“只能靠机理推演的”分开;
2. 对可识别的总功率, 按物理形式 (电场功率 $\propto U^2$ 、振打功率 $\propto 1/T$) 建立并检验模型;
3. 对不可识别的排放层, 以**记录值本身**为锚点建立 Deutsch-Anderson 灰箱, 不作任何未经证实的量纲改写;

4. 在历史运行范围内求解优化问题，并把“结果在哪些方面超出了历史经验”当作结论的一部分如实报告。

第3点是本文与常见做法的关键差别。附件记录的出口浓度约 50 mg/Nm³，题目要求的却是 10 与 5 mg/Nm³，相差 5~10 倍。承认这个差距，题目就自然变为：控制参数应如何由现状调整至超低排放目标，相应的电耗代价是多少。反过来，若为了让数据靠近目标区间而假设记录值点错了小数点，第二问就会变成“把排放放宽到 10 mg/Nm³ 能省多少电”——与题意正好相反。

2 数据审计与关系可识别性分析

2.1 数据完整性与派生变量

附件覆盖 2024 年 5 月 1 日 0 时至 5 月 7 日 23 时 59 分，共 10080 条记录，采样间隔 1 min，时间戳无重复、无缺口。原始字段包括入口温度 H_t 、入口浓度 $C_{in,t}$ 、烟气流量 Q_t 、四电场电压 $U_{i,t}$ 、四电场振打周期 $T_{i,t}$ 、出口浓度 $C_{out,t}$ 和总功率 P_t 。出口浓度缺失 50 条 (0.50%)，其余字段完整。

为表达过程负荷和振打风险，构造粉尘质量负荷

$$L_t = C_{in,t} Q_t / 1000 \quad (1)$$

其中 C_{in} 单位为 g/Nm³， Q 单位为 Nm³/h，故 L 单位为 kg/h。另构造电压平方和、振打频率 $60/T_i$ 与每周期积尘代理 $L_t T_i / 3600$ 。所有聚类与回归参数仅在训练期（5 月 1—5 日）估计，验证期（5 月 6 日）与回顾性测试期（5 月 7 日）不参与拟合，划分按时间顺序而非随机打乱。

2.2 逐列可识别性结论

对 14 列逐一审计的结果见表 1：出口浓度一列无法用于识别目标区间内的排放响应，其余各列则能为功率建模、工况分层或历史操作规律分析提供有效信息。

表 1 附件逐列可识别性审计结论

数据块	审计证据	可用性
C_{out} 出口浓度	有效值 10030 条全部位于 48.74~50.00 mg/Nm ³ ，5491 条恰等于 50；与全部 12 个变量的同期相关系数绝对值均 < 0.017，1~120 min 滞后同样无稳定关系；封顶分类 AUC=0.5098，未封顶子样本回归验证 $R^2 = -0.019$	不可识别，不可修复
P 总功率	由控制量近乎确定，残差标准差约 6 kW（见 §5.2）	精确可辨识
H, C_{in}, Q 工况变量	连续波动并含两段持续异常事件（§4.3）	工况分层与扰动分析
U_i, T_i 控制设定	与工况变量存在稳定相关结构（§4.2）	可提取历史操作规律

在对出口浓度作任何拟合之前，必须先确认一件事：这一列记录是否覆盖了题目要求的目标区间，以及它与控制量之间是否存在可学习的关系。若不先做这一步，任何回归模型都可能在事实上只是复现传感器读数。图 2 给出该列的分布与截顶诊断。已有研究曾用宽负荷下的机组负荷与各电场二次电流训练出口浓度模型 [8]；相比之下，本附件既没有二次电流，出口记录本身也几乎不变化，不具备同样的条件。附录 B 给出直接以 C_{out} 为目标建模的对照结果，可与其他方案交叉比较。

图 2(a) 显示有效记录高度集中于 50 mg/Nm³ 附近，5 与 10 mg/Nm³ 的目标区间内没有任何训练样本；图 2(b) 显示恰好等于 50 的记录占 54.5%。结合表 1 中相关系数绝对值均小于 0.017 这一结果，可以得到一条完整的推论链：目标区间无样本，且控制量与出口记录之间不存在稳定关系，因此无法用黑箱回归直接建立目标区间内的排放模型。这一判断直接决定了后文的分层结构——排放层只能改用显式标注的条件式灰箱模型（§4.4），而不能与功率层一样按数据拟合。

需要强调的是，以上结果只说明该记录不能提供可供学习的信号，不能据此反推物理变量对真实排放没有影响。作为对照，图 3 给出首 24 h 各过程量的时序，用以说明“信息量不足”是出口浓

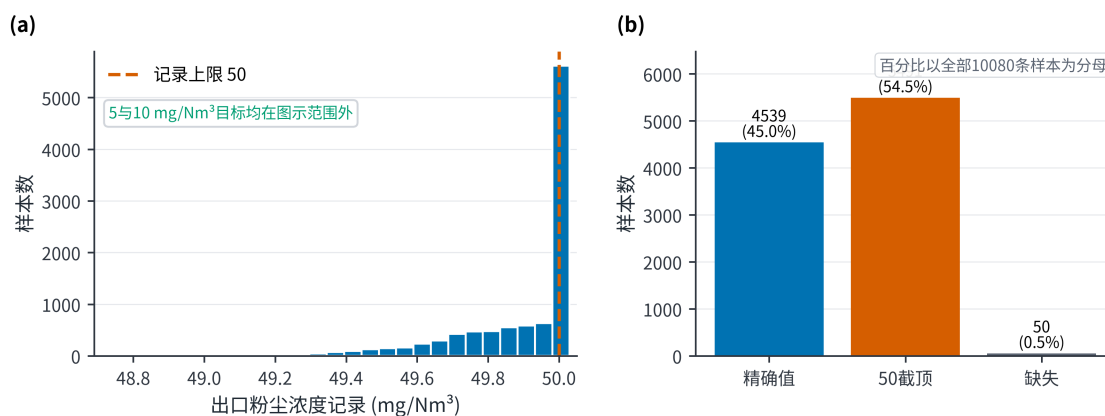


图2 出口浓度数据诊断

度一系列的特有的问题，而非整份数据的普遍状况。

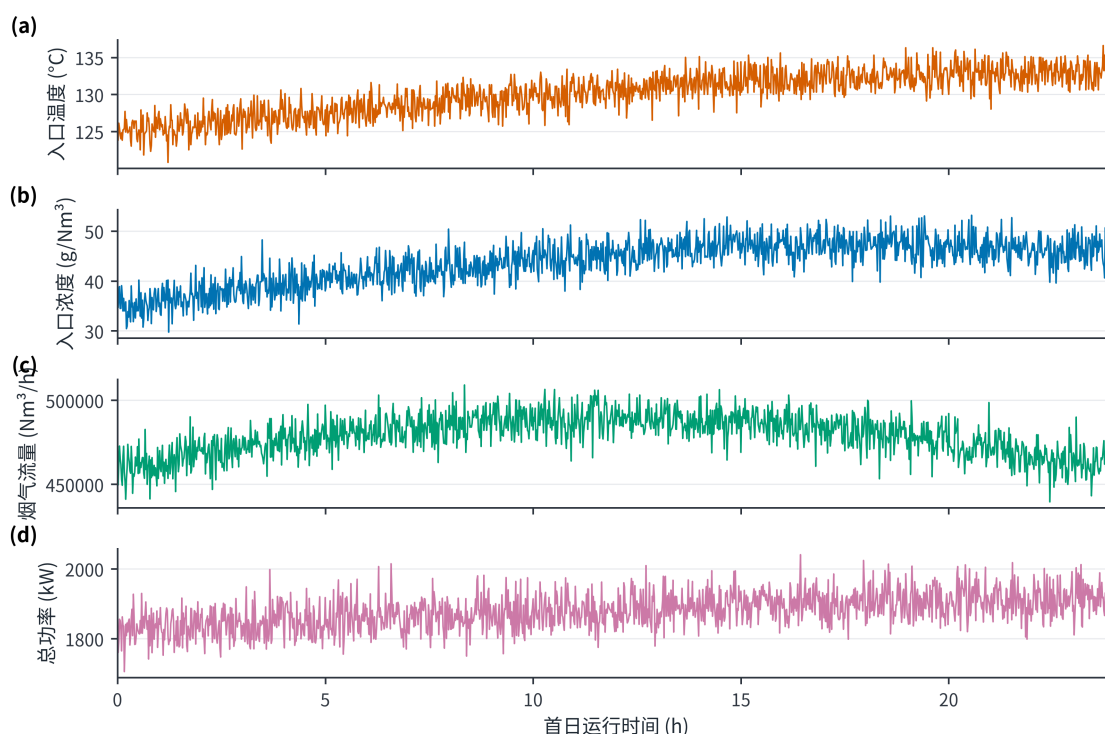


图3 首24 h过程量时序

图3中入口温度、入口浓度、流量与总功率均呈现连续变化和工况迁移特征，波动幅度足以支撑建模；与之相比，出口浓度记录几乎被压缩在单点附近。两者的反差说明：附件整体质量良好，问题仅出在出口浓度这一列，故后文对其余13列仍按实证方式使用。

2.3 排放锚点的取法

本文不对出口记录作任何缩放，直接把各工况出口浓度的记录中位数（都是 50 mg/Nm³）当作机理模型的起算点（下称锚点）。由此确立的问题定位是：设备当前距超低排放水平尚远，本文求解的是由现状降至 10 与 5 mg/Nm³ 的控制路径及其代价。这一取法带来三个后果，后文均予明示：最优点必然落在历史电压范围的上端 (§5.5)；10→5 mg/Nm³ 的功率增幅明显小于从更低起点出发时的估计 (§7)；排放层的数值仍属机理推演，须在补采未截顶的排放数据后重新标定（附录 C）。

3 模型假设、符号与总体框架

3.1 模型假设

本文将假设分为数据事实、情景假设、优化边界与运行约束四类，并逐条标注其证据状态，如表 2 所示。

表 2 模型假设及证据状态

类别	假设	证据状态
数据事实	7 天记录按分钟连续，除 50 条出口浓度缺失外不填造目标值	附件可核验
数据事实	总功率按 U_i^2 与 $1/T_i$ 的物理形式建模，按时间滚动验证	可由验证与测试结果检查
数据事实	排放锚点取出口浓度记录中位数 50 mg/Nm ³ ，不作缩放	附件记录值
情景假设	去除指数按 $K = K_0 S(U)$ 随电压幂次 p 缩放，中心取 $p = 2$	Deutsch 迁移速度形式，未经现场标定
情景假设	电场贡献权重 α_i 取等权、弱前级与中心前级三种形状	工程先验与消融
情景假设	振打峰值指数 $\gamma = 1$ 由积灰量正比关系推导，扫描 1.00/1.30/1.60	本文推导+结构情景
优化边界	控制量位于各工况训练期实测最小—最大范围内	历史运行包络，不等同设备安全边界
优化边界	5 mg/Nm ³ 档允许电压上界外推至历史最大值的 1.05 倍	显式声明的外推，须经容量核算
运行约束	前级电压高于后级至少 3 kV，后级振打周期长于前级至少 100 s	工程排序约束

3.2 主要符号

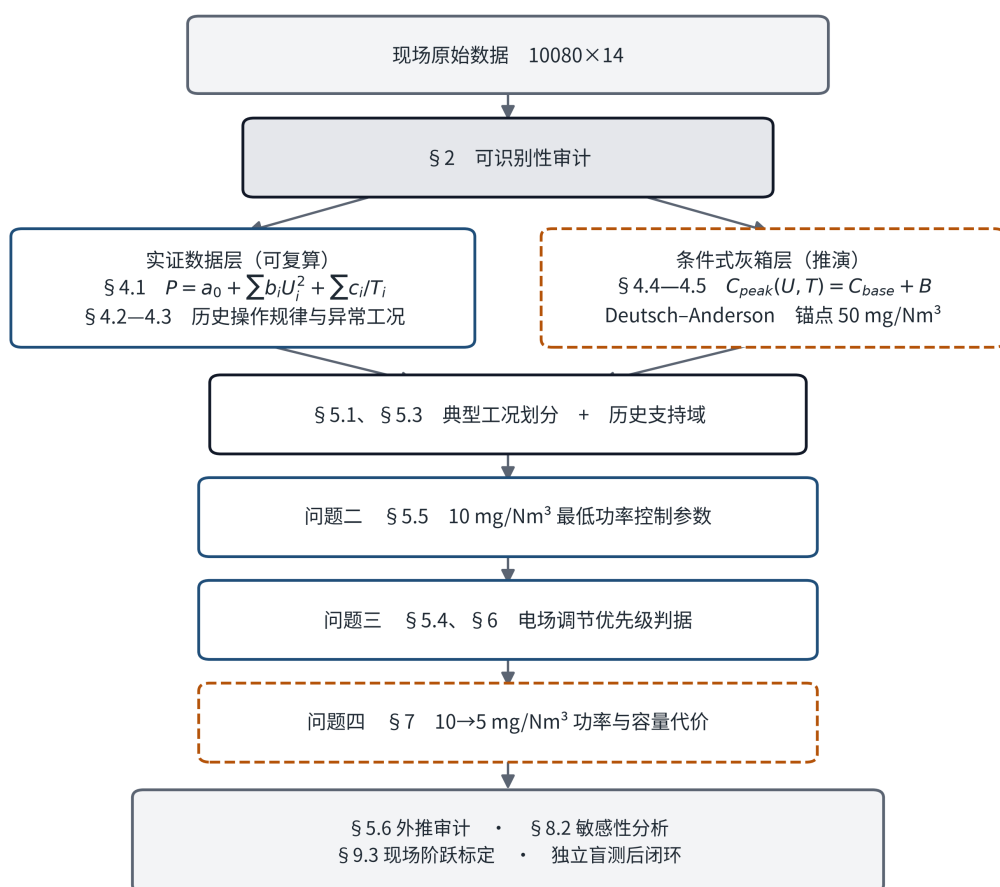
表 3 主要符号与单位

符号	含义	单位
U_i	第 i 电场电压	kV
T_i	第 i 电场振打周期	s
C_{in}, C_{out}	入口、出口粉尘浓度	g/Nm ³ 、mg/Nm ³
Q	烟气流量	Nm ³ /h
L	粉尘质量负荷	kg/h
$P, \hat{P}$	实际、预测总功率	kW
b_i, c_i	电场功率系数、振打功率系数	kW/kV ² 、kW·s
K, K_0	去除指数及其历史锚点值	—
$S(U)$	电压强度因子， $U = U^{ref}$ 时为 1	—
p	去除指数对电压的幂次， $K \propto U^p$	—
$B(T)$	振打引起的峰值增量代理	mg/Nm ³
ρ, γ	振打相位叠加尺度、峰值—周期指数	—

3.3 总体技术路线

前两节分别给出了假设与符号，但四问之间如何衔接、哪些结论可由数据复算、哪些只能条件式采信，仍需一张图整体说明。图 4 给出全文技术路线。

图 4 的关键在于 §2 的可识别性审计把论证分成了两条证据强度不同的路径：实线一侧（实证数据层）的功率律、历史操作规律与异常工况均可由附件直接复算；虚线一侧（条件式灰箱层）的排放模型只能由机理骨架加显式先验构成，须经现场标定后方可采信。两条路径仅在“典型工况+支持域”处合并，此后四问的结论按其所属路径分别标注：问题二、三的功率与优先级结论以实证层为主，问题四的排放代价则属条件式推演。图末的外推审计、敏感性分析与现场标定构成结论的适用边界，说明本文给出的是可检验的试验起点而非可直接下发的设定值。



— 实证辨识：可由附件复算 - - - 条件式推演：须现场标定后方可采信

图 4 全文技术路线

4 问题一：可识别的影响关系与振打峰值

第一问要求解释各变量与出口浓度的关系，并分析振打峰值。本节所用模型与方法为：逐列可识别性审计（相关分析、滞后相关、截顶诊断与对照建模）、物理形式最小二乘功率模型、历史操作规律的相关分析、Deutsch-Anderson 灰箱骨架，以及由积灰质量守恒推导的振打峰值标度关系。按 §2 的审计结论，本节分两层作答：先给出附件能够支持的三类实证结论 (§4.1—§4.3)，再给出只能靠机理推演的排放部分 (§4.4—§4.5)。

4.1 总功率的生成律（实证）

在训练期以物理形式拟合，得到

$$P = 73.09 + \sum_{i=1}^4 b_i U_i^2 + \sum_{i=1}^4 \frac{c_i}{T_i} \quad (2)$$

其中 $b = (0.0849, 0.0851, 0.0847, 0.0853)$ kW/kV²、 $c = (5.43, 5.37, 5.49, 5.26) \times 10^4$ kW·s。四个电场的 b_i 彼此相差不到 0.8%，说明四台高压电源的电气特性一致；写成合并形式即 $P \approx 73.0 + 0.0850 \sum U_i^2 + 5.39 \times 10^4 \sum T_i^{-1}$ 。这个式子的物理含义很直白：**电场功率与电压平方成正比，振打功率与振打频率 $60/T_i$ 成正比**。模型的检验结果见 §5.2。

从这个式子还能直接读出一个此前被忽略的事实。在训练期的中位设定下，电场功率为 1020.3 kW，振打功率为 693.7 kW，即**振打环节占总功率的 38.8%**。振打周期虽非题面强调的主要电气变量，但其对电耗的贡献与电压同一量级，不可忽略。相应地，把四个电场的周期从中位延长到历史 P95 可省下 56.1 kW，延长到历史最大值可省下 109.1 kW，代价是单次积灰量分别增加 8.4% 和 17.9%（图 6b）。§7.3 将说明，该可回收功率与排放限值收紧一档所增加的功率代价处于同一量级——这正是振打周期不宜为节电而无限延长的定量依据。

还有一点值得指出：把入口温度、浓度、流量也加进这个式子后，验证集与测试集的 R^2 变化都不到 10^{-3} （见表 9 末行）。**在现有数据与模型精度下，入口温度、入口浓度与烟气流量对总功率未显示出额外解释力**。后文按工况分别优化所得的功率差异，全部来自各工况不同的排放锚点与取值范围，而非工况本身直接作用于功率。

4.2 历史操作规律（实证）

出口浓度不可识别，并不意味着附件不含操作策略方面的信息。图 5(c) 给出整周记录中工况变量与控制设定的相关系数，其中两类规律值得报告：

规律一（稳定）：入口浓度升高时，四个电场的振打周期一律缩短（ $r = -0.208, -0.211, -0.277, -0.278$ ）。该方向在训练期、验证期与测试期三个时窗内符号一致，是附件中可复现的操作规律，与“高负荷下不宜延长周期”的工程直觉相符。

规律二（不稳定）：入口浓度升高时，前两个电场的电压随之升高（整周 $r = +0.172, +0.166$ ），后两个电场基本不变（ $r = +0.019$ ）。但该规律并不贯穿整周： $r(C_{in}, U_1)$ 在前 5 天为 -0.052 ，第 6、7 天分别为 $+0.507$ 与 $+0.492$ 。即入口浓度升高时抬升前级电压这一做法，仅出现于最后两天。本文如实报告该非平稳性：前后级响应的不对称在整周与最后两天均成立，却不能视为贯穿全周的稳定规律。这也表明历史数据中至少发生过一次操作策略调整，故第三问只将其作为参照，而非定论。

4.3 两段持续异常工况（实证）

出口浓度虽不可识别，但附件仍可能保留控制策略、异常工况响应与功率响应等信息，因此有必要对其余变量继续分析；§4.2 的相关结构与本节的两段异常工况合并呈现于图 5。附件含两段持续

表 4 工况变量与控制设定的历史相关系数（全记录期）

控制量	与入口温度 H	与入口浓度 C_{in}	与烟气流量 Q
U_1	-0.130	+0.172	+0.074
U_2	-0.127	+0.166	+0.070
U_3	-0.112	+0.019	-0.068
U_4	-0.113	+0.019	-0.063
T_1	-0.075	-0.208	-0.180
T_2	-0.064	-0.211	-0.175
T_3	-0.159	-0.277	-0.130
T_4	-0.151	-0.278	-0.135

数十分钟以上的异常工况，其控制端响应方向相反，构成理解设备行为的两个直接入口（图 5a、5b）。

表 5 两段持续异常工况及控制端响应

事件	时间窗	特征	控制端响应
入口浓度尖峰	05-02 14:02—15:49，其中 41 min 超过 60 g/Nm ³	入口浓度峰值 72.47 g/Nm ³ （基线中位 37.41）	U_1 均值 63.7 kV（峰值 67.8），振打周期同步缩短，总功率均值 1924 kW、峰值 2020 kW
入口温度尖峰	05-06 13:00—14:59，持续 120 min	入口温度峰值 158.2 °C（基线中位 126.5），同期入口浓度降至均值 25.9 g/Nm ³	U_1 均值降至 55.5 kV，总功率均值降至 1668 kW

第二段事件在工程上更值得注意：**140~160 °C 正是水泥窑尾粉尘比电阻最高的温度区间**。比电阻过高会引发反电晕，使粉尘向极板迁移的速度下降、火花率上升，是电除尘最难运行的一段工况 [3-4]。附件在这段时间里的记录显示，操作端选择了降压而不是升压，这与“高比电阻区应限制电流密度以抑制反电晕”的常规处置一致；但附件没有二次电流和火花率记录，无法判断这是否为最优处置。由此可得两点：第一，第二问的工况划分把温度作为主轴之一是必要的；第二，本文四类工况的温度中心约为 120.7 与 130.4 °C，**都没有覆盖这段高比电阻区间**，因此本文结论不能外推到该区间，须单独补采数据重新标定 (§9.3)。

综合图 5 可得三点。其一，**稳定的规律**是图 5(c) 中入口浓度与四场振打周期的负相关 ($r = -0.208 \sim -0.278$)，该方向在四个时窗内一致。其二，**不稳定的规律**是入口浓度与前两级电压的正相关，它只在最后两天成立 (§4.2)。其三，图 5(a)(b) 的两段事件说明控制端对不同扰动的处置方向相反：入口浓度尖峰时抬压增功率，入口温度尖峰时反而降压降功率。

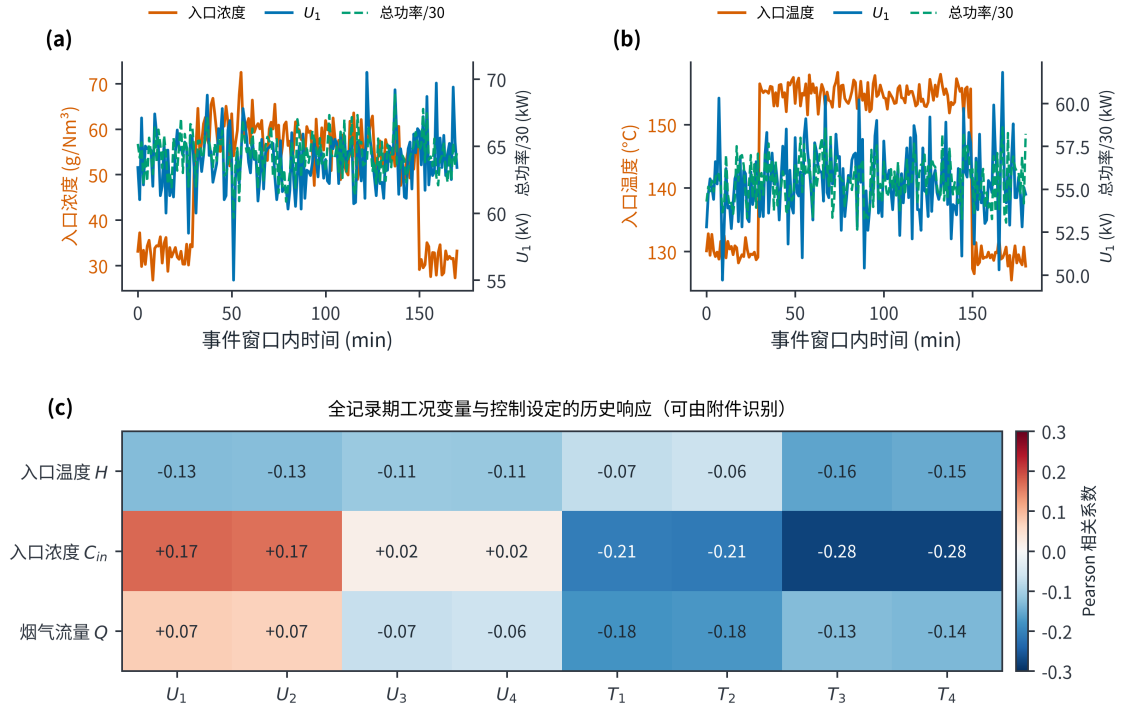
必须指出，以上全部是**历史操作留下的相关关系**，只能说明操作者当时如何响应工况，不能直接翻译为控制量如何影响排放的因果结论——因为排放侧的响应本身不可识别 (§2.2)。故第三问只把它作为经验对照，判据本身仍由模型结构导出 (§5.4)。

4.4 排放响应的灰箱骨架（条件式推演）

Deutsch-Anderson 关系用指数形式描述电除尘效率。它的理想化形式忽略了再飞扬、气流分布和粒径变化，因此只能当作**骨架**：形式取自机理，系数则须另行给定（本文称“灰箱”） [2-4]。White 的经典专著 [12] 与 Parker 的电除尘器电气运行专著 [13] 都强调，电气运行、粉尘性质与振打三者是相互耦合的。定义去除指数

$$K = \ln \frac{1000C_{in}}{C_{out}}, \quad C_{out} = 1000C_{in}e^{-K} \quad (3)$$

K 越大表示除尘越彻底： K 每增加 $\ln 2$ ，出口浓度就减半。对工况 k ，以历史中位电压 $U_{i,k}^{ref}$ 和出口浓度记录中位数 $C_{0,k} = 50 \text{ mg/Nm}^3$ 为起算点，得 $K_{0,k} = \ln(1000C_{in,k}/C_{0,k})$ ，四类工况依次为 6.257、6.514、6.702、6.797。粉尘向极板迁移的速度与电场强度的平方近似成正比，故取



注：振打周期规律（ $C_{in} \uparrow \Rightarrow T \downarrow$ ）在四个时窗内一致；电压规律并不平稳—— $r(C_{in}, U_1)$ 在前5天为 -0.05，在末日为 +0.49。

图 5 附件中可识别的运行证据

$$K_k(U, T) = K_{0,k} S_k(U) - \{G_k(T) - G_k(T^{ref})\}, \quad S_k(U) = \frac{\sum_i \alpha_i \left(\frac{U_i}{U_{i,k}^{ref}} \right)^p}{\sum_i \alpha_i} \quad (4)$$

$$G_k(T) = \sum_{i=1}^4 \beta_{i,k} \left(\frac{T_i}{T_{i,k}^*} + \frac{T_{i,k}^*}{T_i} - 2 \right), \quad \beta_{i,k} = \beta_i \left(\frac{L_k}{L_{med}} \right)^{1/2} \quad (5)$$

这样构造有一个好处：当 $U = U^{ref}$ 、 $T = T^{ref}$ 时式子严格还原为 $K = K_{0,k}$ ，即模型天然复现历史运行点，因此不再需要一个额外的整体比例参数 η ，全部电压敏感性都归结到一个幂次 p 上（中心取 $p = 2$ ）。 $G_k(T)$ 在 $T_i = T_{i,k}^*$ 处取最小值，表示周期过短或过长都不利：过短会频繁引起再飞扬，过长则积灰加重、单次脱落量变大。试验研究也确实观察到再飞扬与最大允许周期之间存在这种权衡 [6]。再加一点安全裕度 s ，连续排放的基础值为 $C_{base,k} = 1000C_{in,k} \exp[-K_k(U, T) + s]$ 。

本节全部排放系数均为工程先验，不是由附件出口浓度估计得到的参数。

表 6 灰箱排放模型参数来源与证据状态

参数	中心取值	含义	来源	是否由附件识别
p	2.0	去除指数对电压的幂次	Deutsch 迁移速度形式	否
α_i	(1.15, 1.10, 0.95, 0.80)	电场去除贡献权重	工程先验与消融	否
β_i	(0.30, 0.27, 0.20, 0.17)	周期偏离损失	工程情景	否
b_0	1.2	峰值尺度	工程情景	否
γ	1.0	周期—峰值指数	本文由积灰量推导	否
ρ	1.0	相位叠加尺度	同相保守情景	否
s	0.08	安全裕度	工程情景	否

4.5 振打峰值：由积灰量推导而非假定

附件只给出振打周期的设定值，没有实际振打时刻、相位和秒级浓度，所以逐次峰值本身无法辨认。但峰值随周期怎样变化可以由质量守恒直接推出，不必假定指数：设单位时间的积灰速率正比于粉尘负荷 L ，那么一个周期内积下的灰饼质量就正比于 $L \cdot T$ ；一次振打抛落的粉尘量又正比于灰饼质量，于是

$$B_k(T; \rho, \gamma) = b_0 \rho \left(\frac{L_k}{L_{med}} \right) \sum_{i=1}^4 w_i \left(\frac{T_i}{T_{i,k}^*} \right)^\gamma, \quad \gamma = 1 \quad (6)$$

振打周期是本题中唯一同时牵动两个结果量的控制量：缩短周期提高振打频率、直接增加电耗，延长周期虽省电却使每次抛落的灰量增大。由上式可得两条互补的结论，也正是第一问关于振打峰值的完整回答：

- **单次瞬时峰值随周期近似线性放大**：周期延长 10% 与 20% 时，单次峰值代理相应放大 10% 与 20%；
- **对小时均值的贡献在一阶近似下与周期无关**：在单位面积灰量所对应的再飞扬比例近似不随振打周期变化这一前提下，单位时间的总再飞扬量正比于 $L \cdot T \times (1/T) = L$ ，只取决于粉尘负荷。该结论是上述前提下的一阶近似，而非无条件成立的物理定律；若周期显著延长使灰饼结构改变、再飞扬比例本身随之变化，该近似即不再适用。

换言之，延长振打周期并不改变再飞扬对小时均值的贡献，只是把等量粉尘集中于次数更少、单次幅度更高的峰值中释放。这条结论有两个直接用途：一是解释了为什么“延长周期省电”必须和瞬时峰值一起考虑 (§7.3)；二是由于现行标准按小时均值考核，本文把瞬时峰值直接卡到限值的做法偏保守 (§7.4)。中心情景取 $\rho = 1$ ，即假设四个电场同时振打、峰值完全叠加，这是最不利的上界；若错开相位则取 $\rho = 0.80$ 或 0.65 。敏感性分析仍扫描 $\gamma \in \{1.00, 1.30, 1.60\}$ ，以检验推导值附近是否稳健。

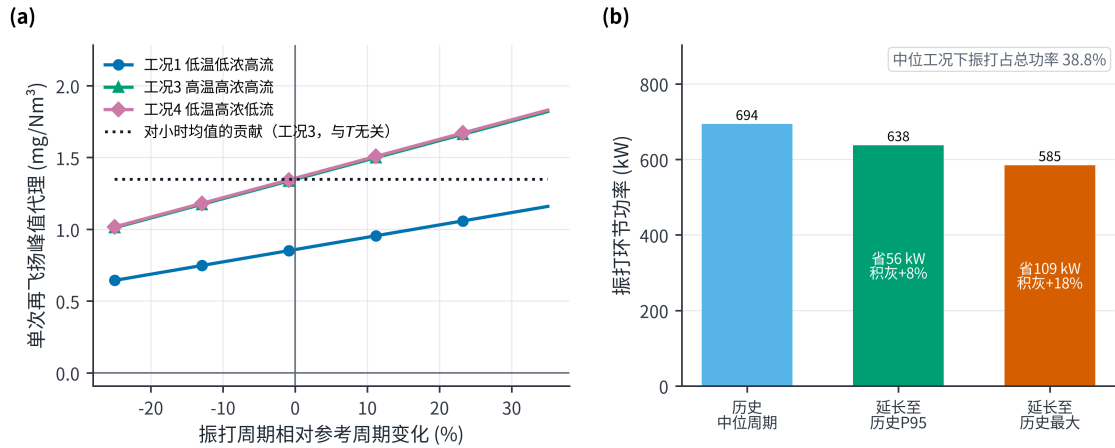


图 6 振打峰值机理与振打功率权衡

图 6(a) 给出周期延长时单次峰值的线性放大，图 6(b) 给出同一动作在功率侧的收益：周期由中位延长至历史极值可回收约 109 kW。二者合起来给出本题的一条关键判断——**振打周期并非越长越省电越好，而是在电耗与瞬时峰值风险之间取舍的权衡变量**。该权衡在第四问中被定量使用：§7.3 将说明振打侧可回收的功率与排放收紧一档的代价处于同一量级，但其额度受峰值裕度限制。

4.6 各因素方向性小结

在上述骨架中，其他条件不变时各因素的作用方向如下。提高 U_i 会增大 K 、降低连续排放，但总功率按 U^2 上升。 K 不变时，入口浓度升高会使出口浓度同比升高。流量增大会缩短烟气在电场中的停留时间、同时抬高粉尘负荷，因而需要更强的捕集能力。温度则通过气体性质、粉尘比电阻和放电状态影响除尘效果，其中 $140\sim160\text{ }^{\circ}\text{C}$ 的高比电阻区间最为不利 (§4.3)。这些方向来自机理与文献 [2-4]，不是从附件回归出来的。为便于对照，表 7 把第一问涉及的五类因素、它们对排放与电耗的作用及各自的证据性质汇总如下。

表 7 各因素对排放与电耗的作用及证据性质

因素	对排放的主要作用	对电耗的作用	证据性质
入口浓度 C_{in}	抬高排放压力：固定 K 时出口浓度同比升高	不直接进入功率模型	机理+历史操作数据 (§4.2)
烟气流量 Q	缩短有效停留时间、抬高粉尘负荷，需更强捕集	未显示额外解释力	工程机理
入口温度 H	经粉尘比电阻与放电状态影响除尘， $140\sim160\text{ }^{\circ}\text{C}$ 最不利	无直接功率项	文献 [3-4] + 异常事件 (§4.3)
电场电压 U_i	提高去除指数 K 、降低连续排放	功率按 U_i^2 上升	排放灰箱（推演）+ 功率实证
振打周期 T_i	过长抬高单次峰值，偏离最优区间增大排放风险	功率按 $1/T_i$ 下降	机理推导+功率实证

该表同时显示了本文分层的必要性：电耗一列全部有实证支撑，排放一列则以机理与文献为主，二者的证据强度并不对等。

5 问题二：典型工况划分与 10 mg/Nm^3 最低功率

第二问要求划分典型工况，并在出口浓度不超过 10 mg/Nm^3 的约束下使总电耗最低。本节所用模型与方法为：以入口三变量作 K-means 聚类并由轮廓系数与 Calinski-Harabasz 指标选择簇数 (§5.1)；以四种设定的滚动时间外推比较确定功率模型形式 (§5.2)；以功率模型为目标函数、以灰箱排放式与历史支持域为约束建立 8 变量非线性规划，用多起点 SLSQP 确定性求解并以随机搜索对照 (§5.3)；由模型结构求得电场优先级的解析判据 (§5.4)；最后以马氏距离作外推审计 (§5.6)，并逐档收紧限值得到排放—功率权衡前沿 (§5.7)。

5.1 工况聚类

聚类只用入口温度、入口浓度和烟气流量三个变量——它们由上游工艺决定，不受控制策略影响，因此不会出现“用结果去定义工况”的问题。三者标准化后作 K-means 聚类 [9]。取 $k = 2, 3, 4, 5, 6$ 比较，轮廓系数 [10]（衡量类内紧凑程度与类间分离程度，取值越大越好）依次为 0.2963、0.3682、0.4084、0.3958、0.3934，在 $k = 4$ 最高。另一项 Calinski-Harabasz 指标依次为 3204.2、4399.0、4721.2、4835.5、4712.0，最大值却出现在 $k = 5$ ，两项判据并不一致。

两项指标的分歧是本节必须交代的问题，图 7 把它并列呈现。

图 7(a) 显示轮廓系数在 $k = 4$ 取最大值，图 7(b) 显示 Calinski-Harabasz 指标却在 $k = 5$ 取最大值。两项统计判据结论不一致，因此 $k = 4$ 并非唯一的“数学最优”聚类数。本文最终取 $k = 4$ ，依据有四条：其一，轮廓系数在 $k = 4$ 最高；其二，四类中心恰好构成温度两档（约 120.7 与 $130.4\text{ }^{\circ}\text{C}$ ）与流量两档（约 44.0 与 $48.1\text{ 万 Nm}^3/\text{h}$ ）的 2×2 组合，物理含义清楚；其三，四类比五类更便于控制解释与后续按工况优化；其四，由此本文把这四类理解为统计意义上的工况分层，而非唯一的物理状态划分。此外如 §4.3 所述，两段异常工况都落在四类中心之外，不被该分层覆盖。

在确定 $k = 4$ 之后，图 8 给出四类工况在入口浓度—温度平面上的实际分布，用以核对分层结果是否与工况的物理含义相符。

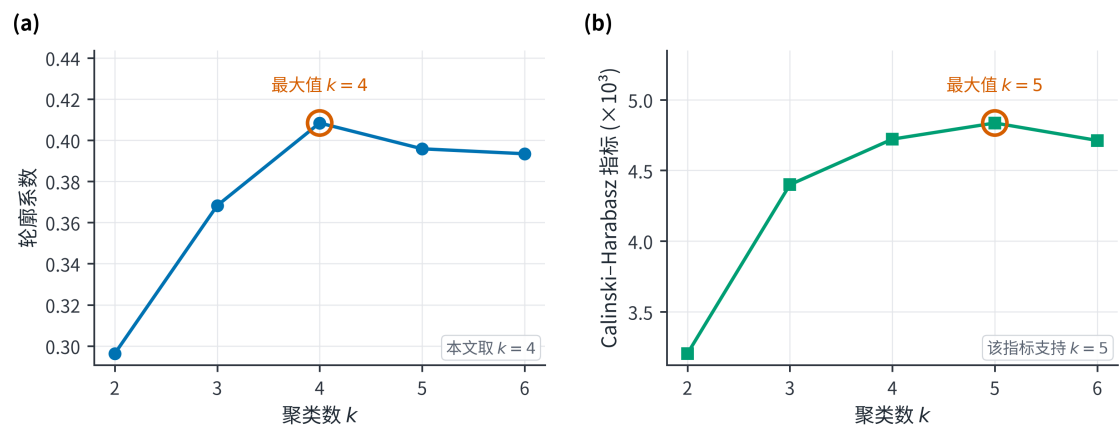


图 7 聚类数选择结果

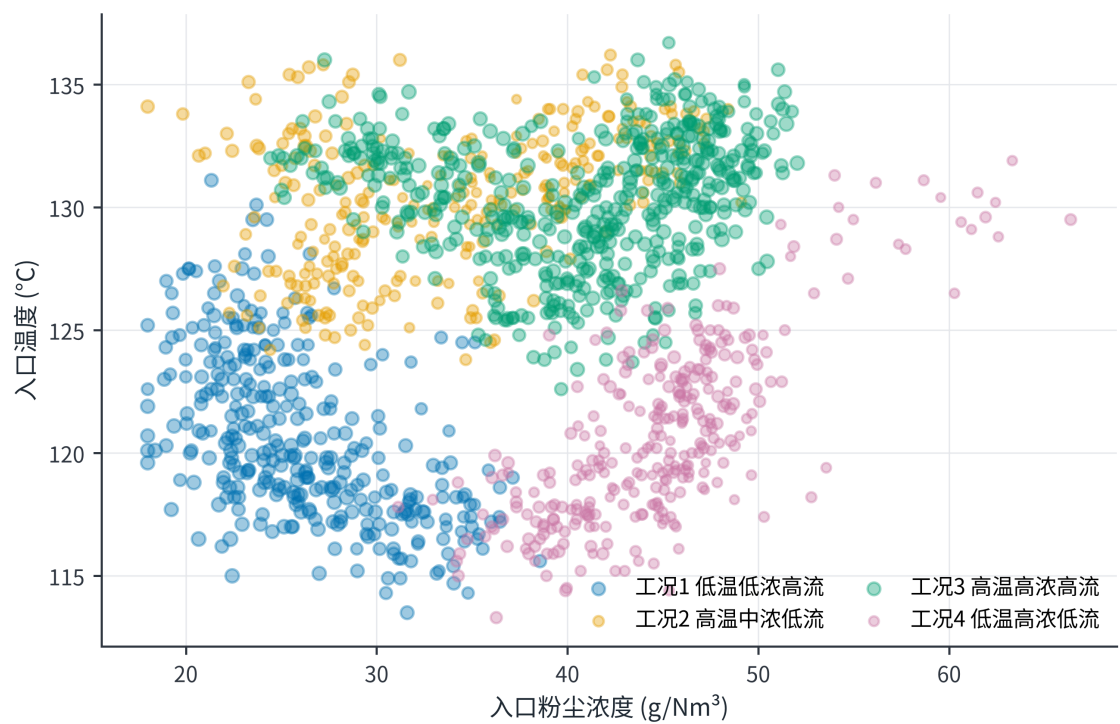


图 8 训练期典型工况分布

图 8 中四类样本在浓度—温度平面上分区清晰，点的大小（流量）进一步区分了同温度带内的两类工况，与上述 2×2 结构一致。四类工况的统计画像见表 9，其中的历史功率与锚点 K_0 将分别用于 §5.5 的对照基准与 §4.4 的排放起算点。

表 8 四类典型工况画像

工况	占比	温度/°C	入口浓度/(g/Nm³)	流量/(Nm³/h)	粉尘负荷/(kg/h)	历史总功率/kW	锚点 K_0
1 低温 低浓度 高流量	23.35%	120.65	26.08	478257	12460	1818.35	6.257
2 高温 中浓度 低流量	20.06%	130.32	33.72	443591	14922	1870.89	6.514
3 高温 高浓度 高流量	34.93%	130.41	40.71	480851	19560	1763.08	6.702
4 低温 高浓度 低流量	21.67%	120.69	44.75	439990	19673	1696.55	6.797

5.2 功率模型的设定与检验

§4.1 采用的形式并不是唯一可选的。为说明“选什么形式”本身有多重要，本文在同一时间划分下比较四种设定：只用均值的基线、电压一次项+周期一次项的岭回归 [11]、电压二次项+周期一次项的岭回归，以及本文的 $U_i^2 + 1/T_i$ 形式。岭回归的正则强度按训练期 3 个逐步加长的窗口滚动选择，本文形式则没有可调超参数。结果见表 9。

表 9 功率模型设定比较与时间外推结果

设定	验证 R^2	验证 RMSE/kW	测试 R^2	测试 RMSE/kW	测试平均偏差/kW
均值基线	-5.3259	140.18	-0.3733	83.51	-43.54
电压一次+周期一次（岭）	0.9798	7.93	0.9180	20.41	-16.65
电压二次+周期一次（岭）	0.9837	7.12	0.9571	14.76	-10.78
本文 $U_i^2 + 1/T_i$	0.9885	5.97	0.9930	5.98	-0.02
本文设定+工况变量	0.9885	5.97	0.9929	5.98	+0.08

上表只回答了“哪种设定更好”，接下来的三幅图分别回答三个递进的问题：图 9 回答最终模型预测得有多准，图 10 回答为什么必须把振打周期写成 $1/T$ ，图 11 回答改正之后残差是否还残留系统性结构。

这张表有三点值得强调。

第一，**该系统性偏差来自模型设定误差，而非数据漂移**。线性 T 的设定在测试日留下 -10.78 kW 的平均偏差，而且偏差在高功率区还会扩大；改成 $1/T$ 后，偏差降到 -0.02 kW，测试 RMSE 从 14.76 kW 降到 5.98 kW， R^2 从 0.957 升到 0.993（图 10）。若沿用线性形式，该偏差很容易被误判为数据本身的问题而额外计入误差预算；实际上，它主要由错误的函数形式引入。

第二， T 是优化中的决策变量。线性形式意味着周期每延长单位时间所节省的功率恒定，而真实的边际节电按 c_i/T_i^2 衰减，即周期越长、再延长的收益越小。因此线性周期设定会使最优振打周期系统性地偏离真实值。

第三，再加入工况变量后各项指标不再改善，印证了 §4.1 的判断：在现有数据精度下，工况变量对总功率没有额外解释力。

验证集上绝对误差的 90% 分位数为 9.78 kW，本文把它作为做功率预算时的余量。它不改变各方案的优劣排序，也不保证在今后的日期里仍能覆盖 90% 的情形。

首先确认最终模型的整体预测水平。

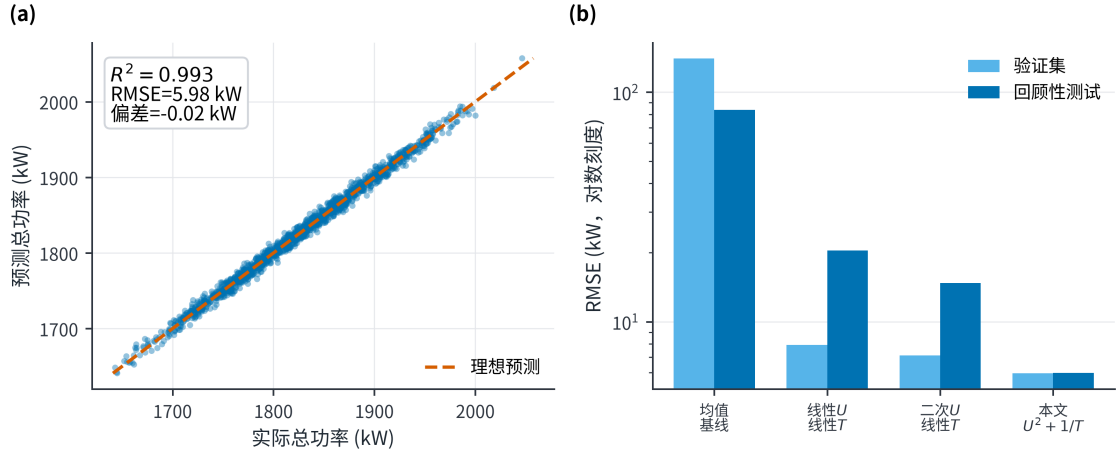


图9 功率模型预测结果

图 9(a) 中测试日的预测值紧贴理想线，图 9(b) 以对数纵轴比较四种设定的 RMSE：本文设定在验证与测试两段均最低，且是唯一在测试段不发生劣化的设定。这确认了 $P = a_0 + \sum b_i U_i^2 + \sum c_i / T_i$ 可以作为后续优化的目标函数使用。

但“准”还不足以说明“对”。图 10 进一步说明设定形式本身为何必须修正。

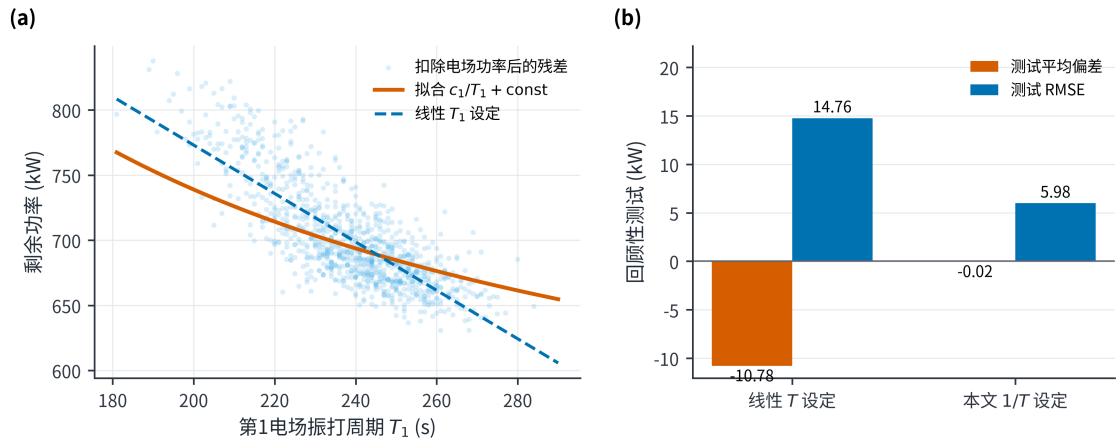


图 10 振打周期的设定形式对比

图 10(a) 把扣除电场功率后的剩余功率对 T_1 展开：数据呈明显的双曲形状，线性设定（虚线）在周期的两端系统性偏离。图 10(b) 则给出后果——线性 T 设定在测试段留下约 -10.8 kW 的平均偏差，改为 $1/T$ 后降至约 0。这是本文修正模型设定的关键证据：最初的系统性负偏差并非数据漂移，而主要由 T 的函数形式设定错误引起。若不作此修正，该偏差会被误计入误差预算，且会使 §5.3 优化中的最优周期系统性偏移。

修正是否彻底，需由残差结构本身确认。

图 11(a) 显示残差围绕零线对称散布，原先在高功率区扩大的负偏差已消失；图 11(b) 的残差分布以零为中心，测试均值 -0.02 kW 。至此三幅图形成完整链条：设定比较（表 9）→ 整体精度（图 9）→ 形式修正的依据（图 10）→ 修正后无残留结构（图 11），据此确定本文的功率模型为 $P = a_0 + \sum b_i U_i^2 + \sum c_i / T_i$ 。

5.3 优化模型

设控制向量 $z = (U_1, \dots, U_4, T_1, \dots, T_4)$ 。对每类工况 k 与限值 C_{lim} 求解

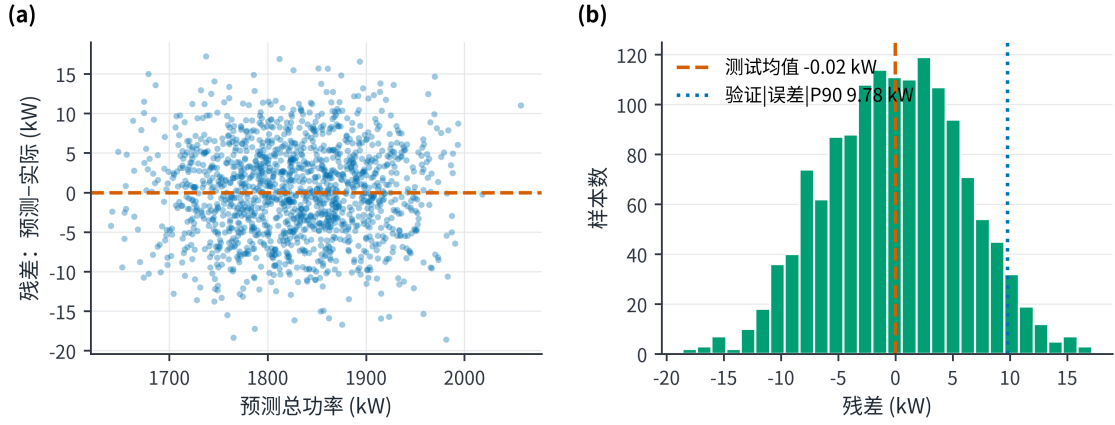


图 11 回顾性测试残差诊断

$$\min_z \hat{P}(z) = a_0 + \sum_i b_i U_i^2 + \sum_i \frac{c_i}{T_i} \quad (7)$$

$$\text{s.t. } C_{base,k}(z) + B_k(z) \leq C_{lim}, \quad U_1 \geq U_3 + 3, U_2 \geq U_4 + 3, T_3 \geq T_1 + 100, T_4 \geq T_2 + 100, \quad z \in \Omega_k \quad (8)$$

取值范围 Ω_k 取该工况训练期内各控制量实际出现过的最小值到最大值。5 mg/Nm³ 一档另外允许电压上界放宽到历史最大值的 1.05 倍，放宽了多少在表 12 中逐条列出。已有的多电场节能研究同样采用“排放受限、分场优化”的思路 [5,7]，但它们的排放模型都由现场实测标定，条件与本文不同，方法对照见附录 A。

关于求解方法，此处说明本文未采用凸优化求解器的原因。以电压 U 为变量时，目标函数 $\sum b_i U_i^2$ 本身是凸的，但排放约束要求这样一个凸函数取值**足够大**——这类约束把可行域挖成了非凸的形状，整体不是凸规划。若改用 $v_i = U_i^2$ 作变量，目标和排放约束都变成线性，可惜排序约束 $U_1 \geq U_3 + 3$ 在 v 下不再是凸集。即在任一坐标下均无法化为凸问题，故不宜宣称凸性。

本文因此采用**多起点 SLSQP 确定性求解**：每个“工况—限值”组合从 53 个起点分别求解（5 个有结构的起点：历史中位、上界、下界、中点、 T^* 点；再加 48 个随机起点），以检验各起点是否收敛至同一解。此外，在同一可行域内均匀随机抽取 4×10^5 个候选点作随机搜索，用作独立对照 (§8.1)。这一做法取代了早期版本的百万级随机采样：问题结构清楚之后，确定性求解既更快，也更容易复算。

5.4 电场优先级的解析判据

在给出数值解之前，先由模型结构推导各电场的调节优先级。把去除指数与电场功率

$$K_k = K_{0,k} \frac{\sum_i \alpha_i \left(U_i / U_{i,k}^{ref} \right)^2}{\sum_i \alpha_i}, \quad P_{field} = \sum_i b_i U_i^2 \quad (9)$$

同时对 U_i^2 求导，得到**单位电功率的去除指数增益**

$$\left. \frac{dK}{dP} \right|_i = \frac{K_{0,k} \alpha_i}{\left(\sum_j \alpha_j \right) b_i \left(U_{i,k}^{ref} \right)^2} \quad (10)$$

该比值越大，表示该电场单位功率带来的除尘收益越高。须说明，这是**灰箱模型内部的条件式**

边际判据：功率系数 b_i 由附件辨识，而去除贡献权重 α_i 取自工程先验，其稳健性由 §8.2 的场权重消融进一步检验；该排序并非由实测出口浓度直接证实的物理事实。由于四个电场的 b_i 几乎相同，排序完全取决于 $\alpha_i/(U_{i,k}^{ref})^2$ ，而不是取决于 α_i 。原因在于功率按电压平方增长，参考电压较低的电场按相同比例抬升时，所需电功率显著更少。代入中心权重与各工况的历史中位电压，得到表 10。可以看到，第三电场的去除权重 $\alpha_3 = 0.95$ 虽然低于前两级，但它的参考电压低约 10 kV，平方之后使它的单位功率收益在四类工况中都排第一。

表 10 电场优先级解析判据 ($\times 10^{-3} \text{ kW}^{-1}$)

工况	电场 1	电场 2	电场 3	电场 4	优先次序
1 低温低浓度高流量	5.768	5.504	6.668	5.573	E3 > E1 > E4 > E2
2 高温中浓度低流量	5.556	5.269	7.023	5.870	E3 > E4 > E1 > E2
3 高温高浓度高流量	6.630	6.327	8.024	6.706	E3 > E4 > E1 > E2
4 低温高浓度低流量	7.551	7.207	7.717	6.502	E3 > E1 > E2 > E4

这也解释了本文早期版本“前级优先”这一结论的真实来源：**它是排序约束 $U_1 \geq U_3 + 3$ 带出来的，而不是权重 α_i 决定的**。由于抬升 U_3 必须同时抬升 U_1 ，该排序约束使二者的调节之间存在耦合，故最优解中 U_1 与 U_3 总是同步走高。若排除该约束，单位功率收益最高的仍是第三电场。

振打侧同理： $\partial P/\partial T_i = -c_i/T_i^2$ 。前级周期短（中位 240 s），后级周期长（中位 441 s），故在相同的周期延长量下，前级的节电量约为后级的 **3.4 倍**。这与电压侧由前级承担主要捕集并不矛盾：电压用于提高捕集能力，周期用于降低电耗，二者的最优着力点分别位于不同电场。

5.5 10 mg/Nm³ 下的最优控制参数

四类工况的中心情景解如表 11（5 mg/Nm³ 行为第四问提供参数）。

表 11 四工况两档目标的最优控制参数

工况	目标/(mg/Nm ³)	U_1/kV	U_2/kV	U_3/kV	U_4/kV	T_1/s	T_2/s	T_3/s	T_4/s	峰值排放/(mg/Nm ³)	预测总功率/kW
1	10	70.2	68.6	57.6	57.4	290	289	492	493	9.948	2045.55
2	10	70.9	69.8	59.2	58.2	280	278	494	498	9.962	2104.30
3	10	68.0	60.3	56.3	57.3	275	280	506	510	9.971	1924.67
4	10	70.5	61.8	57.2	42.9	285	283	495	498	9.991	1851.15
1	5	73.7	71.6	60.5	60.3	290	289	492	493	4.993	2182.25
2	5	74.4	72.9	62.2	61.1	280	278	494	498	4.998	2245.51
3	5	71.4	63.4	59.1	60.4	275	280	506	510	4.991	2056.06
4	5	74.0	66.4	60.1	42.9	285	283	495	498	4.988	1973.10

第二问的答案：在 10 mg/Nm³ 下，四类工况的最低总功率为 1851.15~2104.30 kW，按训练期各工况占比加权为 **1973.0 kW**，折合日电耗 47.35 MWh。作为对照，历史实际加功率是 1783.2 kW（日电耗 42.80 MWh）。也就是说，达到 10 mg/Nm³ 意味着**功率上升 10.6%、日电耗多用 4.55 MWh**。

这一方向是必然的：历史排放约 50 mg/Nm³，将出口浓度压至其五分之一必须提高捕集能力，而在线可调的手段只有电压，电压又按平方计入功率。因此，**本文所称“节能”是指在同一排放目标下相对其他可行控制方案的功率降低，而非相对历史运行工况的降低**。

解的结构与表 10 的判据一致。振打周期 T 在两档下都被推至历史区间上端，因为延长周期直接降低振打功率；电压则按单位功率收益由高到低分配——工况 3 与工况 4 中 U_2 明显低于 U_1 ，工况 4 的 U_4 更被压至历史下界 42.9 kV，正是由于该电场在此工况下单位功率收益最低（表 10 末行）。可见四场同步抬升并非最优。

表中数值为点预测值，编制功率预算时可另计 9.78 kW 的误差余量。电压无须按 0.1 kV 精度执行：多起点求解所得功率完全一致，但控制量本身存在若干功率近乎等价的组合（§8.1）。

5.6 目标点已越出历史运行经验

上表的解能否直接下发给设备？换一个更具体的问法：**若每个电场的电压都没有超过历史最大值，是否就意味着该控制组合落在历史正常运行的范围内？**答案是否定的，且理由可由数据定量给出。

本文引入马氏距离，以定量评价推荐控制组合偏离历史运行经验的程度。仅逐个检查各电压是否越限并不足够，因为八个控制量之间彼此相关：历史上从未出现过 U_1 很高而 U_3 很低的组合，即便二者各自都落在允许范围内。马氏距离平方 $d_M^2 = (z - \mu_k)^T \Sigma_k^{-1} (z - \mu_k)$ 正是把这种相关性一并计入的距离，其中 μ_k 和 Σ_k 是该工况训练期控制量的均值与协方差。将其与历史上 97.5% 的时间均不超过的水平 $q_{k,0.975}$ （四类工况依次为 20.55、18.72、16.85、19.09）相比，即可量化推荐解偏离历史经验的程度，结果见表 12。

表 12 推荐解相对历史运行经验的外推审计

工况与限值	d_M^2	$q_{0.975}$	倍数	最高电压超出历史最大值	判定
工况 1, 10 mg	49.39	20.55	2.40×	0.0%	外推
工况 2, 10 mg	46.46	18.72	2.48×	0.0%	外推
工况 3, 10 mg	77.69	16.85	4.61×	0.0%	外推
工况 4, 10 mg	102.79	19.09	5.38×	0.0%	外推
工况 1, 5 mg	65.84	20.55	3.20×	+5.05%	外推
工况 2, 5 mg	60.31	18.72	3.22×	+5.07%	外推
工况 3, 5 mg	106.26	16.85	6.31×	+5.00%	外推
工况 4, 5 mg	133.43	19.09	6.99×	+5.07%	外推

结论有三层。

其一，最低功率解全都落在历史经验之外， d_M^2 达到该水平的 2.40~6.99 倍。因此本文把 d_M^2 作为**外推风险指标报告**，而不把它当作必须满足的约束——否则求出来的就不是“最低功率”，而是“最低功率且最像以前的操作”，这是两个不同的问题。

其二，如果把 $d_M^2 \leq q_{k,0.975}$ 真的**加成硬约束**，八个“工况—限值”组合里仍有七个有解 (results/mahalanobis_constrained_feasibility.csv)，代价是功率上升：10 mg/Nm³ 档四类工况分别多耗 1.46%~4.38%（加权 2026.5 kW，比最低功率解高 2.7%），5 mg/Nm³ 档三个有解的工况多耗 2.22%~6.07%，只有工况 1 在 5 mg/Nm³ 下无解。这就给出了一个可选的**保守方案**：若现场要求控制组合严格落在历史经验支持的范围内，可以略增电耗为代价采用该方案。本文以最低功率解为主报告结果，同时保留该保守方案，供现场按风险偏好选用。

其三，从电源容量看，10 mg/Nm³ 的四个解在**单变量电压幅值上未显著外推**：各电场电压均不超过该工况历史出现过的最大值。但如上所述，其控制组合仍偏离历史联合运行经验，故实际容量、绝缘与联锁方面的可行性仍须现场核验，不能仅凭幅值合规判定该目标可达。而 5 mg/Nm³ 需要各电场电压比历史最大值再高，最多高出 5.07%；四类工况所需的最小裕度分别是 5%、4%、1%、0%。**这说明 5 mg/Nm³ 首先是容量问题，其次才是电耗问题**：若现场高压电源不具备该裕度，仅靠优化运行参数无法达标，须通过增设电场、改造电源或前端减尘等措施解决。

图 12 把两件不同的事并列：图 12(b) 显示 10 mg/Nm³ 解的各场电压均未超出历史最大值（单变量幅值合规），图 12(a) 却显示同一组解的 d_M^2 已达经验外壳的 2.40~5.38 倍（联合组合越界）。这正说明三个概念必须区分：

单变量历史区间 $\neq$ 历史联合运行支持域 $\neq$ 设备安全域 (11)

前两者的差别由图 12 量化；而第三者——设备安全域——本文无法从附件判断，因为附件不含二次电流、火花率、绝缘状态与联锁信息。因此，“历史上出现过该电压”不能推出“该状态对设备安全可达”。

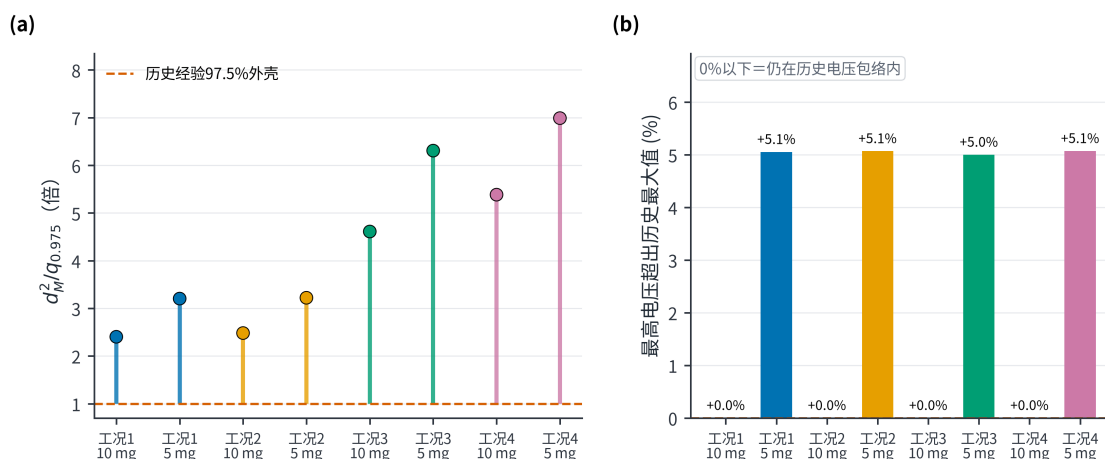


图 12 推荐解相对历史运行经验的外推审计

5.7 排放—功率权衡前沿

只给出 10 与 5 mg/Nm^3 两个孤立点，无法判断二者之间的关系是单调的还是存在拐点，也无法看出限值每收紧一档的边际代价如何变化。为此把排放限值由 10 逐档收紧到 5 mg/Nm^3 并重复求解，得到“排放—功率”权衡曲线（图 13）。四类工况的最优功率随限值收紧单调上升，工况加权功率由 1973.0 kW（10 mg/Nm^3 ）升至 2105.5 kW（5 mg/Nm^3 ），中间各档依次为 1991.3、2012.7、2046.3、2067.7 kW；所需电压裕度同步由 0% 增至 5%（每收紧 1 mg/Nm^3 约增 1 个百分点）。全部档位均高于历史实际加权功率 1783.2 kW。该图把限值与功率的关系整体呈现出来：沿曲线向左（限值更严）必然增加电耗，向右（限值更松）才降低电耗；历史运行点则位于该图右侧更远处。

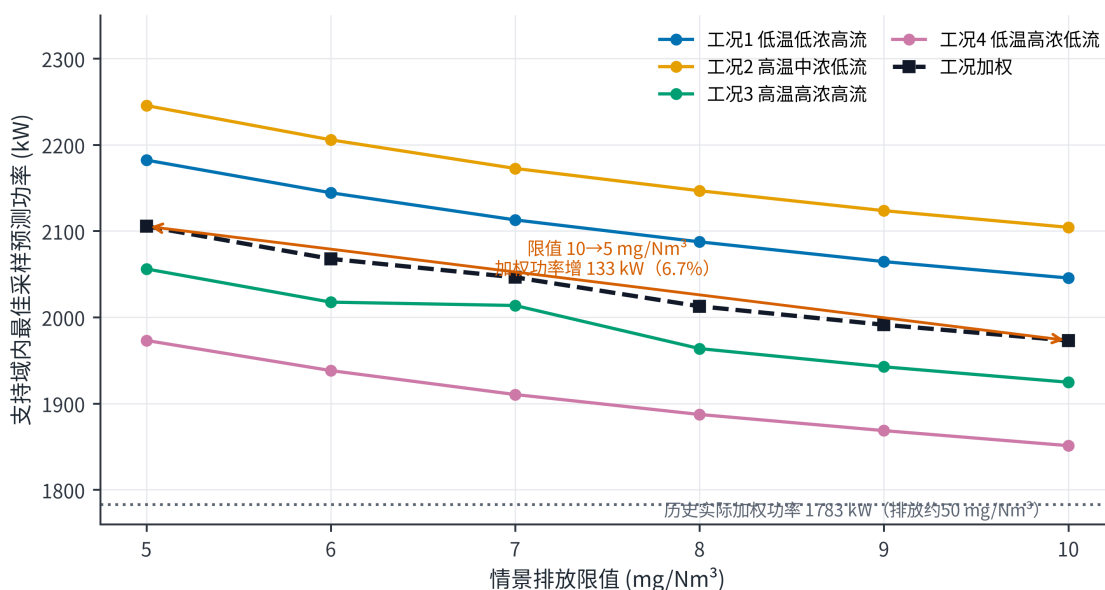


图 13 排放—功率权衡前沿

图 13 给出两点结论。第一，限值越严，最优功率越高，且四类工况与加权曲线均单调，中间不存在“更严反而更省电”的区段。第二，全部档位都位于历史实际加权功率 1783.2 kW 之上。由此可再次明确本文“节能”一词的口径：它只指在同一排放目标下相对其他可行控制方案的功率降低，不能理解为在维持历史排放水平的同时还能额外省电。历史运行点对应的排放水平约为 50 mg/Nm^3 ，位于该图右侧更远处，与图中任一档位都不可直接比较。

6 问题三：代表工况的控制策略比较

第三问要求选取差异显著的两类工况，比较其最优操作参数并解释电压与振打周期的优先级变化。**本节所用模型与方法为：**以 §5.4 的单位电功率去除收益判据 $\alpha_i/(b_i U_i^{ref2})$ 与振打侧的边际节电判据 c_i/T_i^2 为解析依据，对照 §5.5 的数值解读出两类工况的策略差异，并结合 §4.2 的历史操作规律作经验校核，最后整理为可现场执行的控制优先级决策表与三状态运行方案。

选取工况 1 和工况 3。二者流量都较高，但入口浓度和粉尘负荷差异明显：工况 1 平均负荷 12460 kg/h，工况 3 为 19560 kg/h，可用于比较低负荷与高负荷策略。

低负荷工况 1 采用四场同步升压策略（即四个电场按相近幅度同步抬升）：10 mg/Nm³ 下 $U = (70.2, 68.6, 57.6, 57.4)$ kV，四个电场相对各自参考值的抬升幅度接近。原因是该工况锚点 $K_0 = 6.257$ 最低，达标所需的强度因子 $S = 1.288$ 最大，四个电场均须出力，无从取舍。

高负荷工况 3 采用择优升压策略： $U = (68.0, 60.3, 56.3, 57.3)$ kV，其中 U_1 、 U_3 、 U_4 都接近历史最大值，唯独 U_2 停在 60.3 kV（它的历史最大值是 69.3 kV）。这正是表 10 判据的直接体现——工况 3 的优先次序为 $E3 > E4 > E1 > E2$ ，第二电场单位功率收益最低，故被留在低位。限值由 10 收紧到 5 mg/Nm³ 时，各场增量为 $\Delta U = (+3.4, +3.1, +2.8, +3.1)$ kV，这一格局保持不变。

因此，第三问不能简单归结为固定的“前级优先”，而应按单位功率的除尘收益确定调节顺序。以下三条判据可直接计算：

- **电压侧：**按 $\alpha_i/(b_i U_i^{ref2})$ 降序抬升，受排序约束 $U_1 \geq U_3 + 3$ 牵引时同步抬升 U_1 ；
- **振打侧：**按 c_i/T_i^2 降序延长周期，即优先延长周期较短的前级；
- **两者分工：**电压用于提高捕集能力，周期用于降低电耗，二者的最优着力点分别位于不同电场，不应混为一谈。

历史数据对上述判据只提供了部分支持。§4.2 的规律一（入口浓度升高时缩短各场周期）在全部时段方向一致，与高负荷下不宜为节电而延长周期相符；规律二（入口浓度升高时抬升前两级电压）仅在最后两天成立，故本文只将其列作参照。

为便于现场执行，按排放构成与容量状态把上述比较整理成一张决策表，见表 13。

表 13 控制优先级决策规则

状态特征	优先动作	原因
连续排放基值接近限值，振打峰值占比较低	按 $\alpha_i/(b_i U_i^{ref2})$ 降序小步提高电压	主要矛盾是连续捕集强度不足，应优先动用性价比最高的电场
峰值增量占比较高，周期明显长于参考周期	优先缩短过长周期并实施错相振打	主要矛盾是单次积灰脱落和峰值叠加
周期过短、振打频繁，且电耗压力大	优先延长前级周期	前级 c_i/T_i^2 最大，单位延长的节电最多
高粉尘负荷且基值和峰值均高	先错相、再调整周期，最后小步提升电压	先控制峰值，再补充连续捕集能力
某电场电压已达历史最大值	转向次优电场；若各场均达上限则停止优化	数学可行不等于电源容量可行
入口温度进入 140~160 °C 高比电阻带	退出优化，按现场反电晕处置预案运行	该带未被本文工况覆盖，模型外推无效
超出历史支持域或出现火花、连锁风险	停止优化，回退现场安全设定	数学可行不等于设备安全

上述两类策略的差异可由图 14 直接读出。

图 14 显示四个电场并非机械同步抬升：工况 1 四场抬升幅度接近，工况 3、4 则出现明显的高低分化。这一分配由两个因素共同决定——单位功率去除收益 $\alpha_i/(b_i U_i^{ref2})$ 决定优先次序，排序约束 $U_1 \geq U_3 + 3$ 、 $U_2 \geq U_4 + 3$ 则把部分电场的取值绑定在一起。两者叠加的结果是：性价比最高的第

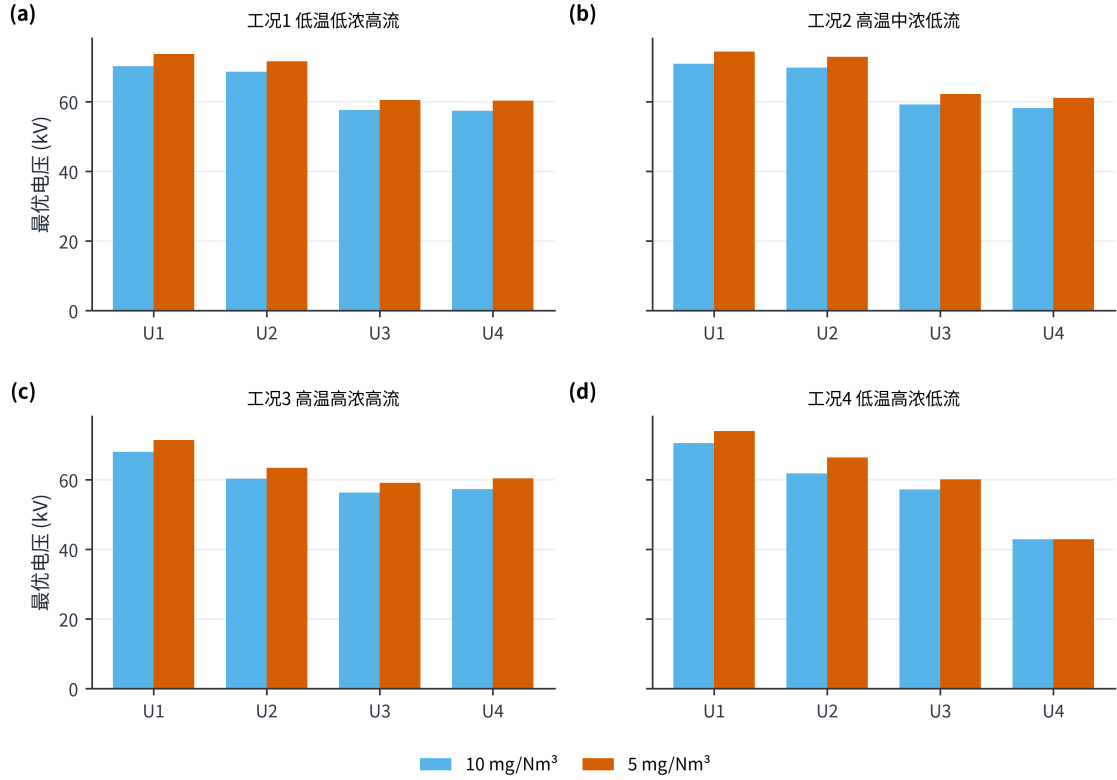


图 14 两档排放限值下的电压配置

三电场被抬到接近上限，与之绑定的 U_1 随之走高，而性价比最低的电场（如工况 4 的 U_4 ）被压至历史下界。各档的振打周期取值见表 11，均被推至历史区间上端。

在线运行设 **TARGET-10**、**TARGET-5**、**FALLBACK** 三种状态。排放目标由上级指令或管理规定选定，不设自动切换阈值——本文排放模型未经现场标定，尚不足以在线自动判定当前排放水平。**TARGET-10** 与 **TARGET-5** 分别读取表 11 中对应工况、对应目标的控制参数。控制量按斜坡逐步逼近目标值，试运行阶段暂定电压变化不超过 0.5 kV/min、振打周期不超过 5 s/min；若现场 DCS、联锁或设备制造商规定了更严的限制，以更严者为准。四场振打先按 $\rho = 1$ （同相振打、峰值完全叠加）这一最不利情形核验，再以 $\rho = 0.80$ 和 0.65 评估错相运行可带来的裕度。传感器失效、火花率超限、工况超出取值范围或联锁触发时，即切换至 **FALLBACK**。

7 问题四：10 收紧至 5 mg/Nm³ 的功率代价

第四问要求把排放约束由 10 mg/Nm³ 收紧到 5 mg/Nm³，估计功率代价并给出高浓度工况建议。**本节所用模型与方法为：**在四场电压同比缩放的简化下导出功率增量的解析式 (§7.1)，与 §5.3 的八变量数值最优解互为验证 (§7.2)，再把该代价与 §4.1 的振打功率结构并列，给出振打侧可回收功率的定量对照 (§7.3)，最后说明峰值约束口径的选取及其偏保守的方向 (§7.4)。

7.1 同比缩放下的解析估计

第四问可在一个简化假设下导出解析式，其解释力强于纯数值优化结果。该假设是：四个电场的电压同比缩放。它并非原八变量非凸优化问题的精确闭式解，但 §7.2 将验证二者相差不足 1%。设在 10 mg/Nm³ 的最优点上，电场功率为 P_{field} 、去除指数为 K_{10} 。把四个电场的电压同时放大 f 倍，则去除指数变成 $K_{10}f^p$ ，电场功率变成 $P_{field}f^2$ 。要让排放减半（计入峰值项后，所需的去除指数增量是 $\Delta K = \ln \frac{10-B}{5-B}$ ），解得 $f = (1 + \Delta K / K_{10})^{1/p}$ ，于是

$$\Delta P = P_{field} \left[\left(1 + \frac{\Delta K}{K_{10}} \right)^{2/p} - 1 \right] \approx \frac{2}{p} \cdot \frac{P_{field}}{K_{10}} \Delta K \quad (12)$$

该式含义清晰：**限值每减半所增加的电耗，等于电场功率乘以所需的去除指数增量、除以当前去除指数，再乘以 $2/p$** 。由此可见，设备当前排放水平越低（ K 越大），进一步减半的相对代价越小。

7.2 分工况结果

表 14 中心情景下 10 收紧至 5 mg/Nm³ 的分工况功率代价

工况	10 mg 功率/kW	5 mg 功率/kW	K_{10}	ΔK	解析 ΔP /kW	数值 ΔP /kW	功率增幅
1 低温低浓度高流量	2045.55	2182.25	8.059	0.804	137.7	136.7	6.68%
2 高温中浓度低流量	2104.30	2245.51	8.338	0.832	142.4	141.2	6.71%
3 高温高浓度高流量	1924.67	2056.06	8.594	0.903	131.3	131.4	6.83%
4 低温高浓度低流量	1851.15	1973.10	8.690	0.906	123.1	122.0	6.59%

解析估计与数值最优解的相对差异不超过 1%（图 15），二者互为验证：数值优化给出结果，解析式给出机理解释。按训练期工况占比加权，

$$\bar{P}_{10} = 1973.0 \text{ kW}, \quad \bar{P}_5 = 2105.5 \text{ kW}, \quad \Delta P / \bar{P}_{10} = 6.72\% \quad (13)$$

按相同运行时长折算，两档日电耗分别约为 47.35 与 50.53 MWh/d，收紧限值的日增电耗约 3.18 MWh/d；按年运行 8760 h 计，年增电耗约 1.16×10^3 MWh。加上 9.78 kW 验证期误差附加量后，两档加权功率分别为 1982.8 与 2115.3 kW；由于两者使用同一附加量，该比例不作为严格上界，只用于功率预算。

值得注意的是，四类工况的增幅高度一致（6.59%~6.83%）。这不是巧合，而是上述解析式的直接推论：增幅 $\approx (2/p)(P_{field}/P)(\Delta K/K)$ ，而四类工况的电场功率占比和当前去除指数都相差不大，所以增幅几乎不随工况变化。

上述闭式估计建立在四场电压同比缩放这一简化之上，而完整优化允许八个控制量各自独立调整。因此有必要检验：这一简化是否会显著偏离八变量数值最优解？

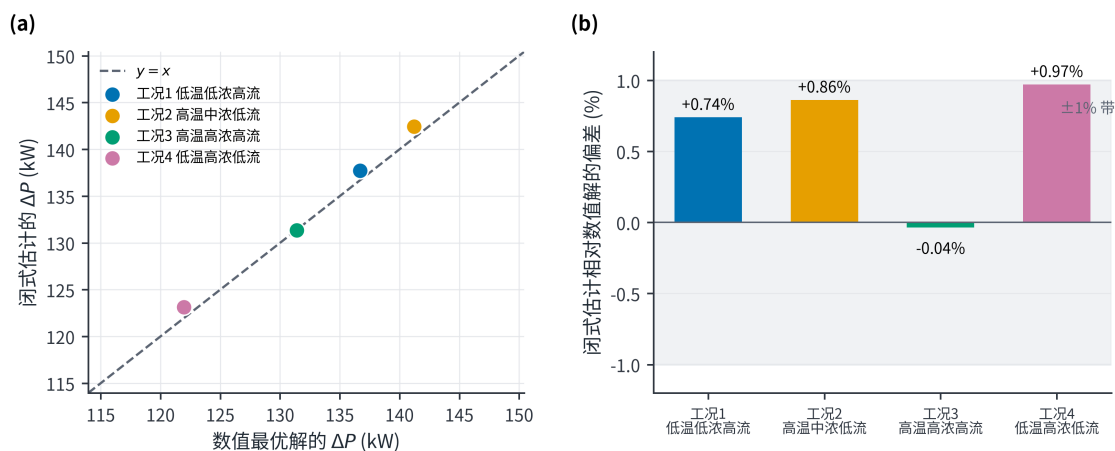


图 15 闭式估计与数值最优解的一致性

图 15(a) 中四类工况的点均紧贴 $y = x$ 线，图 15(b) 给出相对偏差为 $-0.04\% \sim +0.97\%$ ，全部落在

$\pm 1\%$ 带内。这说明按同比缩放导出的解析式虽是近似，却能以不足 1% 的误差复现完整的八变量数值优化结果，因而可以作为第四问功率代价的解析解释使用：数值优化给出数值，解析式给出机理。

7.3 真正的权衡：振打周期能买回多少

将第四问的结果与 §4.1 的振打功率结构并列，可得到本文最具操作价值的一组对照。按工况加权计算，限值由 10 收紧到 5 mg/Nm³ 要多耗约 132.6 kW；而把四个电场的振打周期从历史中位延长到历史 P95 可省下 56.1 kW，延长到历史最大值可省下 109.1 kW。即振打侧可挖掘的全部潜力（约 109 kW），相当于排放收紧一档所增电耗（约 133 kW）的 82%，二者处于同一量级。

但上述节电收益必须与振打峰值排放的代价一并权衡。按 §4.5 的推导，把周期由中位延长至历史最大值（平均延长 17.9%）时，单次振打的灰饼质量与瞬时峰值同比放大约 17.9%。而在表 11 的解中，峰值项本就已经占掉限值的 0.95~1.61 mg/Nm³，相当于 5 mg/Nm³ 这一档约五分之一到三分之一的余量。峰值再放大 17.9%，将进一步占用 0.17~0.29 mg/Nm³ 的排放裕度，该部分必须由电压侧补偿，而补偿电压本身又要付出功率代价。这正是振打周期不宜为节电而无限延长的定量表述：振打周期是本设备上成本最低的节电手段，但其额度有限，且以瞬时排放峰值为代价。

7.4 峰值约束口径的说明

在中心情景的解中，10 mg/Nm³ 方案的连续排放基础值为 8.38~9.00 mg/Nm³，振打带来的峰值增量为 0.95~1.61 mg/Nm³；5 mg/Nm³ 方案的基础值则降到 3.38~4.05 mg/Nm³。可见仅令平均值或基础值低于限值并不充分，否则振打时刻仍可能超限。

需要说明的是，本文把瞬时峰值直接卡到限值，而现行排放标准对颗粒物通常按小时均值考核。按 §4.5 的推导，一次持续数秒到数十秒的再飞扬峰值，对小时均值的贡献正比于 L 而与周期无关。因此本文口径系统性偏保守，会高估达标所需功率。若改用小时均值达标、峰值仅作软性惩罚的口径，所得功率增幅将低于本文报告值。在缺少逐次振打数据的条件下，本文选取保守口径，但不应将其视为唯一的合规解释。

仅约束连续排放并不充分，因为振打瞬间的峰值同样占用限值裕度。图 16 给出两档解中限值裕度的实际构成。

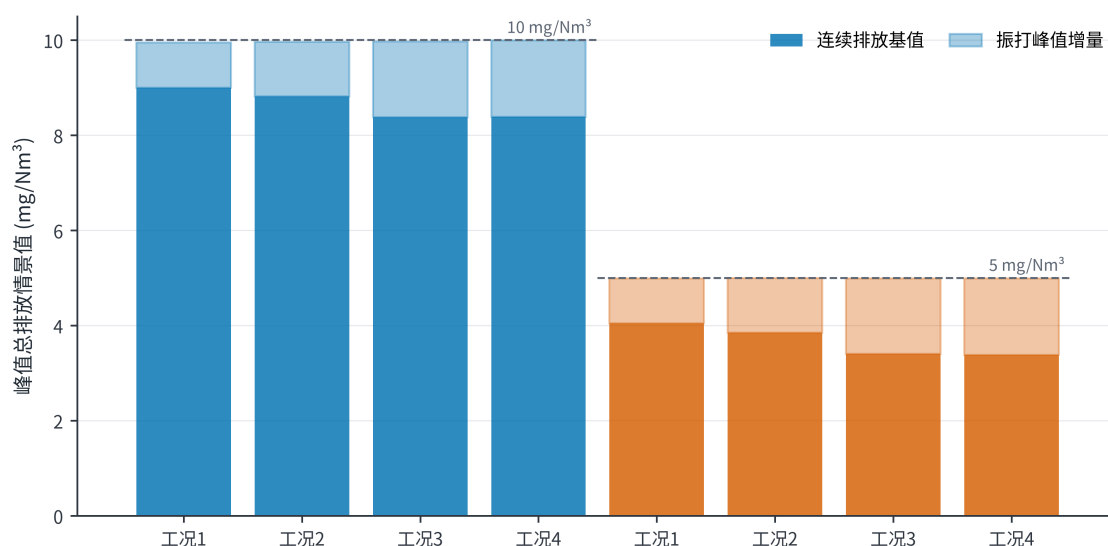


图 16 排放约束分解

图 16 显示峰值项在 10 mg/Nm³ 档占用 0.95~1.61 mg/Nm³，在 5 mg/Nm³ 档更相当于限值的五分之一至三分之一。这正是本文把约束写成 $C_{peak} = C_{base} + B$ 并整体施加、而非只约束 C_{base} 的原因：

若只令连续基值达标，振打时刻仍可能超限。

8 模型检验、敏感性与稳健性

8.1 求解质量

§5.3 已说明本问题不是凸规划，因此单次求解可能只收敛到局部解，必须检验不同初值是否给出不同结果。图 17 给出两项独立检验。八个“工况—限值”组合中，多起点 SLSQP 有 47~53 个起点成功收敛，**所有收敛起点给出的最优功率一致到 10^{-7} kW 以内**，表明各起点均收敛至同一解。作为独立对照，在同一可行域内均匀随机抽取 4×10^5 个候选点的随机搜索，在全部八个组合上均未找到更优解，其最优值反而高出 0.97%~5.17%；其中工况 1 的 10 mg/Nm³ 组合甚至未采到任何可行样本。原因在于可行域被排放约束与排序约束夹成一条窄带，均匀随机采样的命中效率很低。

该对照说明：在问题结构明确之后，确定性求解优于大规模随机搜索。但仍须指出，对非凸问题而言，多起点收敛至同一点是很强的证据，而非数学证明。因此本文只称“最优解”，不宣称“全局最优”。

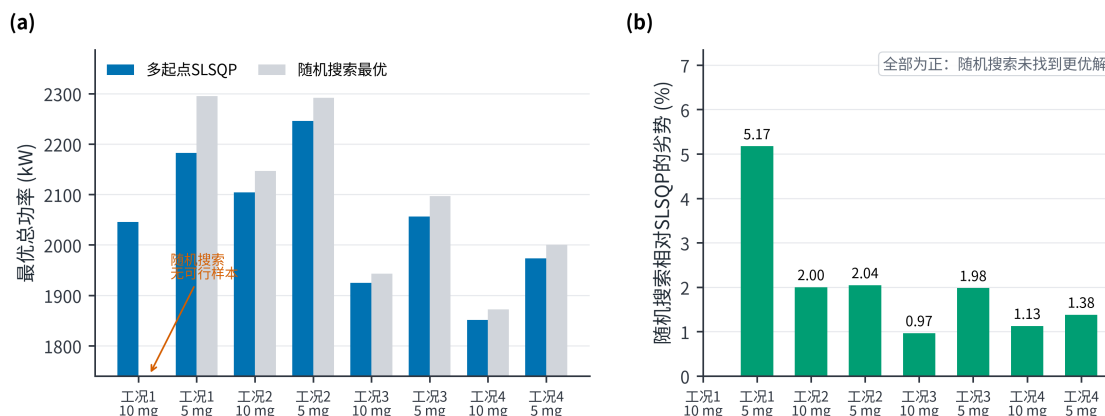


图 17 求解方法对照

图 17(a) 显示两种方法在八个组合上的最优功率，图 17(b) 给出随机搜索相对多起点 SLSQP 的劣势。三点结论：47~53 个起点全部收敛到同一功率； 4×10^5 点随机搜索未找到任何更优解；因此该解在数值上是稳定的。但仍须重申，**这是数值稳定性的强证据，不构成全局最优的数学证明——**对非凸问题，多起点一致只能降低遗漏更优解的可能性，不能排除它。

8.2 先验敏感性

排放层含 $p, \alpha_i, \beta_i, b_0, \gamma, \rho, s$ 共七类先验参数，逐一扫描既无必要也不便阅读。本节按三个层次组织：先看结构设定变化后核心结论是否稳定（层次一），再定位真正主导功率代价与可达性的参数（层次二），最后检验电场优先级是否依赖权重设定（层次三）。

层次一：结构情景下核心结论是否稳定。扫描电场权重形状（3 种） $\times$ 峰值指数 $\gamma \in \{1.00, 1.30, 1.60\} \times$ 相位尺度 $\rho \in \{0.65, 0.80, 1.00\} \times$ 参考周期负荷指数（3 种），共 81 组。

图 18 显示 81 组情景全部可行，加权增幅为 5.13%~7.55%、中位数 6.40%，中心值 6.72% 位于该区间内。其中峰值指数在推导值 1.00 附近变化时增幅几乎不变，这与 §4.5 一致——峰值项占用的是限值余量，本身不是主导项。**即核心结论对灰箱的形状设定并不敏感。**

层次二：什么参数真正主导功率代价与可达性。既然形状设定影响有限，就需要找出真正起决定作用的参数。由 §7.1 的解析式，功率增幅近似正比于 $2/p$ ，即幂次 p 以显式方式进入结论；其余先验只通过峰值裕度间接影响结果。这正是本文把 p 单独提出的原因。表 15 给出 p 的参考表。

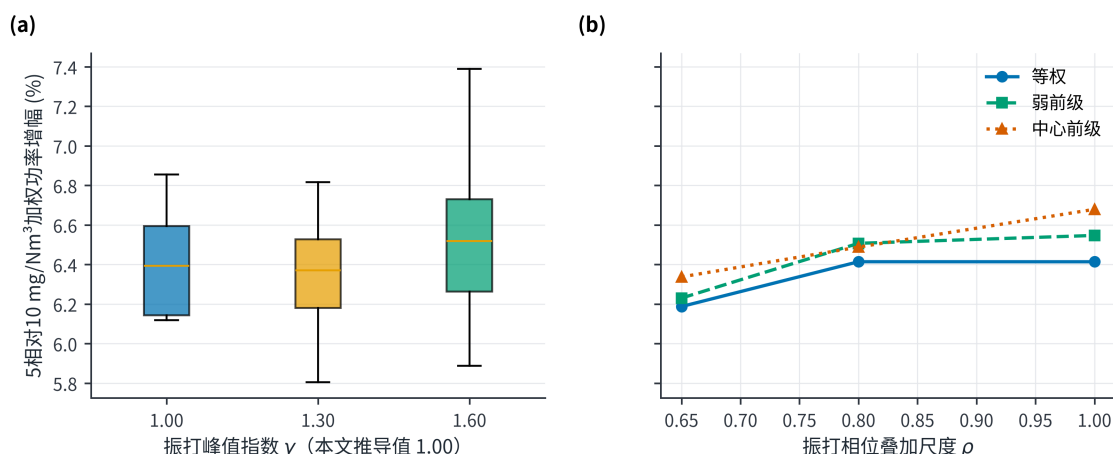


图 18 结构情景下的功率增幅分布

表 15 功率增幅对幂次 p 的依赖

p	物理含义	10→5 mg/Nm³ 加权增幅	5 mg/Nm³ 在 5% 电压裕度内是否可达
1.0	去除指数正比于电压	14.2%	否, 需更大电压裕度
1.5	中间情形	9.2%	否, 需更大电压裕度
2.0	Deutsch 迁移速度形式 (中心)	6.8%	是
2.5	强电压响应	5.4%	是
3.0	极强电压响应	4.4%	是

表中增幅为闭式估计值；对 $p \geq 2$ 的三档另作数值重解校验（图 19 叉号），与解析值相差不超过 0.5 个百分点。

图 19 表明 p 同时决定两件事：**功率增幅**（由 14.2% 降至 4.4%，跨度超过三倍）与**目标可达性**（ $p = 1$ 与 $p = 1.5$ 在 5% 电压裕度内根本无解）。作为对照，另一组扫描—— $p \in \{1.5, 2.0, 2.5\} \times$ 峰值尺度 $b_0 \in \{0.9, 1.2, 1.5\} \times$ 安全裕度 $s \in \{0, 0.08, 0.15\}$ 共 27 组——中，15 组在 5% 电压裕度内四类工况全部可行，不可行的 12 组也全部来自 $p = 1.5$ ；有解各组的加权增幅为 3.99%~7.04%、中位数 5.85%。

因此，就 10→5 mg/Nm³ 的功率增幅与目标可达性而言， p 是主导性的待标定参数，且一次电压阶跃试验即可确定：在工况稳定时把四个电场的电压同时抬高几个百分点，记录未截顶的出口浓度如何变化，再把 $\ln C_{out}$ 对 $\ln U$ 作回归，斜率即为 $-K_0 p$ 。但须明确，这不意味着 p 是唯一需要标定的参数： $\alpha_i, \beta_i, b_0, \gamma, \rho, s$ 同样未由附件识别，它们影响峰值结构与分场控制的具体分配，仍须由后续现场数据一并标定 (§9.3)。

层次三：电场优先级是否依赖权重设定。第三问的判据含先验权重 α_i ，需检验结论是否由该先验单独写入。

图 20 显示，在等权、弱前级与中心前级三种权重形状下，10→5 mg/Nm³ 所需的加权电压增量分布保持一致的格局，与按 $\alpha_i / (b_i U_i^{ref 2})$ 排序的解析优先级相符。这说明优先级结论并非由中心权重单独决定，但其具体数值仍依赖同一灰箱结构与支持域。

需要强调，以上区间均为先验参数网格上的情景范围，而非统计意义上的置信区间。

8.3 稳健性结论

稳健性证据分四层。第一，轮廓系数支持把工况分成四类。第二，四种设定的比较与时间外推支持所选的功率模型，其中 $1/T$ 形式在完全未参与拟合的测试日上表现最好，这是本文最强的实证证据。第三，多起点收敛一致以及随机搜索的对照，支持数值解的质量。第四，解析估计与数值最优解互相验证，情景扫描与权重消融则检验了结论对建模选择的依赖程度。

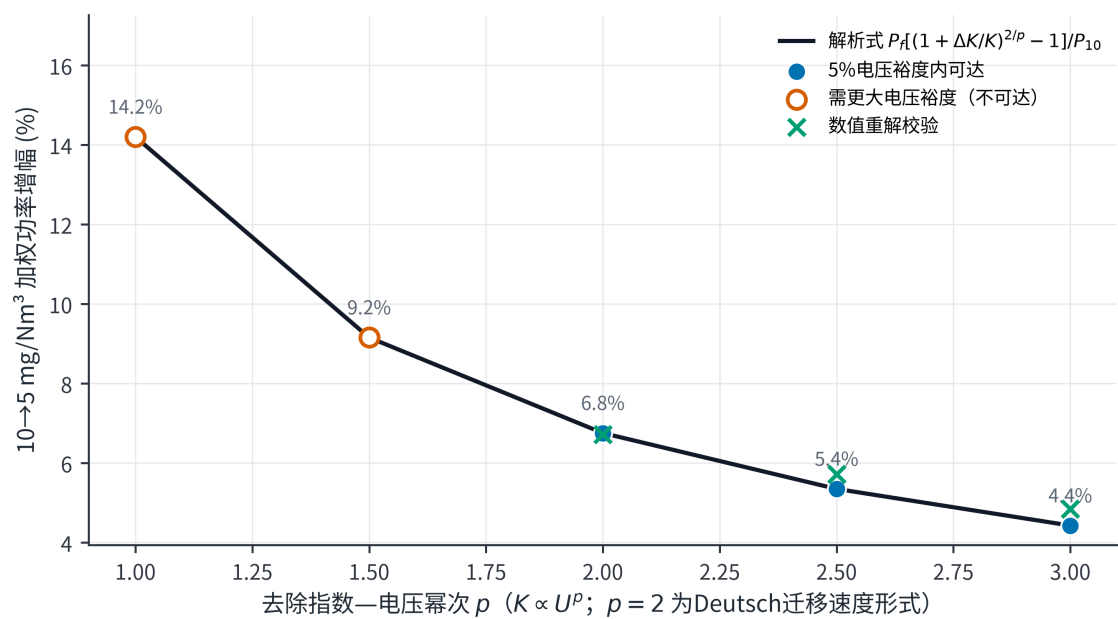


图 19 功率增幅对幂次 p 的依赖

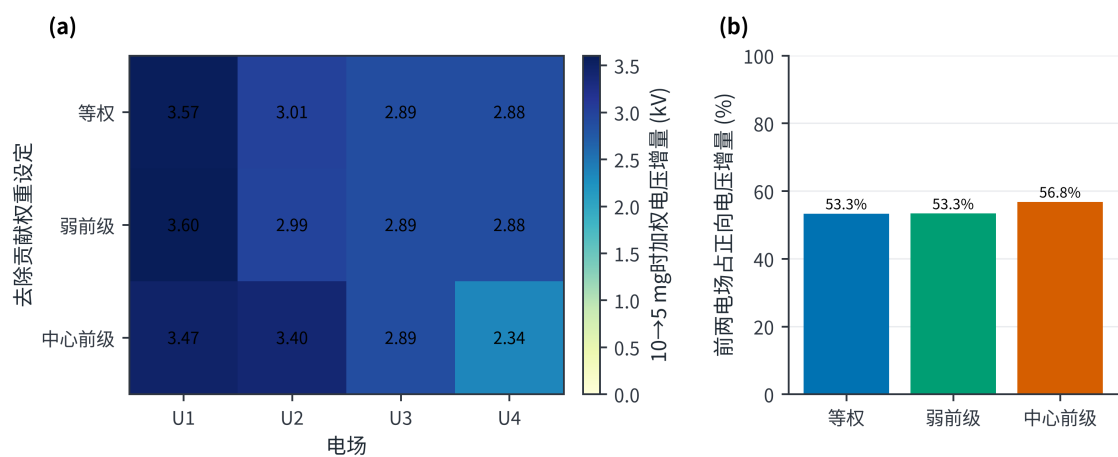


图 20 电场贡献权重消融结果

但仍须强调：排放层的全部数值都是机理推演的结果。无论是误差余量还是情景范围，都不能把一个机理模型变成经过实验验证的模型。

9 模型评价、实施方案与局限

9.1 模型优点

第一，先逐列判定数据能够支持与不能支持的结论，既避免以复杂算法拟合截顶噪声，也未因一列不可用而放弃其余 13 列的信息。

第二，功率模型采用有物理含义的形式而非通用多项式，在未参与拟合的测试日上达到 $R^2 = 0.993$ 、偏差 -0.02 kW ，并由此发现“振打占电耗近四成”这一可直接使用的结论。

第三，排放起算点直接取记录值，不引入未经证实的量纲假设，使第二问回答的是题目实际提出的问题。

第四，电场优先级由公式推出，而不是从某一次求解的现象里归纳。

第五，第四问给出解析式，把功率代价的主要不确定性归结到一个可现场标定的幂次 p ，并同时说明其余灰箱参数仍待标定。

第六，把推荐解偏离历史运行经验的程度作为结论的一部分报告，并给出严格落在历史经验之内的保守方案及其电耗代价，而非回避该问题。

9.2 模型局限

最主要的局限是，排放层的全部系数 $(p, \alpha_i, \beta_i, b_0, \gamma, \rho, s)$ 都无法从附件中辨认，只能依据机理和文献设定；表 15 给出的区间是人为参数网格上的情景范围，不是统计置信区间。

其次，历史运行范围并不等于设备的安全边界， 5 mg/Nm^3 档所需的 5% 电压外推更须经高压电源容量与绝缘裕度核算。模型也没有二次电流、分场功率、火花率、极板积灰状态和联锁信息；分钟级数据不足以观察逐次振打峰值，因此相位叠加尺度 ρ 无法标定。四类工况的温度中心都没有覆盖 $140 \sim 160^\circ\text{C}$ 的高比电阻区间，本文结论在该区间无效。此外，测试日的数据曾被前期处理流程接触过，不能算作完全未触碰的盲测。

因此，本文给出的控制参数只能作为现场试验的起点，不得直接下发生产设备。

9.3 上线与补采方案

建议分四步部署。

第一阶段：影子运行。仅在后台计算、不改变设备设定，用以验证工况识别与 TARGET-10/TARGET-5 参数表的加载逻辑；同期完成高压电源容量核算，确认 5 mg/Nm^3 所需的电压裕度是否存在。

第二阶段：小步阶跃标定。在每类工况内执行单因素小步阶跃，重点是标定幂次 p (§8.2)，并由现场确定电压与周期的实际变化速率上限。

第三阶段：重新标定。以 $1 \sim 5 \text{ s}$ 的未截顶出口浓度、实际振打事件、二次电流、火花率与分场功率，重新标定排放层参数及相位尺度 ρ ，并单独补采 $140 \sim 160^\circ\text{C}$ 高比电阻区间的数据。

第四阶段：独立盲测与闭环投用。冻结模型，在全新日期上盲测，通过后方允许投入闭环。任何超出取值范围、传感器失效或联锁事件均应触发 FALLBACK。EPA 技术资料与相关实验都强调气流分布、收尘面积和再飞扬对性能评价的重要性 [2-3,6]，因此补采数据时应优先补齐这几项。

10 结论

1. **问题一：分层作答。**可由数据确定的部分：总功率几乎完全由控制量决定， $P = 73.1 + 0.0850 \sum U_i^2 + 5.39 \times 10^4 \sum T_i^{-1}$ （测试日 $R^2 = 0.993$ ），其中振打环节在中位工况下占总功

率的 38.8%，而在现有数据精度下工况变量对总功率无额外解释力；历史操作规律中，入口浓度升高时缩短各场振打周期这一规律在四个时段方向一致，而入口浓度升高时抬升前级电压则仅在最后两天成立；附件还含两段持续异常工况，其中 05-06 的 158.2 °C 事件正落在粉尘比电阻的高值区。不可由数据确定的部分：出口浓度记录在 50 mg/Nm³ 附近呈现明显的上限截顶特征，与全部 12 个变量的相关系数绝对值均小于 0.017，无法实证识别控制量对排放的作用。振打峰值则由积灰量守恒推得：单次峰值随周期近似线性放大，而其对小时均值的贡献与周期无关。

2. **问题二：4 类工况与 10 mg/Nm³ 方案。**基于入口三变量识别出 4 类典型工况（轮廓系数 0.4084）。以记录值 50 mg/Nm³ 为锚点，四类工况满足 10 mg/Nm³ 的最低功率为 1851.15~2104.30 kW，加权 1973.0 kW，日电耗 47.35 MWh，较历史实际的 1783.2 kW 上升 10.6%。收紧排放必然增加电耗，同一排放水平下不存在额外的节能空间。该方案电压仍位于历史范围之内，但已明显偏离历史常规操作（马氏距离达经验 97.5% 水平的 2.4~5.4 倍）；若要求严格落在历史经验之内，四类工况仍均可行，加权功率升至 2026.5 kW，多耗 2.7%。
3. **问题三：优先级可由解析判据确定。**单位电功率的除尘收益为 $K_0\alpha_i/(\sum \alpha \cdot b_i U_i^{ref2})$ 。第三电场参考电压最低，该收益在四类工况中均居首；“前级优先”的结论实由排序约束 $U_1 \geq U_3 + 3$ 导出。振打侧按 c_i/T_i^2 排序，应优先延长周期较短的前级。低负荷工况须四场同步升压，高负荷工况则可择优升压，使单位功率收益最低的电场留在低位。
4. **问题四：功率代价与容量约束。**限值由 10 收紧至 5 mg/Nm³ 时，同比缩放假设下的解析估计 $\Delta P = P_{field} \ln[(10 - B)/(5 - B)]/K_{10}$ 给出 123.1~142.4 kW，与八变量数值最优解相差不足 1%；加权功率由 1973.0 升至 2105.5 kW，增幅 6.72%，日电耗由 47.35 增至 50.53 MWh。增幅与 $2/p$ 成正比；就功率增量与目标可达性而言， p 是主导性的待标定参数，但 $\alpha_i, \beta_i, b_0, \gamma, \rho, s$ 同样未由附件识别，仍须现场一并标定。更关键的约束不是电耗而是电源容量：5 mg/Nm³ 要求各电场电压超出历史最大值最多 5.07%，四类工况所需的最小裕度分别为 5%、4%、1%、0%。若既有电源无法提供该裕度，仅靠优化运行参数无法达标，应考虑电源改造、增设电场或前端减尘等措施。以上排放数值均属机理推演，必须在补采未截顶的排放数据、振打事件与电气状态后重新标定。

参考文献

- [1] 中华人民共和国生态环境部. 水泥工业大气污染物排放标准：GB 4915—2013（含 2025 年修改单）[EB/OL]. (2013-12-27)[2026-08-06]. https://www.mee.gov.cn/ywgz/fgbz/bz/bzwb/dqjhjbh/dqgdwrywrwpfbz/201312/t20131227_265765.htm.
- [2] OGLESBY S Jr, NICHOLS G B. *A Manual of Electrostatic Precipitator Technology: Part I—Fundamentals*[R/OL]. Birmingham: Southern Research Institute, 1970[2026-08-06]. <https://nepis.epa.gov/Exe/ZyPURL.cgi?Dockey=9101FJFA.TXT>.
- [3] SZABO M F, SHAH Y M, SCHLIESSER S P. *Inspection Manual for Evaluation of Electrostatic Precipitator Performance*[R/OL]. Washington, DC: U.S. Environmental Protection Agency, 1981[2026-08-06]. <https://nepis.epa.gov/Exe/ZyPURL.cgi?Dockey=9400034C.TXT>.
- [4] MIZUNO A. Electrostatic precipitation[J]. IEEE Transactions on Dielectrics and Electrical Insulation, 2000, 7(5): 615-624. DOI: 10.1109/94.879357.
- [5] LI Z, SHAO C, AN Y, et al. Energy-saving optimal control for a factual electrostatic precipitator with multiple electric-field stages based on GA[J]. Journal of Process Control, 2013, 23(8): 1041-1051. DOI: 10.1016/j.jprocont.2013.06.007.
- [6] NAVARRETE B, VILCHES L F, RODRIGUEZ-GALAN M, et al. A pilot scale study of the rapping

reentrainment and fouling in electrostatic precipitation[J]. Environmental Progress & Sustainable Energy, 2015, 34(1): 7-14. DOI: 10.1002/ep.11934.

[7] ZHENG C, ZHAO Z, GUO Y, et al. A real-time optimization method for economic and effective operation of electrostatic precipitators[J]. Journal of the Air & Waste Management Association, 2020, 70(7): 708-720. DOI: 10.1080/10962247.2020.1767227.

[8] FU Y, SU Z. Deep neural network based concentration prediction model of dry electrostatic precipitator with a modified PID optimizer[J]. Journal of Physics: Conference Series, 2022, 2189: 012018. DOI: 10.1088/1742-6596/2189/1/012018.

[9] MACQUEEN J. Some methods for classification and analysis of multivariate observations[C]//Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability. Berkeley: University of California Press, 1967, 1: 281-297.

[10] ROUSSEEUW P J. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis[J]. Journal of Computational and Applied Mathematics, 1987, 20: 53-65. DOI: 10.1016/0377-0427(87)90125-7.

[11] HOERL A E, KENNARD R W. Ridge regression: Biased estimation for nonorthogonal problems[J]. Technometrics, 1970, 12(1): 55-67. DOI: 10.1080/00401706.1970.10488634.

[12] WHITE H J. *Industrial Electrostatic Precipitation*[M]. Reading, MA: Addison-Wesley, 1963.

[13] PARKER K R. *Electrical Operation of Electrostatic Precipitators*[M]. London: Institution of Electrical Engineers, 2003.

附录 A 与既有多电场节能优化研究的对照

本文与文献 [5]、[7] 同属“在排放约束下对多个电场分别优化”这一类工作，但两点不同：排放模型怎么标定，以及可行范围怎么划。对照见表 A1。

表 A1 与既有多电场节能优化的方法对照

维度	文献 [5] (GA 优化)	文献 [7] (实时优化)	本文
排放模型标定	由现场实测出口浓度回归	由在线仪表实时校正	出口浓度不可识别，改以记录值为锚点的 Deutsch 灰箱，全程标注为推演
功率模型	分场电源功率实测	分场功率与二次电流	仅有总功率，按 $U^2 + 1/T$ 物理设定辨识，测试 $R^2 = 0.993$
振打处理	未纳入优化	作为固定运行参数	纳入决策变量，并由积灰量守恒推导峰值标度
求解方法	遗传算法	滚动实时优化	多起点 SLSQP + 解析优先级判据 + 随机搜索对照
可行域	设备额定范围	在线安全约束	历史运行包络，越界量作为结论显式报告

上述工作的共同前提是排放模型可由现场数据标定。本文缺少这一条件，其处理方式是把不可标定的部分全部收敛到单一幂次 p ，并给出具体的标定方案 (§8.2) ——这是本文与上述工作的主要差别。

附录 B 以出口浓度为直接目标的对照建模

为便于与其他方案交叉比较，这里给出直接以 C_{out_mgNm3} 为目标建模的结果（训练/验证仍按时间划分）：

1. 以是否等于 50 为标签作分类，验证集 AUC=0.5098，与随机猜测无异；
2. 只用未截顶的样本做回归，验证集 $R^2 = -0.0190$ ；
3. 用全部记录值做回归，验证集 $R^2 = -0.0219$ 。

后两项的 R^2 为负，说明模型表现不及常数预测。因此，任何在该列上报告高 R^2 的模型，其性能几乎必然源于以下两种情形之一：随机划分数据导致的时间泄漏，或仅复现了 5491 条恒为 50 的截顶值。

附录 C 结果适用性声明

本文的功率模型、工况划分、历史操作规律、异常工况识别以及数据质量统计，均直接来自附件，可用 src/ 中的程序复算。

排放层的各项系数、振打峰值系数，以及由此得到的 $5/10 \text{ mg/Nm}^3$ 控制参数和功率增幅，则属于机理推演。若数据提供方给出新的未截顶排放数据，或现场标定得到的幂次与 $p = 2$ 不同，应按 §8.2 重新计算，不得沿用本文的数值。 5 mg/Nm^3 档所需的电压外推，须先经高压电源容量与绝缘裕度核算。

附录 D 支撑材料与 AI 工具使用声明

求解流程：

输入：分钟级数据、排放限值 C_{Lim}

1. 按时间划分训练/验证/回顾性测试；
 2. 逐列可识别性审计；提取历史操作规律与异常工况；
 3. 在训练期对 Temp、 C_{in} 、 Q 标准化，K-means 取 $k=4$ ；
 4. 比较四种功率设定，选定 $P = a_0 + \sum b_i U_i^2 + \sum c_i / T_i$ ；
 5. 以记录值 50 mg/Nm^3 为锚点建立灰箱排放层；由积灰量守恒定 $\gamma=1$ ；
 6. 对每个工况与限值：
 - 6.1 取该工况控制量的历史最小—最大范围为支持域；
 - 6.2 多起点 SLSQP 求解（5 结构起点 + 48 随机起点）；
 - 6.3 舍入至 $0.1 \text{ kV} / 1 \text{ s}$ 后按解析优先级修复可行性；
 - 6.4 以 4×10^5 点随机搜索独立对照；
 7. 报告马氏距离外推审计与所需电压裕度；
 8. 扫描 27 组尺度情景与 81 组结构情景，给出幂次 p 参考表；
- 输出：控制参数、功率代价、外推审计、敏感性、图表与运行摘要。

源程序清单： src/scenario_model.py（功率模型、灰箱排放情景、支持域优化与敏感性）、src/paper_figures.py（21 幅正文图与图注追踪表）、src/plot_utils.py（统一绘图样式）、src/integrity_check.py（确定性核验）、src/compute_frontier.py（排放—功率前沿）、src/build_paper_pdf.py（提交稿装配）。生成本文全部结果与图表的完整可运行源程序见附录 E，与电子支撑材料 src/ 目录逐字一致。

支撑材料清单： data/处理数据包；results/下功率基线、滚动验证、最优控制、外推审计、电压裕度需求、求解对照、结构情景、场权重消融、幂次参考表、动态控制表和运行摘要；figures_paper/下 21 幅正文图（20 幅 PNG/PDF 由绘图程序生成，图 1 为 TikZ 源文件独立编译）及 figure_manifest.csv；integrity/下确定性核验结果。

AI 工具使用声明（提交前由参赛队核验确认）： 本稿坚持参赛队人工主导。题意分析、模型构建、公式推导、参数设定、程序审阅、结果复算和论文定稿由参赛队承担；AI 编码助手仅用于代码排错、图表版式检查、文字校对与格式核验。AI 输出均经人工筛选和复核，未替代现场数据，也未把不可识别的排放关系改写为实证结论。

附录 E 完整源程序

本附录给出生成本文全部结果与图表的完整可运行源程序，与电子支撑材料 src/ 目录逐字一致。按依赖顺序排列：情景模型与优化、排放—功率前沿、绘图样式与图集生成、确定性核验、提交稿装配。

E.1 scenario_model.py

```
1  #!/usr/bin/env python3
2  """Reproducible scenario model for the cement ESP problem.
3
4  Two layers are kept strictly separate:
5
6  * the **power layer** is identified from the supplied data. Total power is a
7    near-deterministic function of the control settings alone, with the physically
8    motivated form  $P = a_0 + \sum_i b_i U_i^2 + \sum_i c_i / T_i$  (field power
9    proportional to voltage squared, rapping power proportional to rapping
10   frequency  $60/T_i$ );
11  * the **emission layer** is an explicitly labelled engineering scenario,
12    because the recorded outlet concentration is capped near 50 mg/Nm3 and
13    carries no learnable signal.
14
15  The historical emission anchor is the *recorded* outlet concentration; no
16  decimal-point rescaling is applied. The control problem solved here is
17  therefore the path from the historical ~50 mg/Nm3 level down to the 10 and
18  5 mg/Nm3 targets set by the problem statement.
19  """
20
21  from __future__ import annotations
22
23  import json
24  import math
25  from pathlib import Path
26
27  import numpy as np
28  import pandas as pd
29  from scipy.optimize import minimize
30
31
32  SEED = 20260805
33  ROOT = Path(__file__).resolve().parents[1]
34  DATA_FILE = ROOT / "data" / "full_timeseries_with_flags.csv"
35  CLUSTER_EVAL_FILE = ROOT / "data" / "condition_cluster_evaluation.csv"
36  OUT_DIR = ROOT / "results"
37
38  U_COLS = [f"U{i}_kV" for i in range(1, 5)]
39  T_COLS = [f"T{i}_s" for i in range(1, 5)]
40  COND_COLS = ["Temp_C", "C_in_gNm3", "Q_Nm3h"]
41
42  # ---- central scenario constants (engineering priors, not fitted to C_out) ----
43  ALPHA_CENTRAL = np.array([1.15, 1.10, 0.95, 0.80]) # per-field removal weights
44  BETA_CENTRAL = np.array([0.30, 0.27, 0.20, 0.17]) # rapping-period deviation loss
45  PEAK_SHARES = np.array([0.30, 0.30, 0.22, 0.18]) # per-field share of the peak proxy
46  P_EXPONENT_CENTRAL = 2.0 #  $K \sim U^p$ ;  $p = 2$  is the Deutsch drift-velocity form
47  PEAK_EXPONENT_CENTRAL = 1.0 # derived: cake mass per rap  $\sim L \cdot T \Rightarrow$  peak  $\sim T^1$ 
48  PEAK_SCALE_CENTRAL = 1.2
49  PHASE_SCALE_CENTRAL = 1.0
50  SAFETY_INDEX_CENTRAL = 0.08
51  # 5 mg/Nm3 needs voltage above every field's historical maximum; the margin is
52  # an explicitly declared extrapolation of the historical envelope, not a fact.
```

```

53 HEADROOM_10 = 0.00
54 HEADROOM_5 = 0.05
55
56 MULTISTART = 48
57 SWEEP_MULTISTART = 16 # sensitivity grids: fewer starts, cross-checked against the central solve
58
59
60 def _r2(y: np.ndarray, pred: np.ndarray) -> float:
61     return float(1.0 - np.sum((y - pred) ** 2) / np.sum((y - y.mean()) ** 2))
62
63
64 def _rmse(y: np.ndarray, pred: np.ndarray) -> float:
65     return float(np.sqrt(np.mean((y - pred) ** 2)))
66
67
68 # ----- power models
69
70 def voltage_polynomial(u: np.ndarray) -> np.ndarray:
71     """Equivalent to PolynomialFeatures(degree=2, include_bias=False) for 4 voltages."""
72     cols = [u[:, i] for i in range(4)]
73     for i in range(4):
74         for j in range(i, 4):
75             cols.append(u[:, i] * u[:, j])
76     return np.column_stack(cols)
77
78
79 class _ScaledOLS:
80     """Least squares with per-column scaling.
81
82     The scaling matters: ``1/T`` is of order 4e-3 while ``U^2`` is of order
83     3e3, and an unscaled ``lstsq`` truncates the rapping columns as though they
84     were rank-deficient, which makes the rapping term look irrelevant.
85     """
86
87     def __init__(self) -> None:
88         self.beta: np.ndarray | None = None
89
90     def _fit_design(self, x: np.ndarray, y: np.ndarray) -> None:
91         xx = np.column_stack([np.ones(len(x)), x])
92         scale = np.abs(xx).max(axis=0)
93         scale[scale == 0] = 1.0
94         beta, *_ = np.linalg.lstsq(xx / scale, y, rcond=None)
95         self.beta = beta / scale
96
97     def _predict_design(self, x: np.ndarray) -> np.ndarray:
98         if self.beta is None:
99             raise RuntimeError("model not fitted")
100         return self.beta[0] + x @ self.beta[1:]
101
102
103 class PhysicalPowerModel(_ScaledOLS):
104     """ $P = a_0 + \sum_i b_i U_i^2 + \sum_i c_i / T_i$  (optionally + condition terms)."""
105
106     name = "physical_voltage_square_plus_rapping_frequency"
107
108     def __init__(self, with_conditions: bool = False) -> None:
109         super().__init__()
110         self.with_conditions = with_conditions
111
112     def design(self, frame: pd.DataFrame) -> np.ndarray:
113         u = frame[U_COLS].to_numpy(float)
114         t = frame[T_COLS].to_numpy(float)

```

```

115     blocks = [u ** 2, 1.0 / t]
116     if self.with_conditions:
117         blocks.append(frame[COND_COLS].to_numpy(float))
118     return np.column_stack(blocks)
119
120 def fit(self, frame: pd.DataFrame, y: np.ndarray) -> "PhysicalPowerModel":
121     self._fit_design(self.design(frame), y)
122     return self
123
124 def predict(self, frame: pd.DataFrame) -> np.ndarray:
125     return self._predict_design(self.design(frame))
126
127 # -- fast path used inside the optimiser -----
128 @property
129 def intercept(self) -> float:
130     return float(self.beta[0])
131
132 @property
133 def b(self) -> np.ndarray:
134     return np.asarray(self.beta[1:5], float)
135
136 @property
137 def c(self) -> np.ndarray:
138     return np.asarray(self.beta[5:9], float)
139
140 def power(self, u: np.ndarray, t: np.ndarray) -> np.ndarray:
141     """Total power for raw control arrays (last axis = 4 fields)."""
142     return self.intercept + np.sum(self.b * u ** 2, axis=-1) + np.sum(self.c / t, axis=-1)
143
144 def field_power(self, u: np.ndarray) -> np.ndarray:
145     return np.sum(self.b * u ** 2, axis=-1)
146
147 def rapping_power(self, t: np.ndarray) -> np.ndarray:
148     return np.sum(self.c / t, axis=-1)
149
150
151 class RidgePowerModel:
152     """Standardised ridge baseline with the rapping period entered *linearly*.
153
154     Retained as the misspecified reference point: it is the specification the
155     first version of this paper used, and the comparison in
156     ``power_model_baselines.csv`` shows what treating ``T`` as a linear term
157     costs on the retrospective test window.
158     """
159
160     def __init__(self, alpha: float = 1e-3, quadratic: bool = True):
161         self.alpha = alpha
162         self.quadratic = quadratic
163         self.mean: np.ndarray | None = None
164         self.std: np.ndarray | None = None
165         self.beta: np.ndarray | None = None
166
167     def design(self, frame: pd.DataFrame) -> np.ndarray:
168         u = frame[U_COLS].to_numpy(float)
169         remainder = frame[T_COLS + COND_COLS].to_numpy(float)
170         voltage = voltage_polynomial(u) if self.quadratic else u
171         return np.column_stack([voltage, remainder])
172
173     def fit(self, frame: pd.DataFrame, y: np.ndarray) -> None:
174         x = self.design(frame)
175         self.mean = x.mean(axis=0)
176         self.std = x.std(axis=0)

```

```

177     self.std[self.std == 0] = 1.0
178     z = (x - self.mean) / self.std
179     zz = np.column_stack([np.ones(len(z)), z])
180     penalty = np.eye(zz.shape[1]) * math.sqrt(self.alpha)
181     penalty[0, 0] = 0.0
182     augmented_x = np.vstack([zz, penalty])
183     augmented_y = np.concatenate([y, np.zeros(zz.shape[1])])
184     self.beta = np.linalg.lstsq(augmented_x, augmented_y, rcond=None)[0]
185
186     def predict(self, frame: pd.DataFrame) -> np.ndarray:
187         if self.mean is None or self.std is None or self.beta is None:
188             raise RuntimeError("model not fitted")
189         x = self.design(frame)
190         z = (x - self.mean) / self.std
191         zz = np.column_stack([np.ones(len(z)), z])
192         return np.einsum("ij,j->i", zz, self.beta)
193
194
195     def evaluate_power_models(
196         train: pd.DataFrame,
197         validation: pd.DataFrame,
198         test: pd.DataFrame,
199     ) -> tuple[PhysicalPowerModel, dict[str, object], pd.DataFrame, pd.DataFrame]:
200         """Fit every candidate specification on the training window only."""
201         alpha_grid = (1e-4, 1e-3, 1e-2, 1e-1, 1.0, 10.0)
202         n = len(train)
203         rolling_bounds = ((0, int(0.40 * n), int(0.60 * n)),
204                          (0, int(0.60 * n), int(0.80 * n)),
205                          (0, int(0.80 * n), n))
206         rolling_rows: list[dict[str, object]] = []
207         selected: dict[str, float] = {}
208         for model_name, quadratic in (("Linear_ridge_T", False), ("quadratic_ridge_T", True)):
209             alpha_scores: list[tuple[float, float]] = []
210             for alpha in alpha_grid:
211                 fold_rmse = []
212                 for fold, (start, fit_end, eval_end) in enumerate(rolling_bounds, start=1):
213                     fit = train.iloc[start:fit_end]
214                     evaluate = train.iloc[fit_end:eval_end]
215                     candidate = RidgePowerModel(alpha=alpha, quadratic=quadratic)
216                     candidate.fit(fit, fit["P_total_kW"].to_numpy(float))
217                     pred = candidate.predict(evaluate)
218                     actual = evaluate["P_total_kW"].to_numpy(float)
219                     rmse = _rmse(actual, pred)
220                     fold_rmse.append(rmse)
221                     rolling_rows.append({
222                         "model": model_name, "alpha": alpha, "fold": fold,
223                         "fit_rows": len(fit), "evaluation_rows": len(evaluate),
224                         "r2": _r2(actual, pred), "rmse_kW": rmse,
225                         "bias_kW": float(np.mean(pred - actual)),
226                     })
227                 alpha_scores.append((float(np.mean(fold_rmse)), alpha))
228             selected[model_name] = min(alpha_scores)[1]
229
230         # the physical specification has no tuning parameter; still report it per fold
231         for fold, (start, fit_end, eval_end) in enumerate(rolling_bounds, start=1):
232             fit = train.iloc[start:fit_end]
233             evaluate = train.iloc[fit_end:eval_end]
234             candidate = PhysicalPowerModel().fit(fit, fit["P_total_kW"].to_numpy(float))
235             pred = candidate.predict(evaluate)
236             actual = evaluate["P_total_kW"].to_numpy(float)
237             rolling_rows.append({
238                 "model": "physical_invT", "alpha": np.nan, "fold": fold,

```

```

239         "fit_rows": len(fit), "evaluation_rows": len(evaluate),
240         "r2": _r2(actual, pred), "rmse_kW": _rmse(actual, pred),
241         "bias_kW": float(np.mean(pred - actual)),
242     })
243
244     y_train = train["P_total_kW"].to_numpy(float)
245     fitted: dict[str, object] = {}
246     for model_name, quadratic in (("Linear_ridge_T", False), ("quadratic_ridge_T", True)):
247         m = RidgePowerModel(alpha=selected[model_name], quadratic=quadratic)
248         m.fit(train, y_train)
249         fitted[model_name] = m
250     fitted["physical_invT"] = PhysicalPowerModel().fit(train, y_train)
251     fitted["physical_invT_plus_conditions"] = PhysicalPowerModel(with_conditions=True).fit(train, y_train)
252
253     mean_power = float(y_train.mean())
254     baseline_rows: list[dict[str, object]] = []
255     for split_name, frame in (("train", train), ("validation", validation), ("retrospective_test", test)):
256         actual = frame["P_total_kW"].to_numpy(float)
257         for model_name in ("mean_predictor", "Linear_ridge_T", "quadratic_ridge_T",
258                           "physical_invT", "physical_invT_plus_conditions"):
259             pred = (np.full(len(frame), mean_power) if model_name == "mean_predictor"
260                     else fitted[model_name].predict(frame))
261             baseline_rows.append({
262                 "model": model_name,
263                 "selected_alpha": selected.get(model_name, np.nan),
264                 "split": split_name,
265                 "r2": _r2(actual, pred),
266                 "rmse_kW": _rmse(actual, pred),
267                 "bias_kW": float(np.mean(pred - actual)),
268             })
269
270     chosen: PhysicalPowerModel = fitted["physical_invT"] # type: ignore[assignment]
271     val_actual = validation["P_total_kW"].to_numpy(float)
272     val_pred = chosen.predict(validation)
273     test_actual = test["P_total_kW"].to_numpy(float)
274     test_pred = chosen.predict(test)
275     guard = float(np.quantile(np.abs(val_pred - val_actual), 0.90))
276     metrics: dict[str, object] = {
277         "model": chosen.name,
278         "formula": "P = a0 + sum_i b_i U_i^2 + sum_i c_i / T_i",
279         "fit_scope": "train_only",
280         "intercept_kW": chosen.intercept,
281         "b_field_kW_per_kV2": chosen.b.tolist(),
282         "c_rapping_kW_s": chosen.c.tolist(),
283         "train_r2": _r2(y_train, chosen.predict(train)),
284         "validation_r2": _r2(val_actual, val_pred),
285         "validation_rmse_kW": _rmse(val_actual, val_pred),
286         "validation_bias_kW": float(np.mean(val_pred - val_actual)),
287         "validation_abs_error_q90_kW": guard,
288         "retrospective_test_r2": _r2(test_actual, test_pred),
289         "retrospective_test_rmse_kW": _rmse(test_actual, test_pred),
290         "retrospective_test_bias_kW": float(np.mean(test_pred - test_actual)),
291         "test_protocol": "retrospective_not_blind_due_to_prior_package_use",
292         "reporting_guard_rule": "point_prediction_plus_validation_absolute_error_q90",
293         "ridge_selected_alphas": selected,
294     }
295     return chosen, metrics, pd.DataFrame(rolling_rows), pd.DataFrame(baseline_rows)
296
297
298 # ----- condition layer
299
300 def condition_profiles(df: pd.DataFrame) -> pd.DataFrame:

```

```

301 train = df[df["split"] == "train"].copy()
302 overall_load = train["dust_load_kg_h"].median()
303 labels = {
304     1: "低温低浓度高流量",
305     2: "高温中浓度低流量",
306     3: "高温高浓度高流量",
307     4: "低温高浓度低流量",
308 }
309 rows: list[dict[str, float | int | str]] = []
310 for cluster, group in train.groupby("condition_cluster"):
311     row: dict[str, float | int | str] = {
312         "condition_cluster": int(cluster),
313         "condition_name": labels[int(cluster)],
314         "count": int(len(group)),
315         "share": float(len(group) / len(train)),
316         "Temp_C": float(group["Temp_C"].mean()),
317         "C_in_gNm3": float(group["C_in_gNm3"].mean()),
318         "Q_Nm3h": float(group["Q_Nm3h"].mean()),
319         "dust_load_kg_h": float(group["dust_load_kg_h"].mean()),
320         "C_out_recorded_median": float(group["C_out_mgNm3"].dropna().median()),
321         "historical_power_kW": float(group["P_total_kW"].mean()),
322     }
323     for col in U_COLS + T_COLS:
324         row[f"{col}_min"] = float(group[col].min())
325         row[f"{col}_p05"] = float(group[col].quantile(0.05))
326         row[f"{col}_median"] = float(group[col].median())
327         row[f"{col}_p95"] = float(group[col].quantile(0.95))
328         row[f"{col}_max"] = float(group[col].max())
329         row["load_ratio_to_train_median"] = float(row["dust_load_kg_h"] / overall_load)
330         row["K0_anchor"] = math.log(1000.0 * row["C_in_gNm3"] / row["C_out_recorded_median"])
331     rows.append(row)
332 return pd.DataFrame(rows).sort_values("condition_cluster").reset_index(drop=True)
333
334
335 def scenario_t_star(profile: pd.Series, global_t_median: np.ndarray,
336                    load_exponent: float = -0.18) -> np.ndarray:
337     load_ratio = float(profile["load_ratio_to_train_median"])
338     raw = global_t_median * load_ratio ** load_exponent
339     lower = np.array([profile[f"{c}_min"] for c in T_COLS], float)
340     upper = np.array([profile[f"{c}_max"] for c in T_COLS], float)
341     return np.clip(raw, lower, upper)
342
343
344 def support_statistics(train_group: pd.DataFrame) -> tuple[np.ndarray, np.ndarray, float]:
345     """Empirical control-support centre, inverse covariance and d^2 cut-off."""
346     hist = train_group[U_COLS + T_COLS].to_numpy(float)
347     mu = hist.mean(axis=0)
348     inv_cov = np.linalg.pinv(np.cov(hist, rowvar=False))
349     delta = hist - mu
350     historical_d2 = np.einsum("ij,jk,ik->i", delta, inv_cov, delta)
351     return mu, inv_cov, float(np.quantile(historical_d2, 0.975))
352
353
354 # ----- emission layer
355
356 def scenario_emission(
357     profile: pd.Series,
358     u: np.ndarray,
359     t: np.ndarray,
360     t_star: np.ndarray,
361     p_exponent: float = P_EXPONENT_CENTRAL,
362     peak_scale: float = PEAK_SCALE_CENTRAL,

```

```

363 safety_index: float = SAFETY_INDEX_CENTRAL,
364 alpha_weights: np.ndarray | None = None,
365 peak_exponent: float = PEAK_EXPONENT_CENTRAL,
366 phase_scale: float = PHASE_SCALE_CENTRAL,
367 ) -> tuple[np.ndarray, np.ndarray, np.ndarray]:
368     """Return continuous base emission, rapping peak excess, and their sum.
369
370     ``K = K0 * S(U) - [G(T) - G(T_ref)]`` with
371     ``S(U) = sum_i alpha_i (U_i/U_ref_i)^p / sum_i alpha_i``.
372
373     At the historical anchor ``U = U_ref`` and ``T = T_ref`` this returns
374     ``K = K0 = ln(1000 C_in / C_out_recorded)`` , i.e. the model reproduces the
375     recorded operating point by construction. Every non-control argument is a
376     scenario input; none is estimated from the recorded outlet series.
377     """
378     u_ref = np.array([profile[f"{c}_median"] for c in U_COLS], float)
379     t_ref = np.array([profile[f"{c}_median"] for c in T_COLS], float)
380     cin = float(profile["C_in_gNm3"])
381     k_anchor = float(profile["K0_anchor"])
382
383     if alpha_weights is None:
384         alpha_weights = ALPHA_CENTRAL
385     alpha = np.asarray(alpha_weights, float)
386     strength = np.sum(alpha * (u / u_ref) ** p_exponent, axis=-1) / alpha.sum()
387
388     load_ratio = float(profile["load_ratio_to_train_median"])
389     beta = BETA_CENTRAL * load_ratio ** 0.5
390
391     def loss(period: np.ndarray) -> np.ndarray:
392         ratio = period / t_star
393         return np.sum(beta * (ratio + 1.0 / ratio - 2.0), axis=-1)
394
395     ref_loss = float(loss(t_ref))
396     k = k_anchor * strength - (loss(t) - ref_loss)
397     base = cin * 1000.0 * np.exp(-k + safety_index)
398
399     # Derived form: cake mass per rap ~ L * T, so the per-event peak grows
400     # linearly with the period (peak_exponent = 1) while the contribution to
401     # the hourly mean, ~ L*T*(1/T) = L, does not depend on the period at all.
402     peak = (peak_scale * phase_scale * load_ratio
403             * np.sum(PEAK_SHARES * (t / t_star) ** peak_exponent, axis=-1))
404     return base, peak, base + peak
405
406
407 def field_priority(profile: pd.Series, model: PhysicalPowerModel,
408                  alpha_weights: np.ndarray | None = None) -> np.ndarray:
409     """Removal gain per unit of electrical power for each field.
410
411     Differentiating ``K = K0 * sum_i alpha_i (U_i/U_ref_i)^2 / sum(alpha)`` and
412     ``P = sum_i b_i U_i^2`` with respect to ``U_i^2`` gives
413
414         
$$dK/dP|_i = K0 * \alpha_i / (\sum(\alpha) * b_i * U_{ref\_i}^2),$$

415
416     so the field to raise first is the one with the largest
417     ``alpha_i / U_ref_i^2`` -- *not* the largest ``alpha_i``.
418     """
419     alpha = ALPHA_CENTRAL if alpha_weights is None else np.asarray(alpha_weights, float)
420     u_ref = np.array([profile[f"{c}_median"] for c in U_COLS], float)
421     return float(profile["K0_anchor"]) * alpha / (alpha.sum() * model.b * u_ref ** 2)
422
423
424 # ----- optimisation

```

```

425
426 def _bounds(profile: pd.Series, voltage_headroom: float) -> tuple[np.ndarray, np.ndarray]:
427     lo = np.concatenate([[profile[f"{c}_min"] for c in U_COLS],
428                          [profile[f"{c}_min"] for c in T_COLS]])
429     hi = np.concatenate([[profile[f"{c}_max"] * (1.0 + voltage_headroom) for c in U_COLS],
430                          [profile[f"{c}_max"] for c in T_COLS]])
431     return lo.astype(float), hi.astype(float)
432
433
434 def _ordering(z: np.ndarray) -> np.ndarray:
435     return np.array([z[0] - z[2] - 3.0, z[1] - z[3] - 3.0,
436                     z[6] - z[4] - 100.0, z[7] - z[5] - 100.0])
437
438
439 def _round_controls(z: np.ndarray) -> np.ndarray:
440     return np.concatenate([np.round(z[:4], 1), np.round(z[4:])])
441
442
443 def solve_condition(
444     profile: pd.Series,
445     model: PhysicalPowerModel,
446     t_star: np.ndarray,
447     limit: float,
448     voltage_headroom: float = 0.0,
449     mahalanobis: tuple[np.ndarray, np.ndarray, float] | None = None,
450     multistart: int = MULTISTART,
451     seed: int = SEED,
452     **emission_kwargs,
453 ) -> dict[str, object]:
454     """Deterministic multi-start SLSQP solve of one condition/limit pair.
455
456     The feasible set is *not* convex in ``U``: the objective ``sum b_i U_i^2``
457     is convex while the emission constraint requires a convex function of
458     ``U`` to be large enough, which is a reverse-convex constraint. A dense
459     multi-start is therefore used, and ``verify_with_random_search`` re-checks
460     the result against a large random sample of the same feasible set.
461     """
462     lo, hi = _bounds(profile, voltage_headroom)
463
464     def objective(z: np.ndarray) -> float:
465         return float(model.power(z[:4], z[4:]))
466
467     def emission_slack(z: np.ndarray) -> float:
468         return float(limit - scenario_emission(profile, z[:4], z[4:], t_star, **emission_kwargs)[2])
469
470     constraints: list[dict[str, object]] = [
471         {"type": "ineq", "fun": emission_slack},
472         {"type": "ineq", "fun": _ordering},
473     ]
474     if mahalanobis is not None:
475         mu, inv_cov, threshold = mahalanobis
476         constraints.append({
477             "type": "ineq",
478             "fun": lambda z: float(threshold - (z - mu) @ inv_cov @ (z - mu)),
479         })
480
481     u_ref = np.array([profile[f"{c}_median"] for c in U_COLS], float)
482     t_ref = np.array([profile[f"{c}_median"] for c in T_COLS], float)
483     rng = np.random.default_rng(seed)
484     starts = [np.concatenate([u_ref, t_ref]), hi.copy(), lo.copy(), 0.5 * (lo + hi),
485                  np.concatenate([hi[:4], t_star])]
486     starts += [lo + (hi - lo) * rng.random(8) for _ in range(multistart)]

```

```

487
488 best_z: np.ndarray | None = None
489 best_val = np.inf
490 converged = 0
491 values: list[float] = []
492 for z0 in starts:
493     result = minimize(objective, np.clip(z0, lo, hi), method="SLSQP",
494                       bounds=list(zip(lo, hi)), constraints=constraints,
495                       options={"maxiter": 600, "ftol": 1e-10})
496     if not result.success:
497         continue
498     z = np.clip(result.x, lo, hi)
499     if emission_slack(z) < -1e-7 or np.any(_ordering(z) < -1e-7):
500         continue
501     converged += 1
502     values.append(float(result.fun))
503     if result.fun < best_val:
504         best_val, best_z = float(result.fun), z
505
506 if best_z is None:
507     return {
508         "condition_cluster": int(profile["condition_cluster"]),
509         "condition_name": str(profile["condition_name"]),
510         "limit_mgNm3": float(limit),
511         "voltage_headroom": float(voltage_headroom),
512         "status": "INFEASIBLE_IN_DECLARED_SUPPORT",
513     }
514
515 # Round to the resolution a DCS can actually hold, then repair feasibility
516 # by nudging the most power-efficient field upward.
517 z = _round_controls(best_z)
518 order = np.argsort(-field_priority(profile, model, emission_kwargs.get("alpha_weights")))
519 for _ in range(400):
520     if emission_slack(z) >= 0 and np.all(_ordering(z) >= -1e-9):
521         break
522     for i in order:
523         if z[i] + 0.1 <= hi[i] + 1e-9:
524             z[i] = round(z[i] + 0.1, 1)
525             break
526     else:
527         break
528 if emission_slack(z) < 0:
529     z = best_z # keep the unrounded solution rather than report an infeasible one
530
531 base, peak, total = scenario_emission(profile, z[:4], z[4:], t_star, **emission_kwargs)
532 row: dict[str, object] = {
533     "condition_cluster": int(profile["condition_cluster"]),
534     "condition_name": str(profile["condition_name"]),
535     "limit_mgNm3": float(limit),
536     "voltage_headroom": float(voltage_headroom),
537     "status": "SCENARIO_FEASIBLE",
538     "scenario_base_emission_mgNm3": float(base),
539     "scenario_peak_excess_mgNm3": float(peak),
540     "scenario_peak_total_mgNm3": float(total),
541     "predicted_power_kw": float(model.power(z[:4], z[4:])),
542     "field_power_kw": float(model.field_power(z[:4])),
543     "rapping_power_kw": float(model.rapping_power(z[4:])),
544     "historical_power_kw": float(profile["historical_power_kw"]),
545     "multistart_converged": converged,
546     "multistart_power_spread_kw": float(np.max(values) - np.min(values)) if values else np.nan,
547 }
548 row["power_change_vs_historical_pct"] = 100.0 * (

```

```

549         row["predicted_power_kw"] / float(profile["historical_power_kw"]) - 1.0)
550     for i, col in enumerate(U_COLS + T_COLS):
551         row[col] = float(z[i])
552     return row
553
554
555 def verify_with_random_search(
556     profile: pd.Series,
557     model: PhysicalPowerModel,
558     t_star: np.ndarray,
559     limit: float,
560     voltage_headroom: float,
561     n: int = 400_000,
562     seed: int = SEED,
563     **emission_kwargs,
564 ) -> float:
565     """Best feasible power found by uniform random sampling of the same set."""
566     lo, hi = _bounds(profile, voltage_headroom)
567     rng = np.random.default_rng(seed)
568     z = lo + (hi - lo) * rng.random((n, 8))
569     u = np.round(z[:, :4], 1)
570     t = np.round(z[:, 4:], 1)
571     keep = ((u[:, 0] >= u[:, 2] + 3.0) & (u[:, 1] >= u[:, 3] + 3.0)
572            & (t[:, 0] >= t[:, 2] + 100.0) & (t[:, 1] >= t[:, 3] + 100.0))
573     u, t = u[keep], t[keep]
574     if not len(u):
575         return float("nan")
576     _, total = scenario_emission(profile, u, t, t_star, **emission_kwargs)
577     feasible = total <= limit
578     if not np.any(feasible):
579         return float("nan")
580     return float(np.min(model.power(u[feasible], t[feasible])))
581
582
583 def aggregate(rows: List[dict[str, object]], key: str = "predicted_power_kw") -> float:
584     weights = np.array([float(r["share"]) for r in rows])
585     return float(np.average([float(r[key]) for r in rows], weights=weights))
586
587
588 def solve_all(profiles: pd.DataFrame, model: PhysicalPowerModel,
589              t_stars: dict[int, np.ndarray], limit: float, headroom: float,
590              multistart: int = MULTISTART, **emission_kwargs) -> List[dict[str, object]]:
591     rows = []
592     for _, profile in profiles.iterrows():
593         cluster = int(profile["condition_cluster"])
594         r = solve_condition(profile, model, t_stars[cluster], limit,
595                             voltage_headroom=headroom, multistart=multistart,
596                             **emission_kwargs)
597         r["share"] = float(profile["share"])
598         rows.append(r)
599     return rows
600
601
602 # ----- data-side diagnostics
603
604 def historical_strategy_correlations(df: pd.DataFrame) -> pd.DataFrame:
605     """Operator response of the control settings to the exogenous conditions.
606
607     Unlike the outlet concentration, this relationship *is* identifiable, and
608     it gives Question 3 an empirical baseline. It is reported per window as
609     well as over the whole record, because the two are not the same: the
610     rapping-period rule holds throughout, while the voltage response to inlet

```

```

611 concentration reverses sign between the first five days and the last two.
612 """
613 windows = [("full_record", df), ("train", df[df["split"] == "train"]),
614            ("validation", df[df["split"] == "validation"]),
615            ("retrospective_test", df[df["split"] == "test"])]
616 rows = []
617 for name, frame in windows:
618     for control in U_COLS + T_COLS:
619         row: dict[str, object] = {"window": name, "rows": int(len(frame)), "control": control}
620         for cond in COND_COLS:
621             row[f"r_{cond}"] = float(np.corrcoef(frame[cond], frame[control])[0, 1])
622         rows.append(row)
623 return pd.DataFrame(rows)
624
625
626 def anomaly_events(df: pd.DataFrame) -> pd.DataFrame:
627     """The two sustained excursions planted in the attachment."""
628     specs = [
629         ("入口浓度尖峰", df["C_in_gNm3"] > 60.0),
630         ("入口温度尖峰", df["Temp_C"] > 150.0),
631     ]
632     baseline = {c: float(df[c].median()) for c in COND_COLS + U_COLS + T_COLS + ["P_total_kW"]}
633     rows = []
634     for name, mask in specs:
635         sub = df[mask]
636         row: dict[str, object] = {
637             "event": name,
638             "start": str(sub["timestamp"].min()),
639             "end": str(sub["timestamp"].max()),
640             "minutes": int(len(sub)),
641         }
642         for col in COND_COLS + U_COLS + T_COLS + ["P_total_kW"]:
643             row[f"{col}_event_mean"] = float(sub[col].mean())
644             row[f"{col}_event_max"] = float(sub[col].max())
645             row[f"{col}_event_min"] = float(sub[col].min())
646             row[f"{col}_baseline_median"] = baseline[col]
647         rows.append(row)
648     return pd.DataFrame(rows)
649
650
651 def rapping_power_tradeoff(train: pd.DataFrame, model: PhysicalPowerModel) -> pd.DataFrame:
652     """How much power the rapping term holds, and what extending it recovers."""
653     u_med = train[U_COLS].median().to_numpy(float)
654     t_med = train[T_COLS].median().to_numpy(float)
655     field = float(model.field_power(u_med))
656     base_rap = float(model.rapping_power(t_med))
657     total = model.intercept + field + base_rap
658     rows = [{
659         "scenario": "历史中位周期",
660         "T_setting": ";".join(f"{v:.0f}" for v in t_med),
661         "field_power_kW": field,
662         "rapping_power_kW": base_rap,
663         "total_power_kW": total,
664         "rapping_share_pct": 100.0 * base_rap / total,
665         "power_recovered_kW": 0.0,
666         "cake_mass_per_rap_change_pct": 0.0,
667     }]
668     for label, t_new in (("延长至历史P95", train[T_COLS].quantile(0.95).to_numpy(float)),
669                         ("延长至历史最大值", train[T_COLS].max().to_numpy(float))):
670         rap = float(model.rapping_power(t_new))
671         rows.append({
672             "scenario": label,

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```

673         "T_setting": ";".join(f"{v:.0f}" for v in t_new),
674         "field_power_kW": field,
675         "rapping_power_kW": rap,
676         "total_power_kW": model.intercept + field + rap,
677         "rapping_share_pct": 100.0 * rap / (model.intercept + field + rap),
678         "power_recovered_kW": base_rap - rap,
679         "cake_mass_per_rap_change_pct": 100.0 * (float(np.mean(t_new / t_med)) - 1.0),
680     })
681     return pd.DataFrame(rows)
682
683
684 # ----- driver
685
686 def main() -> None:
687     OUT_DIR.mkdir(parents=True, exist_ok=True)
688     df = pd.read_csv(DATA_FILE, parse_dates=["timestamp"])
689     df["condition_cluster"] = df["condition_cluster"].astype(int)
690     profiles = condition_profiles(df)
691     profiles.to_csv(OUT_DIR / "condition_profiles.csv", index=False, encoding="utf-8-sig")
692
693     train = df[df["split"] == "train"].copy()
694     validation = df[df["split"] == "validation"].copy()
695     test = df[df["split"] == "test"].copy()
696
697     model, metrics, rolling_metrics, baseline_metrics = evaluate_power_models(train, validation, test)
698     rolling_metrics.to_csv(OUT_DIR / "power_model_rolling_validation.csv", index=False, encoding="utf-8-sig")
699     baseline_metrics.to_csv(OUT_DIR / "power_model_baselines.csv", index=False, encoding="utf-8-sig")
700     (OUT_DIR / "power_model_metrics.json").write_text(
701         json.dumps(metrics, ensure_ascii=False, indent=2), encoding="utf-8")
702     guard = float(metrics["validation_abs_error_q90_kW"])
703
704     pd.read_csv(CLUSTER_EVAL_FILE).to_csv(
705         OUT_DIR / "cluster_selection_metrics.csv", index=False, encoding="utf-8-sig")
706
707     historical_strategy_correlations(df).to_csv(
708         OUT_DIR / "historical_strategy_correlations.csv", index=False, encoding="utf-8-sig")
709     anomaly_events(df).to_csv(OUT_DIR / "anomaly_events.csv", index=False, encoding="utf-8-sig")
710     rapping_power_tradeoff(train, model).to_csv(
711         OUT_DIR / "rapping_power_tradeoff.csv", index=False, encoding="utf-8-sig")
712
713     global_t_median = train[T_COLS].median().to_numpy(float)
714     t_stars = {int(p["condition_cluster"]): scenario_t_star(p, global_t_median)
715                for _, p in profiles.iterrows()}
716
717 # -- field priority criterion (analytic) -----
718 priority_rows = []
719 for _, profile in profiles.iterrows():
720     crit = field_priority(profile, model)
721     u_ref = np.array([profile[f"{c}_median"] for c in U_COLS], float)
722     row = {"condition_cluster": int(profile["condition_cluster"]),
723           "condition_name": profile["condition_name"]}
724     for i in range(4):
725         row[f"U_{i+1}_ref_kW"] = float(u_ref[i])
726         row[f"alpha_{i+1}"] = float(ALPHA_CENTRAL[i])
727         row[f"criteria_{i+1}_per_kW"] = float(crit[i])
728         row["rank_order"] = ">".join(f"E{i+1}" for i in np.argsort(-crit))
729     priority_rows.append(row)
730 pd.DataFrame(priority_rows).to_csv(
731     OUT_DIR / "field_priority_criterion.csv", index=False, encoding="utf-8-sig")
732
733 # -- central solutions -----
734 rows10 = solve_all(profiles, model, t_stars, 10.0, HEADROOM_10)

```

```

735 rows5 = solve_all(profiles, model, t_stars, 5.0, HEADROOM_5)
736 optimum = pd.DataFrame(rows10 + rows5)
737 optimum["validation_error_adjusted_power_kw"] = optimum["predicted_power_kw"] + guard
738 optimum["validation_error_guard_kw"] = guard
739 optimum.to_csv(OUT_DIR / "optimal_controls_central_scenario.csv",
740               index=False, encoding="utf-8-sig")
741
742 # -- support-domain audit (reported, not imposed) -----
743 audit_rows = []
744 for row in rows10 + rows5:
745     cluster = int(row["condition_cluster"])
746     group = train[train["condition_cluster"] == cluster]
747     mu, inv_cov, threshold = support_statistics(group)
748     z = np.array([float(row[c]) for c in U_COLS + T_COLS])
749     d2 = float((z - mu) @ inv_cov @ (z - mu))
750     u_max = np.array([group[c].max() for c in U_COLS], float)
751     audit_rows.append({
752         "condition_cluster": cluster,
753         "condition_name": row["condition_name"],
754         "limit_mgNm3": row["limit_mgNm3"],
755         "mahalanobis_d2": d2,
756         "empirical_q975": threshold,
757         "ratio_to_q975": d2 / threshold,
758         "max_voltage_over_historical_max_pct": float(
759             100.0 * max((float(row[c]) / m - 1.0) for c, m in zip(U_COLS, u_max))),
760         "verdict": "SUPPORTED" if d2 <= threshold else "EXTRAPOLATION",
761     })
762 pd.DataFrame(audit_rows).to_csv(OUT_DIR / "support_boundary_audit.csv",
763                               index=False, encoding="utf-8-sig")
764
765 # Does the historical support ellipsoid admit any feasible point at all?
766 restricted_rows = []
767 for _, profile in profiles.iterrows():
768     cluster = int(profile["condition_cluster"])
769     stats = support_statistics(train[train["condition_cluster"] == cluster])
770     for limit, headroom in ((10.0, HEADROOM_10), (5.0, HEADROOM_5)):
771         r = solve_condition(profile, model, t_stars[cluster], limit,
772                             voltage_headroom=headroom, mahalanobis=stats)
773         restricted_rows.append({
774             "condition_cluster": cluster, "limit_mgNm3": limit,
775             "status_with_mahalanobis_constraint": r["status"],
776             "predicted_power_kw": r.get("predicted_power_kw", np.nan),
777         })
778 pd.DataFrame(restricted_rows).to_csv(
779     OUT_DIR / "mahalanobis_constrained_feasibility.csv", index=False, encoding="utf-8-sig")
780
781 # -- minimum voltage headroom needed at each limit -----
782 headroom_rows = []
783 for _, profile in profiles.iterrows():
784     cluster = int(profile["condition_cluster"])
785     for limit in (10.0, 5.0):
786         need = np.nan
787         for m in np.arange(0.0, 0.31, 0.01):
788             if solve_condition(profile, model, t_stars[cluster], limit,
789                               voltage_headroom=float(m), multistart=16
790                               )["status"] == "SCENARIO_FEASIBLE":
791                 need = float(m)
792                 break
793         headroom_rows.append({
794             "condition_cluster": cluster,
795             "condition_name": profile["condition_name"],
796             "limit_mgNm3": limit,

```

```

797         "min_voltage_headroom_pct": 100.0 * need,
798     })
799 pd.DataFrame(headroom_rows).to_csv(
800     OUT_DIR / "voltage_headroom_requirement.csv", index=False, encoding="utf-8-sig")
801
802 # -- optimiser verification against dense random search -----
803 verify_rows = []
804 for _, profile in profiles.iterrows():
805     cluster = int(profile["condition_cluster"])
806     for limit, headroom, rows in ((10.0, HEADROOM_10, rows10), (5.0, HEADROOM_5, rows5)):
807         solved = next(r for r in rows if r["condition_cluster"] == cluster)
808         if solved["status"] != "SCENARIO_FEASIBLE":
809             continue
810         rnd = verify_with_random_search(profile, model, t_stars[cluster], limit, headroom)
811         verify_rows.append({
812             "condition_cluster": cluster,
813             "condition_name": profile["condition_name"],
814             "limit_mgNm3": limit,
815             "slsqp_power_kw": solved["predicted_power_kw"],
816             "random_search_power_kw": rnd,
817             "random_search_gap_pct": 100.0 * (rnd / float(solved["predicted_power_kw"]) - 1.0),
818             "multistart_converged": solved["multistart_converged"],
819             "multistart_power_spread_kw": solved["multistart_power_spread_kw"],
820         })
821 verification = pd.DataFrame(verify_rows)
822 verification.to_csv(OUT_DIR / "optimization_verification.csv", index=False, encoding="utf-8-sig")
823
824 # -- Question 4 -----
825 q4_rows = []
826 for _, profile in profiles.iterrows():
827     cluster = int(profile["condition_cluster"])
828     a = next(r for r in rows10 if r["condition_cluster"] == cluster)
829     b = next(r for r in rows5 if r["condition_cluster"] == cluster)
830     k10 = float(profile["K0_anchor"]) * float(
831         np.sum(ALPHA_CENTRAL * (np.array([a[c] for c in U_COLS], float)
832             / np.array([profile[f"{c}_median"] for c in U_COLS], float)) ** 2)
833         / ALPHA_CENTRAL.sum())
834     delta_k = math.log((10.0 - float(a["scenario_peak_excess_mgNm3"]))
835         / (5.0 - float(a["scenario_peak_excess_mgNm3"])))
836     q4_rows.append({
837         "condition_cluster": cluster,
838         "condition_name": profile["condition_name"],
839         "share": float(profile["share"]),
840         "power_10_kw": float(a["predicted_power_kw"]),
841         "power_5_kw": float(b["predicted_power_kw"]),
842         "field_power_10_kw": float(a["field_power_kw"]),
843         "K_at_10": k10,
844         "delta_K_10_to_5": delta_k,
845         "analytic_delta_power_kw": float(a["field_power_kw"]) * delta_k / k10,
846         "numeric_delta_power_kw": float(b["predicted_power_kw"]) - float(a["predicted_power_kw"]),
847         "increase_pct": 100.0 * (float(b["predicted_power_kw"]) / float(a["predicted_power_kw"]) - 1.0),
848     })
849 q4 = pd.DataFrame(q4_rows)
850 q4.to_csv(OUT_DIR / "question4_by_condition.csv", index=False, encoding="utf-8-sig")
851 weighted_p10 = float(np.average(q4["power_10_kw"], weights=q4["share"]))
852 weighted_p5 = float(np.average(q4["power_5_kw"], weights=q4["share"]))
853 historical_weighted = float(np.average(profiles["historical_power_kw"], weights=profiles["share"]))
854 q4_summary = {
855     "weighted_power_10_kw": weighted_p10,
856     "weighted_power_5_kw": weighted_p5,
857     "weighted_increase_pct": 100.0 * (weighted_p5 / weighted_p10 - 1.0),
858     "historical_weighted_actual_power_kw": historical_weighted,

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```

859     "power_10_vs_historical_pct": 100.0 * (weighted_p10 / historical_weighted - 1.0),
860     "daily_energy_10_MWh": weighted_p10 * 24.0 / 1000.0,
861     "daily_energy_5_MWh": weighted_p5 * 24.0 / 1000.0,
862     "daily_energy_historical_MWh": historical_weighted * 24.0 / 1000.0,
863     "validation_error_guard_kW": guard,
864 }
865 (OUT_DIR / "question4_summary.json").write_text(
866     json.dumps(q4_summary, ensure_ascii=False, indent=2), encoding="utf-8")
867
868 # -- p reference table: the single scalar that needs site calibration ----
869 # Holding the 10 mg/Nm^3 operating point fixed, scaling every voltage by f
870 # gives K = K_10 f^p and P_field = P_field,10 f^2, so halving the limit
871 # (delta_K) costs exactly P_field,10 [(1 + delta_K/K_10)^(2/p) - 1]:
872 # the penalty is governed by 2/p and nothing else.
873 p_rows = []
874 for p_exp in (1.0, 1.5, 2.0, 2.5, 3.0):
875     analytic = []
876     for row in q4_rows:
877         ratio = 1.0 + row["delta_K_10_to_5"] / row["K_at_10"]
878         analytic.append({"share": row["share"],
879                         "delta": row["field_power_10_kW"] * (ratio ** (2.0 / p_exp) - 1.0),
880                         "p10": row["power_10_kW"]})
881     w = np.array([a["share"] for a in analytic])
882     delta = float(np.average([a["delta"] for a in analytic], weights=w))
883     p10 = float(np.average([a["p10"] for a in analytic], weights=w))
884     # required voltage headroom above the historical maximum, same reasoning
885     f = float(np.mean([(1.0 + r["delta_K_10_to_5"] / r["K_at_10"]) ** (1.0 / p_exp)
886                        for r in q4_rows]))
887     r10 = solve_all(profiles, model, t_stars, 10.0, HEADROOM_10,
888                     multistart=SWEEP_MULTISTART, p_exponent=p_exp)
889     r5 = solve_all(profiles, model, t_stars, 5.0, HEADROOM_5,
890                    multistart=SWEEP_MULTISTART, p_exponent=p_exp)
891     feasible = all(r["status"] == "SCENARIO_FEASIBLE" for r in r10 + r5)
892     p_rows.append({
893         "p_exponent": p_exp,
894         "analytic_delta_power_kW": delta,
895         "analytic_increase_pct": 100.0 * delta / p10,
896         "implied_voltage_scaling_10_to_5": f,
897         "feasible_at_5pct_headroom": feasible,
898         "numeric_weighted_power_10_kW": aggregate(r10) if feasible else np.nan,
899         "numeric_weighted_power_5_kW": aggregate(r5) if feasible else np.nan,
900         "numeric_increase_pct": (100.0 * (aggregate(r5) / aggregate(r10) - 1.0)) if feasible else np.nan,
901     })
902 pd.DataFrame(p_rows).to_csv(OUT_DIR / "p_exponent_reference.csv",
903                             index=False, encoding="utf-8-sig")
904
905 # -- scale sensitivity: p x peak scale x safety index (27 combinations) --
906 sensitivity_rows = []
907 for p_exp in (1.5, 2.0, 2.5):
908     for peak_scale in (0.9, 1.2, 1.5):
909         for safety in (0.0, 0.08, 0.15):
910             kw = dict(p_exponent=p_exp, peak_scale=peak_scale, safety_index=safety)
911             r10 = solve_all(profiles, model, t_stars, 10.0, HEADROOM_10,
912                             multistart=SWEEP_MULTISTART, **kw)
913             r5 = solve_all(profiles, model, t_stars, 5.0, HEADROOM_5,
914                             multistart=SWEEP_MULTISTART, **kw)
915             ok = all(r["status"] == "SCENARIO_FEASIBLE" for r in r10 + r5)
916             sensitivity_rows.append({
917                 "p_exponent": p_exp, "peak_scale": peak_scale, "safety_index": safety,
918                 "all_conditions_feasible": ok,
919                 "weighted_power_10_kW": aggregate(r10) if ok else np.nan,
920                 "weighted_power_5_kW": aggregate(r5) if ok else np.nan,

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921         "weighted_power_increase_pct": (100.0 * (aggregate(r5) / aggregate(r10) - 1.0)) if ok else np.nan,
922     })
923 sensitivity = pd.DataFrame(sensitivity_rows)
924 sensitivity.to_csv(OUT_DIR / "question4_sensitivity.csv", index=False, encoding="utf-8-sig")
925
926 # -- structural sensitivity: 81 grey-box shape combinations -----
927 alpha_profiles = {
928     "equal": np.array([1.00, 1.00, 1.00, 1.00]),
929     "weak_front": np.array([1.08, 1.04, 0.98, 0.90]),
930     "central_front": ALPHA_CENTRAL,
931 }
932 structural_rows = []
933 for alpha_name, alpha_weights in alpha_profiles.items():
934     for peak_exponent in (1.00, 1.30, 1.60):
935         for phase_scale in (0.65, 0.80, 1.00):
936             for tstar_exponent in (-0.30, -0.18, -0.10):
937                 stars = {int(p["condition_cluster"]): scenario_t_star(p, global_t_median, tstar_exponent)
938                     for _, p in profiles.iterrows()}
939                 kw = dict(alpha_weights=alpha_weights, peak_exponent=peak_exponent,
940                     phase_scale=phase_scale)
941                 r10 = solve_all(profiles, model, stars, 10.0, HEADROOM_10,
942                     multistart=SWEET_MULTISTART, **kw)
943                 r5 = solve_all(profiles, model, stars, 5.0, HEADROOM_5,
944                     multistart=SWEET_MULTISTART, **kw)
945                 ok = all(r["status"] == "SCENARIO_FEASIBLE" for r in r10 + r5)
946                 structural_rows.append({
947                     "alpha_profile": alpha_name,
948                     "peak_exponent": peak_exponent,
949                     "phase_scale": phase_scale,
950                     "tstar_load_exponent": tstar_exponent,
951                     "all_conditions_feasible": ok,
952                     "weighted_power_10_kw": aggregate(r10) if ok else np.nan,
953                     "weighted_power_5_kw": aggregate(r5) if ok else np.nan,
954                     "weighted_power_increase_pct": (100.0 * (aggregate(r5) / aggregate(r10) - 1.0)) if ok else np.nan,
955                 })
956 structural = pd.DataFrame(structural_rows)
957 structural.to_csv(OUT_DIR / "structural_sensitivity.csv", index=False, encoding="utf-8-sig")
958
959 # -- field weight ablation -----
960 ablation_rows, ablation_summary_rows = [], []
961 for alpha_name, alpha_weights in alpha_profiles.items():
962     r10 = solve_all(profiles, model, t_stars, 10.0, HEADROOM_10,
963         multistart=SWEET_MULTISTART, alpha_weights=alpha_weights)
964     r5 = solve_all(profiles, model, t_stars, 5.0, HEADROOM_5,
965         multistart=SWEET_MULTISTART, alpha_weights=alpha_weights)
966     per_profile = []
967     for a, b in zip(r10, r5):
968         row = {"alpha_profile": alpha_name, "condition_cluster": a["condition_cluster"],
969             "condition_name": a["condition_name"], "share": a["share"]}
970         for i, col in enumerate(U_COLS, start=1):
971             row[f"delta_U{i}_kV"] = float(b[col]) - float(a[col])
972         per_profile.append(row)
973     ablation_rows.append(row)
974     weights = np.array([x["share"] for x in per_profile])
975     deltas = np.array([[x[f"delta_U{i}_kV"] for i in range(1, 5)] for x in per_profile])
976     weighted = np.average(deltas, axis=0, weights=weights)
977     positive = np.maximum(weighted, 0)
978     ablation_summary_rows.append({
979         "alpha_profile": alpha_name,
980         **{f"weighted_delta_U{i}_kV": weighted[i - 1] for i in range(1, 5)},
981         "front_two_share_of_positive_adjustment": float(positive[:2].sum() / positive.sum())
982         if positive.sum() else np.nan,

```

```

983     })
984     pd.DataFrame(ablation_rows).to_csv(
985         OUT_DIR / "field_priority_ablation_by_condition.csv", index=False, encoding="utf-8-sig")
986     pd.DataFrame(ablation_summary_rows).to_csv(
987         OUT_DIR / "field_priority_ablation_summary.csv", index=False, encoding="utf-8-sig")
988
989     # -- historical comparison and online schedule -----
990     historical_rows, dynamic_rows = [], []
991     for _, profile in profiles.iterrows():
992         cluster = int(profile["condition_cluster"])
993         u_med = np.array([profile[f"{c}_median"] for c in U_COLS], float)
994         t_med = np.array([profile[f"{c}_median"] for c in T_COLS], float)
995         median_power = float(model.power(u_med, t_med))
996         a = next(r for r in rows10 if r["condition_cluster"] == cluster)
997         b = next(r for r in rows5 if r["condition_cluster"] == cluster)
998         historical_rows.append({
999             "condition_cluster": cluster,
1000             "condition_name": profile["condition_name"],
1001             "share": float(profile["share"]),
1002             "historical_mean_actual_power_kw": float(profile["historical_power_kw"]),
1003             "model_power_at_historical_median_controls_kw": median_power,
1004             "scenario_optimal_10_power_kw": float(a["predicted_power_kw"]),
1005             "power_change_vs_model_median_pct": 100.0 * (float(a["predicted_power_kw"]) / median_power - 1.0),
1006         })
1007         dynamic_rows.append({
1008             "condition_cluster": cluster,
1009             "condition_name": profile["condition_name"],
1010             "mode_selection_rule": "EXTERNAL_TARGET_SELECTION",
1011             "available_modes": "TARGET-10/TARGET-5/FALLBACK",
1012             "automatic_emission_threshold_switching": False,
1013             "provisional_voltage_ramp_kv_per_min": 0.5,
1014             "provisional_period_ramp_s_per_min": 5.0,
1015             "phase_scenario_for_online_trial": "0.65/0.80/1.00",
1016             "implementation_status": "PROVISIONAL_REQUIRES_T/R_CAPACITY_CHECK_AND_SITE_INTERLOCK_LIMITS",
1017             **{f"target10_{c}": a[c] for c in U_COLS + T_COLS},
1018             **{f"target5_{c}": b[c] for c in U_COLS + T_COLS},
1019         })
1020     pd.DataFrame(historical_rows).to_csv(
1021         OUT_DIR / "historical_median_power_comparison.csv", index=False, encoding="utf-8-sig")
1022     pd.DataFrame(dynamic_rows).to_csv(
1023         OUT_DIR / "dynamic_control_schedule.csv", index=False, encoding="utf-8-sig")
1024
1025     scenario_params = {
1026         "status": "EMISSION_LAYER_IS_SCENARIO_ONLY_C_OUT_NOT_IDENTIFIABLE",
1027         "emission_anchor": "recorded outlet median, no decimal rescaling",
1028         "p_exponent_central": P_EXPONENT_CENTRAL,
1029         "p_exponent_grid": [1.0, 1.5, 2.0, 2.5, 3.0],
1030         "peak_exponent_central": PEAK_EXPONENT_CENTRAL,
1031         "peak_exponent_grid": [1.00, 1.30, 1.60],
1032         "voltage_weight_profiles": {k: v.tolist() for k, v in alpha_profiles.items()},
1033         "rapping_phase_scales": [0.65, 0.80, 1.00],
1034         "tstar_load_exponents": [-0.30, -0.18, -0.10],
1035         "rapping_loss_coefficients": BETA_CENTRAL.tolist(),
1036         "peak_field_shares": PEAK_SHARES.tolist(),
1037         "support_rule": "cluster-specific full historical range plus engineering ordering; "
1038             "Mahalanobis d2 reported as extrapolation audit, not imposed",
1039         "voltage_headroom_10": HEADROOM_10,
1040         "voltage_headroom_5": HEADROOM_5,
1041         "optimiser": "multi-start SLSQP (%d starts) verified against 400k random samples" % MULTISTART,
1042     }
1043     (OUT_DIR / "scenario_parameters.json").write_text(
1044         json.dumps(scenario_params, ensure_ascii=False, indent=2), encoding="utf-8")

```

```

1045
1046 feasible_structural = structural[structural["all_conditions_feasible"]]
1047 feasible_scale = sensitivity[sensitivity["all_conditions_feasible"]]
1048 summary = {
1049     "power_model": metrics,
1050     "cluster_k": 4,
1051     "central_q4": q4_summary,
1052     "rapping_share_at_median_pct": float(
1053         rapping_power_tradeoff(train, model).iloc[0]["rapping_share_pct"]),
1054     "scale_sensitivity": {
1055         "feasible_combinations": int(len(feasible_scale)),
1056         "total_combinations": int(len(sensitivity)),
1057         "increase_pct_min": float(feasible_scale["weighted_power_increase_pct"].min()),
1058         "increase_pct_median": float(feasible_scale["weighted_power_increase_pct"].median()),
1059         "increase_pct_max": float(feasible_scale["weighted_power_increase_pct"].max()),
1060     },
1061     "structural_sensitivity": {
1062         "feasible_combinations": int(len(feasible_structural)),
1063         "total_combinations": int(len(structural)),
1064         "increase_pct_min": float(feasible_structural["weighted_power_increase_pct"].min()),
1065         "increase_pct_median": float(feasible_structural["weighted_power_increase_pct"].median()),
1066         "increase_pct_max": float(feasible_structural["weighted_power_increase_pct"].max()),
1067     },
1068     "optimiser_verification": {
1069         "max_random_search_gap_pct": float(verification["random_search_gap_pct"].max()),
1070         "max_multistart_spread_kw": float(verification["multistart_power_spread_kw"].max()),
1071     },
1072     "central_all_conditions_feasible": bool((optimum["status"] == "SCENARIO_FEASIBLE").all()),
1073 }
1074 (OUT_DIR / "model_run_summary.json").write_text(
1075     json.dumps(summary, ensure_ascii=False, indent=2), encoding="utf-8")
1076 print(json.dumps(summary, ensure_ascii=False, indent=2))
1077
1078
1079 if __name__ == "__main__":
1080     main()

```

E.2 paper_figures.py

```

1  #!/usr/bin/env python3
2  """Generate the complete publication figure set for the cement ESP paper.
3
4  All empirical plots are regenerated from the supplied processed dataset and
5  model result tables. Scenario-only plots are explicitly labelled in captions.
6  Both PNG previews and vector PDF files are exported.
7  """
8
9  from __future__ import annotations
10
11  import sys
12  import hashlib
13  from pathlib import Path
14
15  import matplotlib.pyplot as plt
16  from matplotlib.patches import FancyArrowPatch, FancyBboxPatch
17  import numpy as np
18  import pandas as pd
19
20  SRC_DIR = Path(__file__).resolve().parent
21  if str(SRC_DIR) not in sys.path:
22      sys.path.insert(0, str(SRC_DIR))

```

```

23
24 from plot_utils import (GRAY, LIGHT_GRAY, NEUTRAL_FILL, PALETTE, add_subpanel_label,
25                          figure_legend, finish_axes, headroom, note, save_figure, set_paper_style)
26 from scenario_model import (ALPHA_CENTRAL, PhysicalPowerModel, RidgePowerModel,
27                             T_COLS, U_COLS)
28
29
30 ROOT = SRC_DIR.parent
31 DATA_FILE = ROOT / "data" / "full_timeseries_with_flags.csv"
32 RESULTS = ROOT / "results"
33 OUT = ROOT / "figures_paper"
34
35 COND_SHORT = {
36     1: "工况1\n低温低浓高流",
37     2: "工况2\n高温中浓低流",
38     3: "工况3\n高温高浓高流",
39     4: "工况4\n低温高浓低流",
40 }
41
42 CAPTIONS = {
43     "01_outlet_data_diagnosis": "出口浓度记录存在显著截顶且不覆盖目标区间。有效记录全部位于48.74~50.00 mg/Nm³，其中5491条恰好
44     ⇨ 等于50，5与10 mg/Nm³目标区间内没有任何训练样本，因而无法由该列直接回归出目标区间附近的控制—排放关系。",
45     "02_process_timeseries": "首24小时入口温度、入口浓度、烟气流量与总功率的分钟级变化。入口变量与功率具有连续变化和工况迁
46     ⇨ 移特征，与图2中被压缩在单点附近的出口浓度形成对照。",
47     "22_technical_roadmap": "全文技术路线。可识别性审计把论证分为两条证据强度不同的路径：实证数据层（实线）可由附件复算，
48     ⇨ 条件式灰箱层（虚线）须经现场标定后方可采信；两条路径仅在设计域优化阶段合并，四问结论按其所属路径分别标注。",
49     "04b_operating_evidence": "附件中可识别的另一半信息。(a)(b)为两段持续异常工况及控制端的同步响应：入口浓度尖峰时抬高
50     ⇨  $\$U_{1\$}$ 、功率随之上冲，入口温度尖峰（比电阻高值区）时反而降压降功率；(c)为整周记录中工况变量与控制设定的相关矩
51     ⇨ 阵，显示入口浓度升高时缩短各场振打周期这一稳定规律。全部为历史操作的相关关系，不构成排放侧的因果证据。",
52     "14_rapping_peak_mechanism": "振打周期同时牵动电耗与瞬时峰值。(a)由每周累积灰量 $\$L \cdot T\$$ 推得单次再飞扬峰值随周
53     ⇨ 期近似线性放大，而在单位面积灰量对应的再飞扬比例近似不随周期变化的前提下，单位时间再飞扬量的一阶近似与周期无
54     ⇨ 关（水平点线）；(b)振打环节在中位工况下占总功率约四成，把周期由中位延长到历史极值可回收约109 kW，代价是单次积
55     ⇨ 灰量增加约18%。",
56     "03_cluster_selection": "两项内部指标并不一致：轮廓系数在 $\$k=4\$$ 取最大值，Calinski-Harabasz指标却在 $\$k=5\$$ 取最大值。因此 $\$k=4\$$ 并
57     ⇨ 非唯一的“数学最优”聚类数，本文取 $\$k=4\$$ 的依据是轮廓系数最高、四类簇心构成清晰的温度×流量结构，且四类比五类更便
58     ⇨ 于控制解释。",
59     "04_condition_map": "四类工况在入口浓度—温度平面上形成清晰分区，点的大小表示烟气流量，从而同时呈现三维工况信息。",
60     "07_power_prediction": "总功率由控制量近乎确定地决定。物理设定 $\$P=a_0+\sum b_i U_i^2+\sum c_i/T_i\$$ 在回顾性测试段的 $\$R^2\$$ 与RMSE
    ⇨ 均显著优于均值基线及把振打周期作为线性项的岭回归设定。",
    "08b_feature_importance": "振打周期必须以 $\$1/T\$$ 进入功率模型。(a)扣除电场功率后的剩余功率对 $\$T_{1\$}$ 呈明显双曲形状，线性设定
    ⇨ （虚线）在两端系统性偏离；(b)线性 $\$T\$$ 设定在回顾性测试段留下约-10.8 kW的平均偏差，改为 $\$1/T\$$ 后偏差降至约0，说明该偏
    ⇨ 差来自设定误差而非数据漂移。",
    "08_power_residuals": "改用 $\$1/T\$$ 设定后，回顾性测试残差围绕零对称分布，原先高功率区负偏差扩大的现象消失，确认修正后的误
    ⇨ 差不再含系统性结构。",
    "05b_boundary_proximity": "单变量幅值合规不等于组合落在历史联合运行经验之内。(a)八组解的马氏距离平方为经验97.5%外壳的2
    ⇨ 倍；(b)10 mg/Nm³解的各场电压均未超出该工况历史最大值，5 mg/Nm³解则需超出最多约5%。",
    "19_emission_power_frontier": "排放—功率权衡前沿。限值由10逐档收紧至5 mg/Nm³时四类工况的最优功率单调上升，工况加权功率
    ⇨ 由约1973升至约2105 kW；两档均高于历史实际加权功率1783.2 kW，说明超低排放改造是以增电耗换减排，而非节能。",
    "11_optimal_voltage": "两档限值下的电压配置并非四场机械同步抬升：分配由单位功率去除收益 $\alpha_i/(b_i U_i)^{\text{ref},2}$ 与排
    ⇨ 序约束共同决定，工况3、4中 $\$U_{2\$}$ 明显低于 $\$U_{1\$}$ ，工况4的 $\$U_{4\$}$ 被压至历史下界。",
    "13_emission_decomposition": "限值裕度由连续排放基值与振打峰值增量共同占用。柱顶虚线分别表示10和5 mg/Nm³情景限值；峰值项
    ⇨ 占用0.95~1.61 mg/Nm³，故约束必须按 $\$C_{\text{peak}}=C_{\text{base}}+B\$$ 整体施加，而不能只约束 $\$C_{\text{base}}\$$ 。",
    "18_condition_energy_penalty": "闭式估计与八变量数值最优解的一致性检验。(a)四类工况的点均落在 $\$y=x\$$ 线附近；(b)相对偏差为
    ⇨ -0.04%~+0.97%，全部落在±1%带内，说明按四场电压同比缩放导出的解析式可作为第四问功率代价的解析解释。",
    "15_search_convergence": "多起点SLSQP与同一可行域内 $4 \times 10^5$ 点随机搜索的对照。八个工况—限值组合中，47~53个起点全部收敛到
    ⇨ 同一解（功率一致到 $10^{-7}$  kW以内）；随机搜索均未找到更优解，其最优值反而高出0.97%~5.17%，工况1的10 mg/Nm³组合更是一
    ⇨ 个可行样本都没采到。这构成数值稳定性的强证据，但不是全局最优的数学证明。",
    "16_sensitivity_distribution": "81组结构情景全部可行。峰值指数在本文推导值1.00附近变化几乎不改变功率增幅，相位叠加尺度
    ⇨ 与场权重形状的影响也有限；加权增幅为5.13%~7.55%，中位数6.40%，说明核心结论对灰箱形状设定不敏感。",
    "20_eta_sensitivity": "功率增幅由幂次 $\$p\$$ 主导。 $\$K \propto U^p\$$ 时增幅近似正比于 $\$2/p\$$ ： $\$p=1\$$ 为14.2%， $\$p=2\$$ （Deutsch迁移速度形
    ⇨ 式）为6.8%， $\$p=3\$$ 为4.4%。其中 $\$p=1\$$ 与 $\$p=1.5\$$ 在5%电压裕度内无解，说明 $\$p\$$ 同时决定代价大小与目标可达性，是首要待标定参
    ⇨ 数。",

```

```

61 "17_sensitivity_heatmap": "场权重消融结果。三种去除贡献权重形状下，10→5 mg/Nm³所需的加权电压增量分布与按 $\alpha_i/(b_i U_{i\{ref\,2\}})$ 排序的解析优先级一致，说明优先级结论不由中心权重单独写入。",
62 }
63
64 TRACE_META = {
65     "01_outlet_data_diagnosis": (str(DATA_FILE), "fig01_outlet_diagnosis", "§ 2.2; 图2", "出口记录为量程截顶；不代表真实排放分布"),
66     "02_process_timeseries": (str(DATA_FILE), "fig02_timeseries", "§ 2.2; 图3", "只展示首24 h，不代表全周期分布"),
67     "22_technical_roadmap": (str(ROOT / "10_修订后完整论文_终稿.md"), "fig22_roadmap", "§ 3.3; 图4", "概念流程图，不含新的经验数据"),
68     "04b_operating_evidence": (str(RESULTS / "historical_strategy_correlations.csv"), "fig04b_operating_evidence", "§ 4.2—4.3; 图5", "相关系数为同期线性关联，不等于因果；两段事件为个案，样本量有限"),
69     "14_rapping_peak_mechanism": (str(RESULTS / "rapping_power_tradeoff.csv"), "fig14_peak", "§ 4.5; 图6", "峰值尺度 $b_0$ 仍为工程先验；小时均值结论依赖再飞扬比例近似恒定的前提"),
70     "03_cluster_selection": (str(RESULTS / "cluster_selection_metrics.csv"), "fig03_cluster_selection", "§ 5.1; 图7", "聚类指标只评价统计分离度，不等于物理工况唯一性"),
71     "04_condition_map": (str(DATA_FILE), "fig04_condition_map", "§ 5.1; 图8", "为可视化抽样点；聚类实际使用全部训练样本"),
72     "07_power_prediction": (str(RESULTS / "power_model_baselines.csv"), "fig07_power_prediction + PhysicalPowerModel", "§ 5.2; 图9", "回顾性测试并非真正未触碰盲测"),
73     "08b_feature_importance": (str(RESULTS / "power_model_baselines.csv"), "fig08b_specification", "§ 5.2; 图10", "残差图按 $T_1$ 展开，其余三场周期固定在中位数"),
74     "08_power_residuals": (str(DATA_FILE), "fig08_residuals + PhysicalPowerModel", "§ 5.2; 图11", "残差接近零均值，但仍非全新日期盲测"),
75     "05b_boundary_proximity": (str(RESULTS / "support_boundary_audit.csv"), "fig05b_boundary_proximity", "§ 5.6; 图12", "经验支持域不是设备安全边界；超出部分必须由现场阶跃试验确认"),
76     "19_emission_power_frontier": (str(RESULTS / "emission_power_frontier.csv"), "fig19_frontier", "§ 5.7; 图13", "各限值下均为灰箱情景解，前沿形状依赖排放先验"),
77     "11_optimal_voltage": (str(RESULTS / "optimal_controls_central_scenario.csv"), "grouped_controls(U_COLS)", "§ 6; 图14", "最优电压是情景试验起点，不是设备安全设定"),
78     "18_condition_energy_penalty": (str(RESULTS / "question4_by_condition.csv"), "fig18_penalty", "§ 7.2; 图15", "一致性只说明解析近似可用，不改变灰箱先验的条件性"),
79     "13_emission_decomposition": (str(RESULTS / "optimal_controls_central_scenario.csv"), "fig13_emission", "§ 7.4; 图16", "全部排放值为情景推演，不是实测"),
80     "15_search_convergence": (str(RESULTS / "optimization_verification.csv"), "fig15_optimizer_verification", "§ 8.1; 图17", "随机搜索对照只能证伪，不构成全局最优的数学证明"),
81     "16_sensitivity_distribution": (str(RESULTS / "structural_sensitivity.csv"), "fig16_structural_sensitivity", "§ 8.2; 图18", "扫描范围仍由显式情景网格决定"),
82     "20_eta_sensitivity": (str(RESULTS / "p_exponent_reference.csv"), "fig20_p_sensitivity", "§ 8.2; 图19", " $p$ 的取值区间为先验网格；需由现场阶跃试验标定"),
83     "17_sensitivity_heatmap": (str(RESULTS / "field_priority_ablation_summary.csv"), "fig17_field_ablation", "§ 8.2; 图20", "消融仍依赖同一灰箱结构与支持域"),
84 }
85
86 MANUSCRIPT_FIGURE = {
87     "01_outlet_data_diagnosis": 2,
88     "02_process_timeseries": 3,
89     "22_technical_roadmap": 4,
90     "04b_operating_evidence": 5,
91     "14_rapping_peak_mechanism": 6,
92     "03_cluster_selection": 7,
93     "04_condition_map": 8,
94     "07_power_prediction": 9,
95     "08b_feature_importance": 10,
96     "08_power_residuals": 11,
97     "05b_boundary_proximity": 12,
98     "19_emission_power_frontier": 13,
99     "11_optimal_voltage": 14,
100     "18_condition_energy_penalty": 15,
101     "13_emission_decomposition": 16,
102     "15_search_convergence": 17,
103     "16_sensitivity_distribution": 18,
104     "20_eta_sensitivity": 19,
105     "17_sensitivity_heatmap": 20,

```

```

106 }
107
108
109 def _label_bars(ax, bars, fmt="{:.0f}", dy=3.0):
110     for bar in bars:
111         h = bar.get_height()
112         ax.text(bar.get_x() + bar.get_width() / 2, h + dy, fmt.format(h), ha="center", va="bottom", fontsize=7)
113
114
115 def fig01_outlet_diagnosis(df: pd.DataFrame) -> None:
116     valid = df["C_out_mgNm3"].dropna()
117     fig, axes = plt.subplots(1, 2, figsize=(8.2, 3.2), gridspec_kw={"width_ratios": [1.15, 1]})
118     ax = axes[0]
119     add_subpanel_label(ax, "a")
120     bins = np.linspace(valid.min() - 0.03, valid.max() + 0.03, 28)
121     ax.hist(valid, bins=bins, color=PALETTE[0], edgecolor="white")
122     ax.axvline(50, color=PALETTE[4], linestyle="--", label="记录上限 50")
123     ax.set_xlim(valid.min() - 0.05, valid.max() + 0.05)
124     ax.text(
125         0.03,
126         0.78,
127         "5与10 mg/Nm³目标均在图示范围外",
128         transform=ax.transAxes,
129         color=PALETTE[2],
130         fontsize=8,
131         bbox={"boxstyle": "round", "facecolor": "white", "edgecolor": LIGHT_GRAY},
132     )
133     ax.set_xlabel("出口粉尘浓度记录 (mg/Nm³)")
134     ax.set_ylabel("样本数")
135     ax.legend(frameon=False, loc="upper left")
136     finish_axes(ax)
137
138     ax = axes[1]
139     add_subpanel_label(ax, "b")
140     total = len(df)
141     categories = ["精确值", "50截顶", "缺失"]
142     values = [int(df["C_out_exact_flag"].sum()), int(df["C_out_cap50_flag"].sum()), int(df["C_out_missing_flag"].sum())]
143     bars = ax.bar(categories, values, color=[PALETTE[0], PALETTE[4], GRAY])
144     for bar, value in zip(bars, values):
145         ax.text(bar.get_x() + bar.get_width()/2, value + total*0.012, f"{value}\n({100*value/total:.1f}%)", ha="center", fontsize=8)
146     ax.set_ylabel("样本数")
147     ax.set_ylim(0, max(values) * 1.18)
148     note(ax, "百分比以全部10080条样本为分母", loc="upper right")
149     finish_axes(ax)
150     fig.tight_layout(w_pad=2.0)
151     save_figure(fig, OUT, "01_outlet_data_diagnosis")
152
153
154 def fig02_timeseries(df: pd.DataFrame) -> None:
155     d = df.iloc[:1440].copy()
156     hours = np.arange(len(d)) / 60
157     fields = [
158         ("Temp_C", "入口温度 (°C)", PALETTE[4]),
159         ("C_in_gNm3", "入口浓度 (g/Nm³)", PALETTE[0]),
160         ("Q_Nm3h", "烟气流量 (Nm³/h)", PALETTE[2]),
161         ("P_total_kw", "总功率 (kW)", PALETTE[3]),
162     ]
163     fig, axes = plt.subplots(4, 1, figsize=(8.2, 5.5), sharex=True)
164     for ax, (col, label, color), tag in zip(axes, fields, ["a", "b", "c", "d"]):
165         add_subpanel_label(ax, tag, x=-0.08, y=1.02)
166         ax.plot(hours, d[col], color=color, linewidth=0.8)
167         ax.set_ylabel(label)

```

```

168     finish_axes(ax)
169     axes[-1].set_xlabel("首日运行时间 (h)")
170     axes[-1].set_xlim(0, 24)
171     fig.tight_layout(h_pad=0.5)
172     save_figure(fig, OUT, "02_process_timeseries")
173
174
175 def fig03_cluster_selection(cluster_eval: pd.DataFrame) -> None:
176     """The two internal indices disagree; the figure has to show that openly."""
177     k = cluster_eval["k"].to_numpy(int)
178     sil = cluster_eval["silhouette"].to_numpy(float)
179     ch = cluster_eval["calinski_harabasz"].to_numpy(float)
180     k_sil, k_ch = int(k[sil.argmax()]), int(k[ch.argmax()])
181
182     fig, axes = plt.subplots(1, 2, figsize=(8.0, 3.2))
183     for ax, y, best, label, color, marker in (
184         (axes[0], sil, k_sil, "轮廓系数", PALETTE[0], "o"),
185         (axes[1], ch / 1000.0, k_ch, "Calinski-Harabasz 指标 ( $\times 10^3$ )", PALETTE[2], "s")):
186         ax.plot(k, y, marker + "-", color=color, markersize=5.5, linewidth=1.7)
187         ax.scatter([best], [y[k == best]], s=150, facecolor="none",
188                 edgecolor=PALETTE[4], linewidth=1.8, zorder=4)
189         ax.annotate("最大值  $k=\%d$ " % best, (best, y[k == best][0]),
190                 textcoords="offset points", xytext=(0, 16), ha="center",
191                 fontsize=8.6, color=PALETTE[4])
192         ax.set_xlabel("聚类数  $k$ ")
193         ax.set_ylabel(label)
194         ax.set_xticks(k)
195         headroom(ax, 0.24)
196         finish_axes(ax, grid_axis="both")
197     add_subpanel_label(axes[0], "a")
198     add_subpanel_label(axes[1], "b")
199     note(axes[0], "本文取  $k=4$ ", loc="lower right")
200     note(axes[1], "该指标支持  $k=5$ ", loc="lower right")
201     fig.tight_layout(w_pad=2.2)
202     save_figure(fig, OUT, "03_cluster_selection")
203
204
205 def fig04_condition_map(df: pd.DataFrame) -> None:
206     train = df[df["split"] == "train"]
207     fig, ax = plt.subplots(figsize=(6.7, 4.4))
208     for cluster, group in train.groupby("condition_cluster"):
209         g = group.iloc[:5]
210         sizes = 8 + 28 * (g["Q_Nm3h"] - train["Q_Nm3h"].min()) / (train["Q_Nm3h"].max() - train["Q_Nm3h"].min())
211         ax.scatter(g["C_in_gNm3"], g["Temp_C"], s=sizes, alpha=0.38, color=PALETTE[int(cluster)-1], label=COND_SHORT[int(cluster)].replace("\
↔ n", " "))
212         ax.set_xlabel("入口粉尘浓度 (g/Nm3)")
213         ax.set_ylabel("入口温度 (°C)")
214         ax.legend(frameon=False, ncol=2)
215         finish_axes(ax, grid_axis="both")
216         fig.tight_layout()
217         save_figure(fig, OUT, "04_condition_map")
218
219
220
221 def fig04b_operating_evidence(df: pd.DataFrame, events: pd.DataFrame,
222                             strategy: pd.DataFrame) -> None:
223     """The identifiable half of Question 1.
224
225     The outlet column carries no signal, but the operator's response to the
226     inlet conditions is plainly visible: two sustained excursions (panels a, b)
227     and a stable correlation structure between conditions and settings (c).
228     """

```

```

229 fig = plt.figure(figsize=(8.4, 5.6))
230 gs = fig.add_gridspec(2, 2, height_ratios=[1, 1.05], hspace=0.60, wspace=0.62)
231
232 windows = [
233     ("2024-05-02 13:30", "2024-05-02 16:20", "C_in_gNm3", "入口浓度 (g/Nm³)", "a"),
234     ("2024-05-06 12:30", "2024-05-06 15:30", "Temp_C", "入口温度 (°C)", "b"),
235 ]
236 for col_idx, (t0, t1, col, ylabel, tag) in enumerate(windows):
237     ax = fig.add_subplot(gs[0, col_idx])
238     add_subpanel_label(ax, tag, x=-0.14)
239     win = df[(df["timestamp"] >= t0) & (df["timestamp"] <= t1)]
240     minutes = (win["timestamp"] - win["timestamp"].iloc[0]).dt.total_seconds() / 60
241     ax.plot(minutes, win[col], color=PALETTE[4], linewidth=1.4, label=ylabel.split(" ")[0])
242     ax.set_ylabel(ylabel, color=PALETTE[4], fontsize=9)
243     ax.tick_params(axis="y", labelcolor=PALETTE[4])
244     ax.set_xlabel("事件窗口内时间 (min)")
245     twin = ax.twinx()
246     twin.plot(minutes, win["U1_kV"], color=PALETTE[0], linewidth=1.4, label="$U_1$")
247     twin.plot(minutes, win["P_total_kW"] / 30.0, color=PALETTE[2],
248                 linewidth=1.2, linestyle="--", label="总功率/30")
249     twin.set_ylabel("$U_1$ (kV) 总功率/30 (kW)", fontsize=8.0)
250     twin.spines["top"].set_visible(False)
251     finish_axes(ax, grid_axis="none")
252     handles = ax.get_lines() + twin.get_lines()
253     ax.legend(handles, [h.get_label() for h in handles],
254               frameon=False, fontsize=7.4, ncol=3, loc="upper center",
255               bbox_to_anchor=(0.5, 1.30))
256
257 ax = fig.add_subplot(gs[1, :])
258 add_subpanel_label(ax, "c", x=-0.055)
259 cond_labels = {"r_Temp_C": "入口温度 $$", "r_C_in_gNm3": "入口浓度 $C_{in}$", "r_Q_Nm3h": "烟气流量 $Q$"}
260 full = strategy[strategy["window"] == "full_record"].set_index("control")
261 matrix = full.loc[U_COLS + T_COLS, list(cond_labels)].to_numpy(float).T
262 im = ax.imshow(matrix, cmap="RdBu_r", aspect="auto", vmin=-0.30, vmax=0.30)
263 ax.set_xticks(range(matrix.shape[1]),
264               [f"$U_{i}$" for i in range(1, 5)] + [f"$T_{i}$" for i in range(1, 5)])
265 ax.set_yticks(range(3), list(cond_labels.values()))
266 for i in range(matrix.shape[0]):
267     for j in range(matrix.shape[1]):
268         ax.text(j, i, f"{matrix[i, j]:+.2f}", ha="center", va="center", fontsize=8.6,
269               color="white" if abs(matrix[i, j]) > 0.20 else "#1F2933")
270 cbar = fig.colorbar(im, ax=ax, fraction=0.028, pad=0.015)
271 cbar.set_label("Pearson 相关系数", fontsize=8.4)
272 ax.set_title("全记录期工况变量与控制设定的历史响应 (可由附件识别)", fontsize=9.2, pad=8)
273 tr = strategy[strategy["window"] == "train"].set_index("control")
274 te = strategy[strategy["window"] == "retrospective_test"].set_index("control")
275 ax.text(1.0, -0.32,
276        "注: 振打周期规律 ($C_{in} \uparrow \rightarrow T \downarrow$) 在四个时窗内一致; "
277        "电压规律并不平稳——$r(C_{in}, U_1)$ 在前5天为 %.2f, 在末日为 %.2f。"
278        % (float(tr.loc["U1_kV", "r_C_in_gNm3"]), float(te.loc["U1_kV", "r_C_in_gNm3"])),
279        transform=ax.transAxes, ha="right", va="top", fontsize=7.8, color=GRAY)
280 save_figure(fig, OUT, "04b_operating_evidence")
281
282
283 def fig22_roadmap() -> None:
284     """One-page technical roadmap.
285
286     The paper's argument splits at the identifiability audit into an
287     empirically identified branch and an explicitly conditional grey-box
288     branch; the two are visually distinguished so a reviewer can see at a
289     glance which conclusions carry which strength of evidence.
290     """

```

```

291 EMP, SCN = "#1F4E79", "#B45309" # empirical / conditional
292 fig, ax = plt.subplots(figsize=(7.4, 6.6))
293 ax.set_xlim(0, 10); ax.set_ylim(0, 12.2); ax.axis("off")
294
295 def box(x, y, w, h, text, color, dashed=False, fill="#FFFFFF", fs=9.0):
296     ax.add_patch(FancyBboxPatch(
297         (x - w / 2, y - h / 2), w, h, boxstyle="round,pad=0.10",
298         linewidth=1.3, edgecolor=color, facecolor=fill,
299         linestyle="--" if dashed else "-", zorder=2))
300     ax.text(x, y, text, ha="center", va="center", fontsize=fs,
301           color="#1F2933", zorder=3, linespacing=1.45)
302
303 def arrow(x1, y1, x2, y2, color=GRAY):
304     ax.add_patch(FancyArrowPatch((x1, y1), (x2, y2), arrowstyle="-|>",
305                                mutation_scale=13, linewidth=1.1,
306                                color=color, zorder=1,
307                                shrinkA=2, shrinkB=2))
308
309 box(5, 11.5, 5.6, 0.78, "现场原始数据 10080×14", GRAY, fill="#F3F4F6")
310 arrow(5, 11.11, 5, 10.62)
311 box(5, 10.2, 4.6, 0.80, "§ 2 可识别性审计", "#111827", fill="#E5E7EB")
312
313 arrow(4.0, 9.80, 2.6, 9.25); arrow(6.0, 9.80, 7.4, 9.25)
314 box(2.6, 8.55, 4.0, 1.30,
315     "实证数据层（可复算）\n § 4.1  $P=a_0+\sum b_i u_i^2+\sum c_i/T_i$ \n"
316     "§ 4.2—4.3 历史操作规律与异常工况", EMP)
317 box(7.4, 8.55, 4.0, 1.30,
318     "条件式灰箱层（推演）\n § 4.4—4.5  $SC_{\{peak\}}(U,T)=C_{\{base\}}+B_{\$}$ \n"
319     "Deutsch-Anderson 锚点 50 mg/Nm³", SCN, dashed=True)
320
321 arrow(2.6, 7.90, 4.2, 7.35); arrow(7.4, 7.90, 5.8, 7.35)
322 box(5, 6.95, 5.2, 0.80, "§ 5.1、§ 5.3 典型工况划分 + 历史支持域", "#111827")
323
324 arrow(5, 6.55, 5, 6.05)
325 box(5, 5.65, 5.4, 0.80, "问题二 § 5.5 10 mg/Nm³ 最低功率控制参数", EMP)
326 arrow(5, 5.25, 5, 4.75)
327 box(5, 4.35, 5.4, 0.80, "问题三 § 5.4、§ 6 电场调节优先级判据", EMP)
328 arrow(5, 3.95, 5, 3.45)
329 box(5, 3.05, 5.4, 0.80, "问题四 § 7 10→5 mg/Nm³ 功率与容量代价", SCN, dashed=True)
330
331 arrow(5, 2.65, 5, 2.15)
332 box(5, 1.55, 6.6, 1.10,
333     "§ 5.6 外推审计 • § 8.2 敏感性分析\n § 9.3 现场阶跃标定 • 独立盲测后闭环",
334     GRAY, fill="#F3F4F6", fs=8.6)
335
336 handles = [
337     plt.Line2D([], [], color=EMP, linewidth=1.6, label="实证辨识：可由附件复算"),
338     plt.Line2D([], [], color=SCN, linewidth=1.6, linestyle="--",
339               label="条件式推演：须现场标定后方可采信"),
340 ]
341 ax.legend(handles=handles, loc="lower center", bbox_to_anchor=(0.5, -0.03),
342          ncol=2, frameon=False, fontsize=8.4)
343 fig.tight_layout()
344 save_figure(fig, OUT, "22_technical_roadmap")
345
346
347 def fig05b_boundary_proximity(audit: pd.DataFrame) -> None:
348     """How far outside historical joint operating experience each solution sits."""
349     ordered = audit.sort_values(["condition_cluster", "limit_mgNm3"],
350                                ascending=[True, False]).reset_index(drop=True)
351     labels = [f"工况{int(c)}\n{int(l)} mg"
352              for c, l in zip(ordered["condition_cluster"], ordered["limit_mgNm3"])]

```

```

353 colors = [PALETTE[int(c) - 1] for c in ordered["condition_cluster"]]
354
355 fig, axes = plt.subplots(1, 2, figsize=(8.4, 3.6), gridspec_kw={"width_ratios": [1.15, 1]})
356 ax = axes[0]
357 add_subpanel_label(ax, "a")
358 x = np.arange(len(ordered))
359 ax.vlines(x, 1.0, ordered["ratio_to_q975"], color=colors, linewidth=2.2, alpha=0.8)
360 ax.scatter(x, ordered["ratio_to_q975"], color=colors, s=45,
361           edgcolor="black", linewidth=0.5, zorder=3)
362 ax.axhline(1.0, color=PALETTE[4], linestyle="--", linewidth=1.2, label="历史经验97.5%外壳")
363 ax.set_xticks(x, labels, fontsize=7.6)
364 ax.set_ylabel(r"$d_M^2/q_{0.975}$ (倍)")
365 ax.legend(frameon=False, loc="upper left", fontsize=8)
366 headroom(ax, 0.16)
367 finish_axes(ax)
368
369 ax = axes[1]
370 add_subpanel_label(ax, "b")
371 bars = ax.bar(x, ordered["max_voltage_over_historical_max_pct"], color=colors, width=0.62)
372 _label_bars(ax, bars, fmt="{:.1f}%", dy=0.12)
373 ax.axhline(0, color=PALETTE[4], linestyle="--", linewidth=1.2)
374 ax.set_xticks(x, labels, fontsize=7.6)
375 ax.set_ylabel("最高电压超出历史最大值 (%)")
376 note(ax, "0%以下=仍在历史电压包络内", loc="upper left")
377 headroom(ax, 0.30)
378 finish_axes(ax)
379 fig.tight_layout(w_pad=1.9)
380 save_figure(fig, OUT, "05b_boundary_proximity")
381
382
383 def fit_power_model(df: pd.DataFrame) -> PhysicalPowerModel:
384     train = df[df["split"] == "train"]
385     return PhysicalPowerModel().fit(train, train["P_total_kW"].to_numpy(float))
386
387
388 def fit_linear_period_model(df: pd.DataFrame) -> RidgePowerModel:
389     """The misspecified reference: rapping period entered as a linear term."""
390     train = df[df["split"] == "train"]
391     metrics = pd.read_json(RESULTS / "power_model_metrics.json", typ="series")
392     alpha = float(metrics["ridge_selected_alphas"]["quadratic_ridge_T"])
393     model = RidgePowerModel(alpha=alpha)
394     model.fit(train, train["P_total_kW"].to_numpy(float))
395     return model
396
397
398 def fig07_power_prediction(df: pd.DataFrame, model: RidgePowerModel, baselines: pd.DataFrame) -> None:
399     test = df[df["split"] == "test"]
400     actual = test["P_total_kW"].to_numpy(float)
401     pred = model.predict(test)
402     fig, axes = plt.subplots(1, 2, figsize=(8.2, 3.5))
403     ax = axes[0]
404     add_subpanel_label(ax, "a")
405     ax.scatter(actual, pred, s=10, alpha=0.42, color=PALETTE[0], edgcolors="none")
406     lo, hi = min(actual.min(), pred.min()), max(actual.max(), pred.max())
407     ax.plot([lo, hi], [lo, hi], "--", color=PALETTE[4], label="理想预测")
408     metrics = pd.read_json(RESULTS / "power_model_metrics.json", typ="series")
409     ax.text(0.04, 0.94,
410           "$R^2$=%.3f$\nRMSE$=%.2f kW\n偏差$=+.2f$ kW" % (float(metrics["retrospective_test_r2"]),
411                                                         float(metrics["retrospective_test_rmse_kW"]),
412                                                         float(metrics["retrospective_test_bias_kW"])),
413           transform=ax.transAxes, va="top",
414           bbox={"boxstyle": "round", "facecolor": "white", "edgecolor": "LIGHT_GRAY"})

```

```

415 ax.set_xlabel("实际总功率 (kW)")
416 ax.set_ylabel("预测总功率 (kW)")
417 ax.legend(frameon=False, loc="lower right")
418 finish_axes(ax, grid_axis="both")
419
420 ax = axes[1]
421 add_subpanel_label(ax, "b")
422 labels = ["均值\n基线", "线性$U$\n线性$T$", "二次$U$\n线性$T$", "本文\n$U^2+1/T$"]
423 model_order = ["mean_predictor", "linear_ridge_T", "quadratic_ridge_T", "physical_invT"]
424 x = np.arange(len(model_order))
425 val = baselines[baselines["split"] == "validation"].set_index("model").loc[model_order, "rmse_kw"]
426 tst = baselines[baselines["split"] == "retrospective_test"].set_index("model").loc[model_order, "rmse_kw"]
427 ax.bar(x-0.18, val, 0.36, color=PALETTE[5], label="验证集")
428 ax.bar(x+0.18, tst, 0.36, color=PALETTE[0], label="回顾性测试")
429 ax.set_xticks(x, labels, fontsize=7.6)
430 ax.set_ylabel("RMSE (kW, 对数刻度)")
431 ax.set_yscale("log")
432 ax.legend(frameon=False, loc="upper right")
433 finish_axes(ax)
434 fig.tight_layout()
435 save_figure(fig, OUT, "07_power_prediction")
436
437
438 def fig08_residuals(df: pd.DataFrame, model: RidgePowerModel) -> None:
439     test = df[df["split"] == "test"]
440     actual = test["P_total_kw"].to_numpy(float)
441     pred = model.predict(test)
442     res = pred - actual
443     fig, axes = plt.subplots(1, 2, figsize=(7.7, 3.1))
444     add_subpanel_label(axes[0], "a")
445     axes[0].scatter(pred, res, s=9, alpha=0.38, color=PALETTE[0], edgecolors="none")
446     axes[0].axhline(0, color=PALETTE[4], linestyle="--")
447     axes[0].set_xlabel("预测总功率 (kW)")
448     axes[0].set_ylabel("残差: 预测-实际 (kW)")
449     finish_axes(axes[0], grid_axis="both")
450     add_subpanel_label(axes[1], "b")
451     axes[1].hist(res, bins=28, color=PALETTE[2], edgecolor="white")
452     metrics = pd.read_json(RESULTS / "power_model_metrics.json", typ="series")
453     guard = float(metrics["validation_abs_error_q90_kw"])
454     axes[1].axvline(res.mean(), color=PALETTE[4], linestyle="--", label=f"测试均值 {res.mean():.2f} kW")
455     axes[1].axvline(guard, color=PALETTE[0], linestyle=":", label=f"验证|误差|P90 {guard:.2f} kW")
456     axes[1].set_xlabel("残差 (kW)")
457     axes[1].set_ylabel("样本数")
458     axes[1].legend(frameon=False)
459     finish_axes(axes[1])
460     fig.tight_layout(w_pad=2.0)
461     save_figure(fig, OUT, "08_power_residuals")
462
463
464 def fig08b_specification(df: pd.DataFrame, model: PhysicalPowerModel,
465                        linear_model: RidgePowerModel, baselines: pd.DataFrame) -> None:
466     """Why the rapping period must enter as 1/T rather than T.
467
468     Panel (a) shows the residual of a voltage-only fit against the period:
469     the curvature is the signature of a 1/T term. Panel (b) contrasts the
470     retrospective-test bias of both specifications.
471     """
472     train = df[df["split"] == "train"]
473     test = df[df["split"] == "test"]
474     fig, axes = plt.subplots(1, 2, figsize=(8.2, 3.4), gridspec_kw={"width_ratios": [1.15, 1]})
475
476     ax = axes[0]

```

```

477 add_subpanel_label(ax, "a")
478 u = train[U_COLS].to_numpy(float)
479 resid = train["P_total_kw"].to_numpy(float) - model.field_power(u) - model.intercept
480 t1 = train["T1_s"].to_numpy(float)
481 order = np.argsort(t1)
482 ax.scatter(t1[order][:7], resid[order][:7], s=7, alpha=0.22,
483            color=PALETTE[5], edgecolors="none", label="扣除电场功率后的残差")
484 grid = np.linspace(t1.min(), t1.max(), 120)
485 rap_other = float(np.sum(model.c[1:] / train[T_COLS[1:]].median().to_numpy(float)))
486 ax.plot(grid, model.c[0] / grid + rap_other, color=PALETTE[4], linewidth=2.0,
487         label=r"拟合  $\mathcal{C}_1/T_1 + \mathrm{const}$ ")
488 slope = np.polyfit(t1, resid, 1)
489 ax.plot(grid, np.polyval(slope, grid), "--", color=PALETTE[0], linewidth=1.6,
490         label=r"线性  $\mathcal{T}_1$  设定")
491 ax.set_xlabel("第1电场振打周期  $\mathcal{T}_1$  (s)")
492 ax.set_ylabel("剩余功率 (kW)")
493 ax.legend(frameon=False, fontsize=8, loc="upper right")
494 finish_axes(ax, grid_axis="both")
495
496 ax = axes[1]
497 add_subpanel_label(ax, "b")
498 names = {"quadratic_ridge_T": "线性  $\mathcal{T}$  设定", "physical_invT": "本文  $\mathcal{I}/\mathcal{T}$  设定"}
499 tst = baselines[baselines["split"] == "retrospective_test"].set_index("model")
500 x = np.arange(2)
501 bias = [float(tst.loc[k, "bias_kw"]) for k in names]
502 rmse = [float(tst.loc[k, "rmse_kw"]) for k in names]
503 ax.bar(x - 0.18, bias, 0.36, color=PALETTE[4], label="测试平均偏差")
504 ax.bar(x + 0.18, rmse, 0.36, color=PALETTE[0], label="测试 RMSE")
505 for xi, (b, r) in enumerate(zip(bias, rmse)):
506     ax.text(xi - 0.18, b + (0.6 if b >= 0 else -1.6), f"{b:+.2f}", ha="center", fontsize=8)
507     ax.text(xi + 0.18, r + 0.6, f"{r:.2f}", ha="center", fontsize=8)
508 ax.axhline(0, color=GRAY, linewidth=0.8)
509 ax.set_xticks(x, list(names.values()))
510 ax.set_ylabel("回顾性测试 (kW)")
511 ax.legend(frameon=False, fontsize=8)
512 headroom(ax, 0.22)
513 finish_axes(ax)
514 fig.tight_layout(w_pad=2.0)
515 save_figure(fig, OUT, "08b_feature_importance")
516
517
518 def grouped_controls(optimum: pd.DataFrame, cols: list[str], stem: str, ylabel: str) -> None:
519     fig, axes = plt.subplots(2, 2, figsize=(8.2, 5.8), sharey=True)
520     for ax, cluster, tag in zip(axes.flat, range(1,5), ["a", "b", "c", "d"]):
521         add_subpanel_label(ax, tag)
522         sub = optimum[optimum["condition_cluster"] == cluster].set_index("limit_mgNm3")
523         x = np.arange(4)
524         ax.bar(x-0.18, sub.loc[10.0, cols], 0.36, color=PALETTE[5], label="10 mg/Nm³")
525         ax.bar(x+0.18, sub.loc[5.0, cols], 0.36, color=PALETTE[4], label="5 mg/Nm³")
526         ax.set_xticks(x, [c.split("_")[0] for c in cols])
527         ax.set_title(COND_SHORT[cluster].replace("\n", " "), fontsize=9)
528         finish_axes(ax)
529     axes[0,0].set_ylabel(ylabel); axes[1,0].set_ylabel(ylabel)
530     handles, labels = axes[0,0].get_legend_handles_labels()
531     fig.tight_layout(h_pad=2.0, rect=(0, 0.06, 1, 1))
532     figure_legend(fig, handles, labels, ncol=2, bottom=0.0)
533     save_figure(fig, OUT, stem)
534
535
536
537 def fig13_emission(optimum: pd.DataFrame) -> None:
538     x = np.arange(8)

```

```

539 ordered = optimum.sort_values(["Limit_mgNm3", "condition_cluster"], ascending=[False, True])
540 base = ordered["scenario_base_emission_mgNm3"].to_numpy()
541 peak = ordered["scenario_peak_excess_mgNm3"].to_numpy()
542 colors = [PALETTE[0]]*4 + [PALETTE[4]]*4
543 fig, ax = plt.subplots(figsize=(8.0, 4.0))
544 ax.bar(x, base, color=colors, alpha=0.82, label="连续排放基值")
545 ax.bar(x, peak, bottom=base, color=colors, alpha=0.35, edgecolor=colors, label="振打峰值增量")
546 ax.hlines(10, -0.5, 3.5, colors=GRAY, linestyle="--", linewidth=1.0)
547 ax.hlines(5, 3.5, 7.5, colors=GRAY, linestyle="--", linewidth=1.0)
548 ax.text(3.45, 10.15, "10 mg/Nm³", ha="right", fontsize=8, color=GRAY)
549 ax.text(7.45, 5.15, "5 mg/Nm³", ha="right", fontsize=8, color=GRAY)
550 ax.set_xticks(x, [f"工况{i}" for i in range(1,5)]*2)
551 ax.set_ylabel("峰值总排放情景值 (mg/Nm³)")
552 ax.legend(frameon=False, ncol=2)
553 finish_axes(ax)
554 fig.tight_layout()
555 save_figure(fig, OUT, "13_emission_decomposition")
556
557
558 def fig14_peak(profiles: pd.DataFrame) -> None:
559     """The rapping trade-off, derived rather than assumed.
560
561     Cake mass per rap grows like L*T, so the per-event peak grows linearly with
562     the period while the contribution to the hourly mean, L*T*(1/T) = L, does
563     not depend on the period at all. Panel (b) prices the same lever in power.
564     """
565     trade = pd.read_csv(RESULTS / "rapping_power_tradeoff.csv")
566     ratio = np.linspace(0.75, 1.35, 200)
567     fig, axes = plt.subplots(1, 2, figsize=(8.3, 3.5), gridspec_kw={"width_ratios": [1.1, 1]})
568
569     ax = axes[0]
570     add_subpanel_label(ax, "a")
571     for cluster in (1, 3, 4):
572         pr = profiles[profiles["condition_cluster"] == cluster].iloc[0]
573         peak = 1.2 * float(pr["load_ratio_to_train_median"]) * ratio
574         ax.plot(100 * (ratio - 1), peak, color=PALETTE[cluster - 1],
575               marker=["o", "s", "A", "D"][cluster - 1], markevery=40,
576               label=COND_SHORT[cluster].replace("\n", " "))
577     ax.axhline(0, color=GRAY, linewidth=0.8)
578     ax.axvline(0, color=GRAY, linewidth=0.8)
579     ax.plot(100 * (ratio - 1), np.full_like(ratio, 1.2 * float(
580         profiles[profiles["condition_cluster"] == 3].iloc[0]["load_ratio_to_train_median"])),
581           ":", color="#111827", linewidth=1.5, label="对小时均值的贡献 (工况3, 与TS无关)")
582     ax.set_xlabel("振打周期相对参考周期变化 (%)")
583     ax.set_ylabel("单次再飞扬峰值代理 (mg/Nm³)")
584     ax.legend(frameon=False, fontsize=7.8, loc="upper left")
585     headroom(ax, 0.18)
586     finish_axes(ax, grid_axis="both")
587
588     ax = axes[1]
589     add_subpanel_label(ax, "b")
590     x = np.arange(len(trade))
591     bars = ax.bar(x, trade["rapping_power_kw"], color=[PALETTE[5], PALETTE[2], PALETTE[4]], width=0.6)
592     _label_bars(ax, bars, fmt="{:.0f}", dy=6)
593     for xi, row in trade.iterrows():
594         if row["power_recovered_kw"] > 0:
595             ax.text(xi, row["rapping_power_kw"] / 2,
596                   "省%.0f kW\n积灰+%.0f%%" % (row["power_recovered_kw"],
597                                             row["cake_mass_per_rap_change_pct"]),
598                   ha="center", va="center", fontsize=7.6, color="white")
599     ax.set_xticks(x, ["历史\n中位周期", "延长至\n历史P95", "延长至\n历史最大"], fontsize=8)
600     ax.set_ylabel("振打环节功率 (kW)")

```

```

601 note(ax, "中位工况下振打占总功率 %.1f%%" % float(trade.iloc[0]["rapping_share_pct"]),
602       loc="upper right")
603 headroom(ax, 0.24)
604 finish_axes(ax)
605 fig.tight_layout(w_pad=2.0)
606 save_figure(fig, OUT, "14_rapping_peak_mechanism")
607
608
609 def fig15_optimizer_verification(verification: pd.DataFrame) -> None:
610     """Multi-start SLSQP against a 4x10^5-point random search of the same set."""
611     ordered = verification.sort_values(["condition_cluster", "limit_mgNm3"],
612                                       ascending=[True, False]).reset_index(drop=True)
613     labels = [f"工况{int(c)}\n{int(l)} mg"
614              for c, l in zip(ordered["condition_cluster"], ordered["limit_mgNm3"])]
615     x = np.arange(len(ordered))
616     slsqp = ordered["slsqp_power_kw"].to_numpy(float)
617     rnd = ordered["random_search_power_kw"].to_numpy(float)
618     gaps = ordered["random_search_gap_pct"].to_numpy(float)
619
620     fig, axes = plt.subplots(1, 2, figsize=(8.4, 3.3), gridspec_kw={"width_ratios": [1.2, 1]})
621
622     ax = axes[0]
623     add_subpanel_label(ax, "a")
624     lo = float(np.nanmin(slsqp)) * 0.94
625     hi = float(np.nanmax([np.nanmax(slsqp), np.nanmax(rnd)])) * 1.04
626     ax.set_ylim(lo, hi)
627     ax.bar(x - 0.18, slsqp, 0.36, color=PALETTE[0], label="多起点SLSQP")
628     ax.bar(x + 0.18, np.where(np.isfinite(rnd), rnd, lo), 0.36,
629           color=LIGHT_GRAY, label="随机搜索最优")
630     for xi, value in enumerate(rnd):
631         if not np.isfinite(value):
632             ax.annotate("随机搜索\n无可行样本", xy=(xi + 0.18, lo),
633                       xytext=(xi + 0.55, lo + 0.30 * (hi - lo)),
634                       fontsize=7.2, color=PALETTE[4], ha="left",
635                       arrowprops={"arrowstyle": "->", "color": PALETTE[4], "linewidth": 0.9})
636     ax.set_xticks(x, labels, fontsize=7.4)
637     ax.set_ylabel("最优总功率 (kW)")
638     ax.legend(frameon=False, fontsize=8, loc="upper left", ncol=2)
639     finish_axes(ax)
640
641     ax = axes[1]
642     add_subpanel_label(ax, "b")
643     finite = np.isfinite(gaps)
644     ax.bar(x[finite], gaps[finite], color=PALETTE[2], width=0.6)
645     for xi, value in zip(x[finite], gaps[finite]):
646         ax.text(xi, value + 0.12, f"{value:.2f}", ha="center", va="bottom", fontsize=7.4)
647     ax.set_ylim(0, float(np.nanmax(gaps)) * 1.42)
648     ax.set_xticks(x, labels, fontsize=7.4)
649     ax.set_ylabel("随机搜索相对SLSQP的劣势 (%)")
650     note(ax, "全部为正: 随机搜索未找到更优解", loc="upper right")
651     finish_axes(ax)
652     fig.tight_layout(w_pad=2.2)
653     save_figure(fig, OUT, "15_search_convergence")
654
655
656 def fig16_structural_sensitivity(structural: pd.DataFrame) -> None:
657     valid = structural[structural["all_conditions_feasible"] == True]
658     fig, axes = plt.subplots(1, 2, figsize=(8.2, 3.5), sharey=True)
659     ax = axes[0]
660     add_subpanel_label(ax, "a")
661     exs = sorted(valid["peak_exponent"].unique())
662     data = [valid[valid["peak_exponent"] == x]["weighted_power_increase_pct"] for x in exs]

```

```

663 boxes = ax.boxplot(data, tick_labels=[f"{x:.2f}" for x in exps],
664                     patch_artist=True, showfliers=False)
665 for i, box in enumerate(boxes["boxes"]):
666     box.set_facecolor(PALETTE[i % len(PALETTE)]); box.set_alpha(0.72)
667 ax.set_xlabel(r"振打峰值指数  $\gamma$  (本文推导值 1.00) ")
668 ax.set_ylabel("5相对10 mg/Nm2加权功率增幅 (%)")
669 finish_axes(ax)
670
671 ax = axes[1]
672 add_subpanel_label(ax, "b")
673 phases = sorted(valid["phase_scale"].unique())
674 for profile, style, label, color in zip(
675     ["equal", "weak_front", "central_front"], ["o-", "s--", "A:"],
676     ["等权", "弱前级", "中心前级"], [PALETTE[0], PALETTE[2], PALETTE[4]]):
677     medians = [valid[(valid["phase_scale"] == ph) & (valid["alpha_profile"] == profile)]
678                 ["weighted_power_increase_pct"].median() for ph in phases]
679     ax.plot(phases, medians, style, color=color, label=label)
680 ax.set_xlabel(r"振打相位叠加尺度  $\rho$ ")
681 ax.legend(frameon=False)
682 finish_axes(ax, grid_axis="both")
683 fig.tight_layout(w_pad=2.0)
684 save_figure(fig, OUT, "16_sensitivity_distribution")
685
686
687 def fig17_field_ablation(ablation: pd.DataFrame) -> None:
688     profiles = ["equal", "weak_front", "central_front"]
689     labels = ["等权", "弱前级", "中心前级"]
690     cols = [f"weighted_delta_U{i}_kV" for i in range(1,5)]
691     d = ablation.set_index("alpha_profile").loc[profiles, cols]
692     fig, axes = plt.subplots(1, 2, figsize=(7.8, 3.3), gridspec_kw={"width_ratios": [1.45, 1]})
693     ax = axes[0]
694     add_subpanel_label(ax, "a")
695     im = ax.imshow(d.to_numpy(), cmap="YlGnBu", aspect="auto", vmin=0)
696     ax.set_xticks(range(4), [f"U{i}" for i in range(1,5)])
697     ax.set_yticks(range(3), labels)
698     ax.set_xlabel("电场")
699     ax.set_ylabel("去除贡献权重设定")
700     for i in range(3):
701         for j in range(4):
702             ax.text(j, i, f"{d.iloc[i,j]:.2f}", ha="center", va="center", color="white" if d.iloc[i,j]>5 else "black", fontsize=8)
703     cbar = fig.colorbar(im, ax=ax, fraction=0.05, pad=0.04)
704     cbar.set_label("10→5 mg时加权电压增量 (kV)")
705
706     ax = axes[1]
707     add_subpanel_label(ax, "b")
708     shares = 100*ablation.set_index("alpha_profile").loc[profiles, "front_two_share_of_positive_adjustment"]
709     bars = ax.bar(labels, shares, color=[PALETTE[0], PALETTE[2], PALETTE[4]])
710     _label_bars(ax, bars, fmt="{:.1f}%", dy=0.7)
711     ax.set_ylabel("前两电场占正向电压增量 (%)")
712     ax.set_ylim(0,100)
713     finish_axes(ax)
714     fig.tight_layout(w_pad=2.0)
715     save_figure(fig, OUT, "17_sensitivity_heatmap")
716
717
718 def fig18_penalty(q4: pd.DataFrame) -> None:
719     """Does the closed-form estimate reproduce the full eight-variable optimum?
720
721     Four near-equal bars of ~6.7% carry little information; what is worth
722     showing is that the analytical approximation and the numerical optimum
723     agree to within 1%.
724     """

```

```

725 q4 = q4.sort_values("condition_cluster")
726 ana = q4["analytic_delta_power_kw"].to_numpy(float)
727 num = q4["numeric_delta_power_kw"].to_numpy(float)
728 rel = 100.0 * (ana - num) / num
729
730 fig, axes = plt.subplots(1, 2, figsize=(8.3, 3.5), gridspec_kw={"width_ratios": [1, 1.05]})
731
732 ax = axes[0]
733 add_subpanel_label(ax, "a")
734 lo, hi = min(ana.min(), num.min()) - 8, max(ana.max(), num.max()) + 8
735 ax.plot([lo, hi], [lo, hi], "--", color=GRAY, linewidth=1.2, label="$y=x$")
736 for idx in range(len(q4)):
737     ax.scatter(num[idx], ana[idx], s=70, color=PALETTE[idx], zorder=3,
738               edgecolor="white", linewidth=0.8,
739               label=COND_SHORT[q4["condition_cluster"].iloc[idx]].replace("\n", " "))
740 ax.set_xlim(lo, hi); ax.set_ylim(lo, hi)
741 ax.set_xlabel("数值最优解的  $\Delta P$  (kW)")
742 ax.set_ylabel("闭式估计的  $\Delta P$  (kW)")
743 ax.legend(frameon=False, fontsize=7.4, loc="upper left")
744 finish_axes(ax, grid_axis="both")
745
746 ax = axes[1]
747 add_subpanel_label(ax, "b")
748 x = np.arange(len(q4))
749 bars = ax.bar(x, rel, color=[PALETTE[i] for i in range(len(q4))], width=0.58)
750 for bar, value in zip(bars, rel):
751     ax.text(bar.get_x() + bar.get_width() / 2,
752            value + (0.04 if value >= 0 else -0.09),
753            f"{value:+.2f}%", ha="center",
754            va="bottom" if value >= 0 else "top", fontsize=8)
755 ax.axhline(0, color=GRAY, linewidth=0.8)
756 ax.axhspan(-1, 1, color=NEUTRAL_FILL, alpha=0.45, zorder=0)
757 ax.text(len(q4) - 0.5, 0.86, " $\pm 1\%$  带", ha="right", va="top", fontsize=7.8, color=GRAY)
758 ax.set_xticks(x, [COND_SHORT[int(c)] for c in q4["condition_cluster"]], fontsize=7.4)
759 ax.set_ylabel("闭式估计相对数值解的偏差 (%)")
760 ax.set_ylim(-1.2, 1.2)
761 finish_axes(ax)
762 fig.tight_layout(w_pad=2.2)
763 save_figure(fig, OUT, "18_condition_energy_penalty")
764
765
766 def _relpath(path: str) -> str:
767     """交付物中一律使用相对路径，避免写入本机绝对路径。"""
768     try:
769         return Path(path).resolve().relative_to(ROOT).as_posix()
770     except ValueError:
771         return Path(path).name
772
773
774 TIKZ_ROWS = [{
775     "figure_id": "01", "manuscript_figure": 1, "stem": "00_esp_schematic",
776     "caption": "四电电场除尘器结构与变量定义",
777     "png": "figures_paper/00_esp_schematic.pdf", "pdf": "figures_paper/00_esp_schematic.pdf",
778     "source_data": "figures_paper/00_esp_schematic.tex",
779     "transformation": "按赛题给定的设备结构示意图以Tikz矢量重绘，xelatex独立编译",
780     "supported_manuscript_claims": "§1.1；图1",
781     "limitations": "结构示意图，不含实测数据；比例非工程真实尺寸",
782 }]
783
784
785 def fig19_frontier(frontier: pd.DataFrame, weighted: pd.DataFrame) -> None:
786     """排放—功率权衡前沿：说明放宽限值换来的功率下降不是等排放下的节能。"""

```

```

787 fig, ax = plt.subplots(figsize=(7.4, 4.0))
788 for idx, (cluster, grp) in enumerate(frontier.groupby("condition_cluster")):
789     grp = grp.sort_values("limit_mgNm3")
790     ax.plot(grp["limit_mgNm3"], grp["predicted_power_kw"], "o-",
791             color=PALETTE[idx], markersize=4.5, linewidth=1.5,
792             label=COND_SHORT[int(cluster)].replace("\n", " "))
793 w = weighted.sort_values("limit_mgNm3")
794 ax.plot(w["limit_mgNm3"], w["weighted_power_kw"], "s--", color="#111827",
795         markersize=5, linewidth=2.0, label="工况加权")
796 hist = float(frontier["historical_power_kw"].iloc[0]) if "historical_power_kw" in frontier else np.nan
797 p10 = float(w[w["limit_mgNm3"] == 10.0]["weighted_power_kw"].iloc[0])
798 p5 = float(w[w["limit_mgNm3"] == 5.0]["weighted_power_kw"].iloc[0])
799 if np.isfinite(hist):
800     ax.axhline(hist, color=GRAY, linestyle=":", linewidth=1.3)
801     ax.text(9.9, hist - 14, f"历史实际加权功率 {hist:.0f} kW (排放约50 mg/Nm³) ",
802            fontsize=8, color=GRAY, ha="right")
803 ax.annotate("", xy=(10.0, p10), xytext=(5.0, p5),
804            arrowprops={"arrowstyle": "<->", "color": PALETTE[4], "linewidth": 1.3})
805 ax.text(7.5, (p5 + p10) / 2 + 10,
806        f"限值 10→5 mg/Nm³\n加权功率增 {p5 - p10:.0f} kW ({100*(p5-p10)/p10:.1f}% )",
807        fontsize=8.2, color=PALETTE[4], ha="center")
808 ax.set_xlabel("情景排放限值 (mg/Nm³)")
809 ax.set_ylabel("支持域内最佳采样预测功率 (kW)")
810 ax.set_xticks(sorted(frontier["limit_mgNm3"].unique()))
811 ax.legend(frameon=False, ncol=2, fontsize=8.4, loc="upper right")
812 headroom(ax, 0.16)
813 finish_axes(ax, grid_axis="both")
814 fig.tight_layout()
815 save_figure(fig, OUT, "19_emission_power_frontier")
816
817
818 def fig20_p_sensitivity(p_ref: pd.DataFrame, prior: pd.DataFrame) -> None:
819     """The increase is set by one calibratable scalar: the exponent p in  $K \sim U^p$ ."""
820     ref = p_ref.sort_values("p_exponent")
821     fig, ax = plt.subplots(figsize=(6.9, 3.9))
822     ax.plot(ref["p_exponent"], ref["analytic_increase_pct"], "-", color="#111827",
823            linewidth=1.8, zorder=2, label=r"解析式  $\$P\_f[(1+\Delta K/K)^{\{2/p\}-1}/P_{\{10\}}\$]")
824     feas = ref[ref["feasible_at_5pct_headroom"] == True]
825     infeas = ref[ref["feasible_at_5pct_headroom"] != True]
826     ax.scatter(feas["p_exponent"], feas["analytic_increase_pct"], s=62, color=PALETTE[0],
827            edgecolor="white", linewidth=0.8, zorder=4, label="5%电压裕度内可达")
828     ax.scatter(infeas["p_exponent"], infeas["analytic_increase_pct"], s=62,
829            facecolor="white", edgecolor=PALETTE[4], linewidth=1.6, zorder=4,
830            label="需更大电压裕度 (不可达) ")
831     num = ref.dropna(subset=["numeric_increase_pct"])
832     ax.scatter(num["p_exponent"], num["numeric_increase_pct"], marker="x", s=52,
833            color=PALETTE[2], zorder=5, label="数值重解校验")
834     for _, row in ref.iterrows():
835         ax.annotate(f"{row['analytic_increase_pct']:.1f}%",
836                (row["p_exponent"], row["analytic_increase_pct"]),
837                textcoords="offset points", xytext=(0, 10), ha="center",
838                fontsize=8.2, color=GRAY)
839     ax.set_xlabel(r"去除指数—电压幂次  $\$p\$$  ( $\$K\propto U^p\$$ ;  $\$p=2\$$  为Deutsch迁移速度形式) ")
840     ax.set_ylabel("10→5 mg/Nm³ 加权功率增幅 (%)")
841     ax.legend(frameon=False, fontsize=8, loc="upper right")
842     headroom(ax, 0.24)
843     finish_axes(ax, grid_axis="both")
844     fig.tight_layout()
845     save_figure(fig, OUT, "20_eta_sensitivity")
846
847
848 def write_manifest() -> None:$ 
```

```

849 script_hash = hashlib.sha256(Path(__file__).read_bytes()).hexdigest()
850 model_hash = hashlib.sha256((SRC_DIR / "scenario_model.py").read_bytes()).hexdigest()
851 rows = []
852 for stem, caption in CAPTIONS.items():
853     source_data, function_name, claims, limitations = TRACE_META[stem]
854     source_data = _relpath(source_data)
855     rows.append({"figure_id": stem.split("_")[0], "manuscript_figure": MANUSCRIPT_FIGURE[stem], "stem": stem, "caption": caption,
856                "png": f"figures_paper/{stem}.png", "pdf": f"figures_paper/{stem}.pdf",
857                "source_data": source_data,
858                "transformation": f"paper_figures.py::{function_name}; plot_sha256={script_hash}; model_sha256={model_hash}",
859                "supported_manuscript_claims": claims,
860                "limitations": limitations})
861 rows = TIKZ_ROWS + rows
862 rows.sort(key=lambda r: r["manuscript_figure"])
863 pd.DataFrame(rows).to_csv(OUT / "figure_manifest.csv", index=False, encoding="utf-8-sig")
864 lines = ["# 论文图组与图注", "", "所有图均提供 PNG 预览和 PDF 矢量版本。", ""]
865 for row in sorted(rows, key=lambda x: x["manuscript_figure"]):
866     lines.extend([
867         f"## 正文图{row['manuscript_figure']} (文件编号{row['figure_id']}) {row['stem']}",
868         f"- 源数据: `{row['source_data']}`",
869         f"- 变换: `{row['transformation']}`",
870         f"- 支持的正文论断: {row['supported_manuscript_claims']}",
871         f"- 局限: {row['limitations']}",
872         f"- PNG: `{row['png']}`", f"- PDF: `{row['pdf']}`", ""])
873 (OUT / "图注清单.md").write_text("\n".join(lines), encoding="utf-8")
874
875
876 def main() -> None:
877     set_paper_style()
878     df = pd.read_csv(DATA_FILE, parse_dates=["timestamp"])
879     profiles = pd.read_csv(RESULTS / "condition_profiles.csv")
880     cluster_eval = pd.read_csv(RESULTS / "cluster_selection_metrics.csv")
881     optimum = pd.read_csv(RESULTS / "optimal_controls_central_scenario.csv")
882     audit = pd.read_csv(RESULTS / "support_boundary_audit.csv")
883     verification = pd.read_csv(RESULTS / "optimization_verification.csv")
884     structural = pd.read_csv(RESULTS / "structural_sensitivity.csv")
885     ablation = pd.read_csv(RESULTS / "field_priority_ablation_summary.csv")
886     baselines = pd.read_csv(RESULTS / "power_model_baselines.csv")
887     q4 = pd.read_csv(RESULTS / "question4_by_condition.csv")
888     frontier = pd.read_csv(RESULTS / "emission_power_frontier.csv")
889     frontier_w = pd.read_csv(RESULTS / "emission_power_frontier_weighted.csv")
890     prior = pd.read_csv(RESULTS / "question4_sensitivity.csv")
891     p_ref = pd.read_csv(RESULTS / "p_exponent_reference.csv")
892     events = pd.read_csv(RESULTS / "anomaly_events.csv")
893     strategy = pd.read_csv(RESULTS / "historical_strategy_correlations.csv")
894     model = fit_power_model(df)
895     linear_model = fit_linear_period_model(df)
896
897     fig01_outlet_diagnosis(df)
898     fig02_timeseries(df)
899     fig22_roadmap()
900     fig04b_operating_evidence(df, events, strategy)
901     fig14_peak(profiles)
902     fig03_cluster_selection(cluster_eval)
903     fig04_condition_map(df)
904     fig07_power_prediction(df, model, baselines)
905     fig08b_specification(df, model, linear_model, baselines)
906     fig08_residuals(df, model)
907     fig05b_boundary_proximity(audit)
908     fig19_frontier(frontier, frontier_w)
909     grouped_controls(optimum, U_COLS, "11_optimal_voltage", "最优电压 (kV)")
910     fig18_penalty(q4)

```

```

911 fig13_emission(optimum)
912 fig15_optimizer_verification(verification)
913 fig16_structural_sensitivity(structural)
914 fig20_p_sensitivity(p_ref, prior)
915 fig17_field_ablation(ablation)
916 write_manifest()
917 print(f"Generated {len(CAPTIONS)} figures in {OUT}")
918
919
920 if __name__ == "__main__":
921     main()

```

E.3 plot_utils.py

```

1  """Shared publication plotting style for the cement ESP paper."""
2
3  from __future__ import annotations
4
5  from pathlib import Path
6
7  import matplotlib.pyplot as plt
8  from cycler import cycler
9
10
11 # Okabe-Ito 色序: 色觉障碍友好, 灰度打印仍可区分
12 PALETTE = ["#0072B2", "#E69F00", "#009E73", "#CC79A7", "#D55E00", "#56B4E9"]
13 GRAY = "#5B6472"
14 LIGHT_GRAY = "#D1D5DB"
15 NEUTRAL_FILL = "#DFE3E8"
16
17
18 def set_paper_style() -> None:
19     plt.rcParams.update(
20         {
21             "font.family": "sans-serif",
22             # Linux 优先 Noto, macOS 回落到系统中文字体
23             "font.sans-serif": [
24                 "Noto Sans CJK SC",
25                 "Source Han Sans SC",
26                 "PingFang SC",
27                 "Arial Unicode MS",
28                 "WenQuanYi Zen Hei",
29                 "DejaVu Sans",
30             ],
31             "axes.unicode_minus": False,
32             "font.size": 10.0,
33             "axes.labelsize": 10.0,
34             "axes.titlesize": 10.0,
35             "xtick.labelsize": 9.0,
36             "ytick.labelsize": 9.0,
37             "legend.fontsize": 9.0,
38             "legend.frameon": False,
39             "legend.handlelength": 1.8,
40             "legend.columnspacing": 1.4,
41             "legend.borderaxespad": 0.4,
42             "axes.linewidth": 0.9,
43             "axes.edgecolor": "#2F3640",
44             "axes.labelcolor": "#1F2933",
45             "axes.prop_cycle": cycler(color=PALETTE),
46             "lines.linewidth": 1.7,
47             "lines.markersize": 5.0,

```

```

48         "xtick.direction": "out",
49         "ytick.direction": "out",
50         "xtick.major.width": 0.9,
51         "ytick.major.width": 0.9,
52         "xtick.color": "#2F3640",
53         "ytick.color": "#2F3640",
54         "grid.color": "#E5E7EB",
55         "grid.linewidth": 0.6,
56         "grid.alpha": 0.9,
57         "figure.dpi": 160,
58         "savefig.dpi": 400,
59         "savefig.bbox": "tight",
60         "savefig.pad_inches": 0.03,
61         "pdf.fonttype": 42,
62         "ps.fonttype": 42,
63     }
64 )
65
66
67 def finish_axes(ax, grid_axis: str = "y") -> None:
68     if grid_axis != "none":
69         ax.grid(True, axis=grid_axis)
70     ax.set_axisbelow(True)
71     ax.spines["top"].set_visible(False)
72     ax.spines["right"].set_visible(False)
73
74
75 def add_subpanel_label(ax, label: str, x: float = -0.12, y: float = 1.05) -> None:
76     """Add a compact (a), (b), ... identifier to a multi-panel figure."""
77     ax.text(
78         x,
79         y,
80         f"({label})",
81         transform=ax.transAxes,
82         fontsize=10.5,
83         fontweight="bold",
84         va="bottom",
85         ha="right",
86     )
87
88
89 def headroom(ax, frac: float = 0.18) -> None:
90     """在数据上方留出空白，避免图例或注释压住柱顶与折线。"""
91     lo, hi = ax.get_ylim()
92     ax.set_ylim(lo, lo + (hi - lo) * (1.0 + frac))
93
94
95 def figure_legend(fig, handles, labels, ncol: int, bottom: float = -0.02) -> None:
96     """把多面板共用图例统一放到画布下方，杜绝与数据重叠。"""
97     fig.legend(
98         handles,
99         labels,
100         loc="lower center",
101         ncol=ncol,
102         bbox_to_anchor=(0.5, bottom),
103         frameon=False,
104     )
105
106
107 def note(ax, text: str, loc: str = "upper left", color: str | None = None) -> None:
108     """角落说明文字，带白底以免压住网格与数据。"""
109     anchor = {

```

```

110     "upper left": (0.02, 0.98, "left", "top"),
111     "upper right": (0.98, 0.98, "right", "top"),
112     "lower left": (0.02, 0.03, "left", "bottom"),
113     "lower right": (0.98, 0.03, "right", "bottom"),
114 }[loc]
115 ax.text(
116     anchor[0],
117     anchor[1],
118     text,
119     transform=ax.transAxes,
120     ha=anchor[2],
121     va=anchor[3],
122     fontsize=8.0,
123     color=color or GRAY,
124     bbox={
125         "boxstyle": "round,pad=0.28",
126         "facecolor": "white",
127         "edgecolor": LIGHT_GRAY,
128         "linewidth": 0.6,
129         "alpha": 0.92,
130     },
131 )
132
133
134 def save_figure(fig, out_dir: Path, stem: str) -> None:
135     out_dir.mkdir(parents=True, exist_ok=True)
136     fig.savefig(out_dir / f"{stem}.png")
137     fig.savefig(out_dir / f"{stem}.pdf")
138     plt.close(fig)

```

E.4 compute_frontier.py

```

1  #!/usr/bin/env python3
2  """Compute the scenario emission-power trade-off frontier.
3
4  The support-domain search is repeated at a range of emission limits, so the
5  paper can show the whole trade-off curve rather than only the 10 and
6  5 mg/Nm^3 end points. Because the historical anchor is ~50 mg/Nm^3, every
7  point on this curve costs power relative to historical operation; the curve
8  also records how much voltage headroom above the historical maximum each
9  limit needs before it becomes reachable at all.
10
11  Usage:  python3 src/compute_frontier.py
12  """
13
14  from __future__ import annotations
15
16  from pathlib import Path
17
18  import numpy as np
19  import pandas as pd
20
21  from scenario_model import (
22      DATA_FILE,
23      HEADROOM_5,
24      T_COLS,
25      condition_profiles,
26      evaluate_power_models,
27      scenario_t_star,
28      solve_condition,
29  )

```

```

30
31 ROOT = Path(__file__).resolve().parents[1]
32 OUT_DIR = ROOT / "results"
33 LIMITS = [5.0, 6.0, 7.0, 8.0, 9.0, 10.0]
34
35
36 def main() -> None:
37     df = pd.read_csv(DATA_FILE, parse_dates=["timestamp"])
38     df["condition_cluster"] = df["condition_cluster"].astype(int)
39     profiles = condition_profiles(df)
40
41     train = df[df["split"] == "train"].copy()
42     validation = df[df["split"] == "validation"].copy()
43     test = df[df["split"] == "test"].copy()
44     model, metrics, _, _ = evaluate_power_models(train, validation, test)
45     guard = float(metrics["validation_abs_error_q90_kw"])
46
47     global_t_median = train[T_COLS].median().to_numpy(float)
48     t_stars = {int(p["condition_cluster"]): scenario_t_star(p, global_t_median)
49                 for _, p in profiles.iterrows()}
50     historical_weighted = float(np.average(profiles["historical_power_kw"],
51                                           weights=profiles["share"])))
52
53     rows: list[dict[str, object]] = []
54     weighted_rows: list[dict[str, object]] = []
55     for limit in LIMITS:
56         # smallest declared headroom that makes every condition reachable
57         headroom = 0.0
58         while headroom <= HEADROOM_5 + 1e-9:
59             trial = [solve_condition(p, model, t_stars[int(p["condition_cluster"])]), limit,
60                     voltage_headroom=headroom, multistart=12)
61             for _, p in profiles.iterrows():
62                 if all(r["status"] == "SCENARIO_FEASIBLE" for r in trial):
63                     break
64             headroom = round(headroom + 0.01, 2)
65
66         solved = [solve_condition(p, model, t_stars[int(p["condition_cluster"])]), limit,
67                 voltage_headroom=headroom)
68         for _, p in profiles.iterrows():
69             for profile_row, r in zip(profiles.itertuples(), solved):
70                 rows.append({
71                     "condition_cluster": r["condition_cluster"],
72                     "condition_name": r["condition_name"],
73                     "limit_mgNm3": limit,
74                     "voltage_headroom_pct": 100.0 * headroom,
75                     "status": r["status"],
76                     "predicted_power_kw": r.get("predicted_power_kw", np.nan),
77                     "validation_error_adjusted_power_kw": r.get("predicted_power_kw", np.nan) + guard,
78                     "share": float(profile_row.share),
79                     "historical_power_kw": historical_weighted,
80                 })
81         weights = np.array([float(p.share) for p in profiles.itertuples()])
82         powers = np.array([float(r.get("predicted_power_kw", np.nan)) for r in solved])
83         weighted_rows.append({
84             "limit_mgNm3": limit,
85             "voltage_headroom_pct": 100.0 * headroom,
86             "weighted_power_kw": float(np.average(powers, weights=weights)),
87             "weighted_daily_energy_MWh": float(np.average(powers, weights=weights)) * 24.0 / 1000.0,
88             "historical_power_kw": historical_weighted,
89         })
90
91     pd.DataFrame(rows).to_csv(OUT_DIR / "emission_power_frontier.csv",

```

```

92         index=False, encoding="utf-8-sig")
93     pd.DataFrame(weighted_rows).to_csv(OUT_DIR / "emission_power_frontier_weighted.csv",
94         index=False, encoding="utf-8-sig")
95     print(pd.DataFrame(weighted_rows).to_string(index=False))
96
97
98 if __name__ == "__main__":
99     main()

```

E.5 integrity_check.py

```

1  #!/usr/bin/env python3
2  """Deterministic integrity checks for manuscript numbers and artifacts."""
3
4  from __future__ import annotations
5
6  import json
7  import re
8  import sys
9  from pathlib import Path
10
11 import numpy as np
12 import pandas as pd
13
14 SRC = Path(__file__).resolve().parent
15 if str(SRC) not in sys.path:
16     sys.path.insert(0, str(SRC))
17
18 from scenario_model import PhysicalPowerModel, _r2, _rmse
19
20
21 ROOT = SRC.parent
22 DATA = ROOT / "data" / "full_timeseries_with_flags.csv"
23 RESULTS = ROOT / "results"
24 FIGURES = ROOT / "figures_paper"
25 MANUSCRIPT = ROOT / "10_修订后完整论文_终稿.md"
26 OUT = ROOT / "integrity"
27
28
29 def close(a: float, b: float, tol: float = 1e-6) -> bool:
30     return bool(abs(a - b) <= tol)
31
32
33 def main() -> None:
34     checks: List[dict[str, object]] = []
35
36     def check(name: str, passed: bool, actual: object, expected: object) -> None:
37         checks.append({"check": name, "passed": bool(passed), "actual": actual, "expected": expected})
38
39     df = pd.read_csv(DATA)
40     check("raw_row_count", len(df) == 10080, len(df), 10080)
41     check("outlet_missing_count", int(df["C_out_missing_flag"].sum()) == 50, int(df["C_out_missing_flag"].sum()), 50)
42     valid = df["C_out_mgNm3"].dropna()
43     check("outlet_valid_count", len(valid) == 10030, len(valid), 10030)
44     check("outlet_min", close(float(valid.min()), 48.74), float(valid.min()), 48.74)
45     check("outlet_max", close(float(valid.max()), 50.0), float(valid.max()), 50.0)
46     check("outlet_cap50_count", int((valid == 50).sum()) == 5491, int((valid == 50).sum()), 5491)
47     check("outlet_le10_count", int((valid <= 10).sum()) == 0, int((valid <= 10).sum()), 0)
48     check("outlet_le5_count", int((valid <= 5).sum()) == 0, int((valid <= 5).sum()), 0)
49
50     metrics = json.loads((RESULTS / "power_model_metrics.json").read_text())

```

```

51 train, val, test = (df[df["split"] == s] for s in ("train", "validation", "test"))
52 model = PhysicalPowerModel().fit(train, train["P_total_kW"].to_numpy(float))
53 vp, tp = model.predict(val), model.predict(test)
54 recomputed = {
55     "validation_r2": _r2(val["P_total_kW"].to_numpy(float), vp),
56     "validation_rmse_kW": _rmse(val["P_total_kW"].to_numpy(float), vp),
57     "retrospective_test_r2": _r2(test["P_total_kW"].to_numpy(float), tp),
58     "retrospective_test_rmse_kW": _rmse(test["P_total_kW"].to_numpy(float), tp),
59     "retrospective_test_bias_kW": float(np.mean(tp - test["P_total_kW"].to_numpy(float))),
60 }
61 for key, value in recomputed.items():
62     check(f"power_metric_{key}", close(value, float(metrics[key]), 1e-9), value, metrics[key])
63 guard = float(np.quantile(np.abs(vp-val["P_total_kW"].to_numpy(float)), 0.90))
64 check("validation_error_guard_q90", close(guard, float(metrics["validation_abs_error_q90_kW"]), 1e-9), guard, metrics["
↔ validation_abs_error_q90_kW"])
65 check(
66     "reporting_guard_rule_named_consistently",
67     metrics.get("reporting_guard_rule") == "point_prediction_plus_validation_absolute_error_q90"
68     and "conservative_reporting_rule" not in metrics,
69     {k: metrics[k] for k in metrics if "reporting" in k or "guard" in k},
70     {"reporting_guard_rule": "point_prediction_plus_validation_absolute_error_q90"},
71 )
72 check(
73     "power_model_is_physical_specification",
74     metrics["model"] == "physical_voltage_square_plus_rapping_frequency",
75     metrics["model"], "physical_voltage_square_plus_rapping_frequency",
76 )
77 # the four field coefficients must be near-identical: the fields share a design
78 b = np.asarray(metrics["b_field_kW_per_kV2"], float)
79 check("field_coefficients_consistent", float(b.std() / b.mean()) < 0.01,
80     float(b.std() / b.mean()), "<1% relative spread")
81
82 baselines = pd.read_csv(RESULTS / "power_model_baselines.csv")
83 tst_base = baselines[baselines["split"] == "retrospective_test"].set_index("model")
84 check(
85     "invT_beats_linear_T_on_test",
86     float(tst_base.loc["physical_invT", "rmse_kW"]) < float(tst_base.loc["quadratic_ridge_T", "rmse_kW"]),
87     tst_base["rmse_kW"].to_dict(), "physical_invT < quadratic_ridge_T",
88 )
89 check(
90     "invT_removes_test_bias",
91     abs(float(tst_base.loc["physical_invT", "bias_kW"])) < 1.0
92     and abs(float(tst_base.loc["quadratic_ridge_T", "bias_kW"])) > 5.0,
93     {"physical_invT": float(tst_base.loc["physical_invT", "bias_kW"]),
94      "quadratic_ridge_T": float(tst_base.loc["quadratic_ridge_T", "bias_kW"])},
95     "|bias| < 1 kW with 1/T, > 5 kW with linear T",
96 )
97 check(
98     "condition_variables_add_nothing",
99     abs(float(tst_base.loc["physical_invT_plus_conditions", "r2"])
100         - float(tst_base.loc["physical_invT", "r2"])) < 1e-3,
101     {"with": float(tst_base.loc["physical_invT_plus_conditions", "r2"]),
102      "without": float(tst_base.loc["physical_invT", "r2"])},
103     "delta R2 < 1e-3",
104 )
105 rolling = pd.read_csv(RESULTS / "power_model_rolling_validation.csv")
106 check("rolling_fold_count", rolling["fold"].nunique() == 3, int(rolling["fold"].nunique()), 3)
107
108 cluster = pd.read_csv(RESULTS / "cluster_selection_metrics.csv")
109 selected = cluster[cluster["selected"]].iloc[0]
110 check("selected_cluster_k", int(selected["k"]) == 4, int(selected["k"]), 4)
111 check("silhouette_k4", close(float(selected["silhouette"]), 0.4083689, 1e-6), float(selected["silhouette"]), 0.4083689)

```

```

112
113 optimum = pd.read_csv(RESULTS / "optimal_controls_central_scenario.csv")
114 check(
115     "central_parameter_table_complete",
116     len(optimum) == 8
117     and set(map(tuple, optimum[["condition_cluster", "limit_mgNm3"]].to_numpy()))
118     == {(cluster, limit) for cluster in range(1, 5) for limit in (10.0, 5.0)},
119     optimum[["condition_cluster", "limit_mgNm3"]].values.tolist(),
120     "4 conditions × 2 targets",
121 )
122 check(
123     "validation_error_columns_named_consistently",
124     {"validation_error_adjusted_power_kw", "validation_error_guard_kw"}.issubset(optimum.columns)
125     and {"conservative_power_kw", "power_guard_kw"}.isdisjoint(optimum.columns),
126     sorted(optimum.columns),
127     "new validation-error names present; old conservative names absent",
128 )
129 check("central_all_feasible", bool((optimum["status"] == "SCENARIO_FEASIBLE").all()), optimum["status"].unique().tolist(), ["
↳ SCENARIO_FEASIBLE"])
130 check("central_limits_respected", bool((optimum["scenario_peak_total_mgNm3"] <= optimum["limit_mgNm3"] + 1e-9).all()), float((optimum["
↳ scenario_peak_total_mgNm3"] - optimum["limit_mgNm3"]).max()), "<=0")
131 check(
132     "emission_anchor_not_rescaled",
133     bool((pd.read_csv(RESULTS / "condition_profiles.csv")["C_out_recorded_median"] == 50.0).all()),
134     pd.read_csv(RESULTS / "condition_profiles.csv")["C_out_recorded_median"].tolist(),
135     "all 50.0 mg/Nm3 (recorded, unscaled)",
136 )
137 scenario = json.loads((RESULTS / "scenario_parameters.json").read_text())
138 check("no_outlet_rescaling_parameter", "central_outlet_scale" not in scenario,
139     sorted(scenario), "no central_outlet_scale key")
140
141 audit = pd.read_csv(RESULTS / "support_boundary_audit.csv")
142 check("support_audit_rows", len(audit) == 8, len(audit), 8)
143 check(
144     "targets_leave_historical_experience",
145     bool((audit["ratio_to_q975"] > 1.0).all()),
146     float(audit["ratio_to_q975"].min()), ">1 for all eight solutions",
147 )
148 check(
149     "10mg_inside_voltage_envelope",
150     bool((audit[audit["limit_mgNm3"] == 10.0]["max_voltage_over_historical_max_pct"] <= 1e-6).all()),
151     float(audit[audit["limit_mgNm3"] == 10.0]["max_voltage_over_historical_max_pct"].max()),
152     "<=0% over the historical maximum",
153 )
154 restricted = pd.read_csv(RESULTS / "mahalanobis_constrained_feasibility.csv")
155 merged = restricted.merge(
156     optimum[["condition_cluster", "limit_mgNm3", "predicted_power_kw"]],
157     on=["condition_cluster", "limit_mgNm3"], suffixes=("_shell", "_free"))
158 feasible_shell = merged["status_with_mahalanobis_constraint"] == "SCENARIO_FEASIBLE"
159 check(
160     "mahalanobis_shell_variant_mostly_feasible",
161     int(feasible_shell.sum()) == 7,
162     int(feasible_shell.sum()), "7 of 8 combinations feasible inside the shell",
163 )
164 penalty = 100.0 * (merged.loc[feasible_shell, "predicted_power_kw_shell"]
165     / merged.loc[feasible_shell, "predicted_power_kw_free"] - 1.0)
166 check(
167     "shell_variant_costs_power",
168     bool((penalty > 0).all()),
169     [round(float(penalty.min()), 2), round(float(penalty.max()), 2)],
170     "staying inside the shell is never cheaper than the free optimum",
171 )

```

```

172 headroom = pd.read_csv(RESULTS / "voltage_headroom_requirement.csv")
173 check(
174     "headroom_10mg_is_zero",
175     bool((headroom[headroom["Limit_mgNm3"] == 10.0]["min_voltage_headroom_pct"] == 0.0).all()),
176     headroom[headroom["Limit_mgNm3"] == 10.0]["min_voltage_headroom_pct"].tolist(), "all 0%",
177 )
178 check(
179     "headroom_5mg_within_declared_margin",
180     float(headroom[headroom["Limit_mgNm3"] == 5.0]["min_voltage_headroom_pct"].max()) <= 5.0,
181     float(headroom[headroom["Limit_mgNm3"] == 5.0]["min_voltage_headroom_pct"].max()), "<=5%",
182 )
183
184 q4 = pd.read_csv(RESULTS / "question4_by_condition.csv")
185 p10 = float(np.sum(q4["share"] * q4["power_10_kw"]) / q4["share"].sum())
186 p5 = float(np.sum(q4["share"] * q4["power_5_kw"]) / q4["share"].sum())
187 increase = 100 * (p5 / p10 - 1)
188 summary = json.loads((RESULTS / "question4_summary.json").read_text())
189 check("weighted_power_10", close(p10, summary["weighted_power_10_kw"], 1e-9), p10, summary["weighted_power_10_kw"])
190 check("weighted_power_5", close(p5, summary["weighted_power_5_kw"], 1e-9), p5, summary["weighted_power_5_kw"])
191 check("weighted_increase", close(increase, summary["weighted_increase_pct"], 1e-9), increase, summary["weighted_increase_pct"])
192 check(
193     "tightening_costs_power_vs_history",
194     summary["power_10_vs_historical_pct"] > 0 and summary["weighted_increase_pct"] > 0,
195     {"10mg_vs_history_pct": summary["power_10_vs_historical_pct"],
196     "increase_pct": summary["weighted_increase_pct"]},
197     "both positive: tightening the limit costs power",
198 )
199 rel = np.abs(q4["analytic_delta_power_kw"] - q4["numeric_delta_power_kw"]) / q4["numeric_delta_power_kw"]
200 check(
201     "analytic_matches_numeric_delta_power",
202     float(rel.max()) < 0.02, float(rel.max()), "<2% relative difference",
203 )
204
205 verification = pd.read_csv(RESULTS / "optimization_verification.csv")
206 check("verification_rows", len(verification) == 8, len(verification), 8)
207 check(
208     "multistart_agrees_on_optimum",
209     float(verification["multistart_power_spread_kw"].max()) < 1.0,
210     float(verification["multistart_power_spread_kw"].max()), "<1 kW across converged starts",
211 )
212 gaps = verification["random_search_gap_pct"].dropna()
213 check(
214     "random_search_finds_nothing_better",
215     bool((gaps >= -1e-9).all()), float(gaps.min()), ">=0 for every combination",
216 )
217
218 sens = pd.read_csv(RESULTS / "question4_sensitivity.csv")
219 check("sensitivity_grid_size", len(sens) == 27, len(sens), 27)
220 p_ref = pd.read_csv(RESULTS / "p_exponent_reference.csv")
221 check("p_reference_rows", len(p_ref) == 5, len(p_ref), 5)
222 check(
223     "scale_sensitivity_feasible_count",
224     int(sens["all_conditions_feasible"].sum()) == 15,
225     int(sens["all_conditions_feasible"].sum()), 15,
226 )
227 numeric = p_ref.dropna(subset=["numeric_increase_pct"])
228 gap = (numeric["analytic_increase_pct"] - numeric["numeric_increase_pct"]).abs().max()
229 check(
230     "analytic_vs_numeric_p_gap",
231     float(gap) < 0.5, round(float(gap), 3), "<0.5 percentage points",
232 )
233 check(

```

```

234     "p_reference_monotone_decreasing",
235     bool((p_ref.sort_values("p_exponent")["analytic_increase_pct"].diff().dropna() < 0).all()),
236     p_ref.sort_values("p_exponent")["analytic_increase_pct"].round(3).tolist(),
237     "increase falls as p rises (proportional to 2/p)",
238 )
239
240 structural = pd.read_csv(RESULTS / "structural_sensitivity.csv")
241 structural_vals = structural.loc[structural["all_conditions_feasible"] == True, "weighted_power_increase_pct"]
242 check("structural_total_count", len(structural) == 81, len(structural), 81)
243 check("structural_all_feasible", len(structural_vals) == 81, len(structural_vals), 81)
244
245 rapping = pd.read_csv(RESULTS / "rapping_power_tradeoff.csv")
246 check(
247     "rapping_share_reported",
248     30.0 < float(rapping.iloc[0]["rapping_share_pct"]) < 45.0,
249     float(rapping.iloc[0]["rapping_share_pct"]), "between 30% and 45% at median settings",
250 )
251 strategy = pd.read_csv(RESULTS / "historical_strategy_correlations.csv")
252 check(
253     "strategy_windows_reported",
254     set(strategy["window"]) == {"full_record", "train", "validation", "retrospective_test"},
255     sorted(set(strategy["window"])),
256     "four evaluation windows",
257 )
258 period_rule = strategy[strategy["control"].str.startswith("T")]
259 check(
260     "rapping_period_rule_stable_across_windows",
261     bool((period_rule["r_C_in_gNm3"] < 0).all()),
262     float(period_rule["r_C_in_gNm3"].max()), "<0 in every window and field",
263 )
264 events = pd.read_csv(RESULTS / "anomaly_events.csv")
265 check("anomaly_events_documented", len(events) == 2, len(events), 2)
266
267 manuscript = MANUSCRIPT.read_text(encoding="utf-8")
268 # A final submission may print this verifier itself in Appendix D. Restrict
269 # manuscript-content checks to the paper and non-code appendices so string
270 # literals inside the printed source do not become false positives.
271 review_manuscript = manuscript.split("## 附录D 完整源程序", 1)[0]
272 ref_body = review_manuscript.split("## 参考文献", 1)[0]
273 cited = set()
274 for token in re.findall(r"\\([0-9,-]+)\\", ref_body):
275     for part in token.split(","):
276         if "-" in part:
277             a, b = map(int, part.split("-"))
278             cited.update(range(a, b + 1))
279         else:
280             cited.add(int(part))
281 listed = set(map(int, re.findall(r"^\\([0-9]+)\\ ", review_manuscript.split("## 参考文献", 1)[1], flags=re.M)))
282 check("all_references_cited", cited == set(range(1, 14)), sorted(cited), list(range(1, 14)))
283 check("reference_list_complete", listed == set(range(1, 14)), sorted(listed), list(range(1, 14)))
284 clean_manuscript = re.sub(r"<!--.*?-->\\n?", "", review_manuscript)
285 table6_segment = clean_manuscript.split("表11\\u3000四工况两档目标", 1)[1].split("表12", 1)[0]
286 table6_rows: dict[tuple[int, float], list[float]] = {}
287 for line in table6_segment.splitlines():
288     if not re.match(r"^\\[ 1-4] \\| (? :10|5) \\|", line):
289         continue
290     cells = [cell.strip() for cell in line.strip().strip("|").split("|")]
291     table6_rows[(int(cells[0]), float(cells[1]))] = [float(x) for x in cells[2:]]
292 table6_matches = len(table6_rows) == 8
293 for key, displayed in table6_rows.items():
294     expected_row = optimum[
295         (optimum["condition_cluster"] == key[0]) & (optimum["limit_mgNm3"] == key[1])

```

```

296     ]
297     if len(expected_row) != 1:
298         table6_matches = False
299         continue
300     row = expected_row.iloc[0]
301     expected = [
302         *[round(float(row[f"U{i}_kW"]), 1) for i in range(1, 5)],
303         *[round(float(row[f"T{i}_s"]), 0) for i in range(1, 5)],
304         round(float(row["scenario_peak_total_mgNm3"]), 3),
305         round(float(row["predicted_power_kW"]), 2),
306     ]
307     table6_matches &= bool(np.allclose(displayed, expected, atol=5e-3))
308     check(
309         "table6_complete_and_matches_results",
310         table6_matches,
311         {f"condition_{key[0]}_target_{key[1]:g}": value for key, value in table6_rows.items()},
312         "8 rows matching optimal_controls_central_scenario.csv after displayed rounding",
313     )
314     p_table_segment = clean_manuscript.split("表15\u3000功率增幅对幂次", 1)[1].split("\n\n", 3)[1]
315     printed = {}
316     for line in p_table_segment.splitlines():
317         cells = [c.strip().strip("***) for c in line.strip().strip("|").split("|")]
318         if len(cells) == 4 and re.fullmatch(r"[\d.]+", cells[0]):
319             printed[float(cells[0])] = cells[2]
320     expected = {float(r["p_exponent"]): f"{r['analytic_increase_pct']:.1f}%"
321                 for _, r in p_ref.iterrows()}
322     check(
323         "p_table_matches_source_at_ldp",
324         printed == expected, printed, expected,
325     )
326
327     abstract = clean_manuscript.split("## 摘要", 1)[1].split("***关键词", 1)[0]
328     abstract_chars = len(re.sub(r"\s+", "", abstract))
329     check("abstract_fills_one_page", 700 <= abstract_chars <= 1300, abstract_chars, "700-1300 chars (one page)")
330     check("anonymous_manuscript", "作者: " not in clean_manuscript and "学校: " not in clean_manuscript, ["作者: " in clean_manuscript, "
331     ⇨ 学校: " in clean_manuscript], [False, False])
332     check("no_english_abstract", "## Abstract" not in clean_manuscript, "## Abstract" in clean_manuscript, False)
333     check("outlet_rescaling_claim_removed", "乘0.10" not in clean_manuscript and "0.1缩放" not in clean_manuscript, ["乘0.10" in
334     ⇨ clean_manuscript, "0.1缩放" in clean_manuscript], [False, False])
335     check("table_caption_count", len(re.findall(r"表\d+", clean_manuscript, flags=re.M)) == 15, len(re.findall(r"表\d+", clean_manuscript,
336     ⇨ flags=re.M)), 15)
337     check("manuscript_figure_count", len(re.findall(r"^[图", clean_manuscript, flags=re.M)) == 20, len(re.findall(r"^[图",
338     ⇨ clean_manuscript, flags=re.M)), 20)
339     check("ai_disclosure_present", "AI工具使用声明" in clean_manuscript, "AI工具使用声明" in clean_manuscript, True)
340     check("support_list_exists", (ROOT / "支撑材料文件清单.md").exists(), (ROOT / "支撑材料文件清单.md").exists(), True)
341     ai_detail = next(ROOT.parent.glob("*/AI工具使用详情*.md"), None)
342     check("ai_detail_exists", ai_detail is not None, str(ai_detail), "AI工具使用详情*.md present in the process directory")
343
344     manifest = pd.read_csv(FIGURES / "figure_manifest.csv")
345     check("figure_manifest_count", len(manifest) == 20, len(manifest), 20)
346     required_trace = {"source_data", "transformation", "caption", "supported_manuscript_claims", "limitations"}
347     check("figure_trace_fields", required_trace.issubset(manifest.columns), sorted(manifest.columns), sorted(required_trace))
348     missing_artifacts = []
349     for _, row in manifest.iterrows():
350         for col in ("png", "pdf", "source_data"):
351             artifact = Path(row[col])
352             if not artifact.is_absolute():
353                 artifact = ROOT / artifact
354             if not artifact.exists():
355                 missing_artifacts.append(f"{row['stem']}:{col}")
356     check("figure_artifacts_and_sources_exist", not missing_artifacts, missing_artifacts, [])
357

```

```

354     OUT.mkdir(parents=True, exist_ok=True)
355     failed = [x for x in checks if not x["passed"]]
356     report = {"status": "PASS" if not failed else "FAIL", "total_checks": len(checks), "passed": len(checks)-len(failed), "failed": len(
        ↳ failed), "checks": checks}
357     (OUT / "deterministic_integrity_results.json").write_text(json.dumps(report, ensure_ascii=False, indent=2), encoding="utf-8")
358     print(json.dumps({k: report[k] for k in ("status", "total_checks", "passed", "failed")}, ensure_ascii=False, indent=2))
359     if failed:
360         for item in failed:
361             print("FAILED", item)
362             raise SystemExit(1)
363
364
365 if __name__ == "__main__":
366     main()

```

E.6 build_paper_pdf.py

```

1  #!/usr/bin/env python3
2  """Build the CUMCM submission PDF from the final Markdown manuscript.
3
4  Converts ``10_修订后完整论文_终稿.md`` into XeLaTeX and compiles it.
5  Layout follows the CUMCM paper format specification: A4, >=2.5 cm margins,
6  abstract alone on page 1, continuous arabic page numbers centred in the footer.
7
8  Usage: python3 src/build_paper_pdf.py [--no-code | --core-code]
9  """
10
11 from __future__ import annotations
12
13 import re
14 import subprocess
15 import sys
16 from pathlib import Path
17
18 ROOT = Path(__file__).resolve().parent.parent
19 MANUSCRIPT = ROOT / "10_修订后完整论文_终稿.md"
20 BUILD = ROOT / "build"
21 SRC = ROOT / "src"
22 CODE_MODULES = ["scenario_model.py", "paper_figures.py", "plot_utils.py",
23                "compute_frontier.py", "integrity_check.py", "build_paper_pdf.py"]
24
25 # ----- inline text
26
27 _SPECIALS = {
28     "\\": r"\textbackslash{}", "&": r"&", "%": r"%", "$": r"$",
29     "#": r"#", "_": r"_", "{": r"{", "}" : r"}",
30     "~": r"\textasciitilde{}", "^": r"\textasciicircum{}",
31     # symbols the CJK/serif fonts do not carry -> render them as maths
32     "\u2103": r"$^{\circ}$C", "\u221d": r"$\propto$", "\u2260": r"$\neq$",
33     "\u2264": r"$\le$", "\u2265": r"$\ge$", "\u2248": r"$\approx$",
34     "\u207b": r"$^-$", "\u00b9": r"$^1$", "\u2070": r"$^0$",
35     "\u2192": r"$\rightarrow$", "\u2191": r"$\uparrow$", "\u2193": r"$\downarrow$",
36 }
37
38
39 def esc(text: str) -> str:
40     return "".join(_SPECIALS.get(ch, ch) for ch in text)
41
42
43 #: bare links and DOIs are single unbreakable words for TeX; \url can break them
44 _URL_RE = re.compile(r"(https?://\S+)?(?=[\u3002\uFF0C]|$)|(?<DOI: )(\d\.\S+)?(?=[\u3002]|$)")

```

```

45
46
47 def _urlify(piece: str) -> str:
48     r"""Escape text, but wrap any bare URL or DOI in a breakable url command."""
49     out: list[str] = []
50     last = 0
51     for m in _URL_RE.finditer(piece):
52         out.append(esc(piece[last:m.start()]))
53         link = m.group(0).rstrip(".")
54         trailing = m.group(0)[len(link):]
55         out.append(r"\url{" + link + "}" + esc(trailing))
56         last = m.end()
57     out.append(esc(piece[last:]))
58     return "".join(out)
59
60
61 #: Markdown bold may wrap a maths span ("**若把$x$作为约束**"), so the markers
62 #: cannot be resolved inside a single plain-text piece. They are replaced by a
63 #: sentinel first and paired up after every segment has been rendered.
64 _BOLD = "\x00"
65
66
67 def _smart_quotes(piece: str, state: list[bool]) -> str:
68     """Straight double quotes in Chinese prose -> paired curly quotes.
69
70     ``state`` carries the open/close flag across the whole line, so a quoted
71     phrase that wraps a maths span still gets a closing quote at its end.
72     """
73     out = []
74     for ch in piece:
75         if ch == '"':
76             out.append("\u201c" if state[0] else "\u201d")
77             state[0] = not state[0]
78         else:
79             out.append(ch)
80     return "".join(out)
81
82
83 def inline(text: str) -> str:
84     """Convert inline Markdown to LaTeX, leaving  $...$  math untouched."""
85     out: list[str] = []
86     quote_open = [True]
87     for i, seg in enumerate(re.split(r"(\^[^$]*\$)", text)):
88         if i % 2:
89             # math segment - pass through verbatim
90             out.append(seg)
91             continue
92         # code spans next: their content must be escaped but not styled
93         for j, piece in enumerate(re.split(r"('[^']*')", seg)):
94             if j % 2:
95                 out.append(r"\texttt{" + esc(piece[1:-1]) + "}")
96             else:
97                 piece = piece.replace("***", _BOLD)
98                 out.append(_urlify(_smart_quotes(piece, quote_open)))
99     rendered = "".join(out)
100
101     # pair the sentinels; an unmatched trailing marker is dropped
102     parts = rendered.split(_BOLD)
103     if len(parts) == 1:
104         return rendered
105     result = parts[0]
106     for k, chunk in enumerate(parts[1:], start=1):
107         result += (r"\textbf{" if k % 2 else "}") + chunk

```

```

107     if len(parts) % 2 == 0:           # odd number of markers
108         result += "}"
109     return result
110
111
112 # ----- block level
113
114 def _display_width(text: str) -> int:
115     """Rough typeset width in en-units (CJK glyphs are double width)."""
116     text = re.sub(r"\${^$}*\\$", "xx", text)      # math sets compactly
117     text = re.sub(r"[*]", "", text)
118     return sum(2 if ord(ch) > 0x2E80 else 1 for ch in text)
119
120
121 # A row of ``l`` columns cannot wrap, so any table whose natural width exceeds
122 # the text block silently runs off the page. Text-heavy columns therefore
123 # become wrapping ``X`` columns and the table is set to exactly \textwidth.
124 _LINE_BUDGET = 84          # en-units that fit across the 16 cm text block
125 _WRAP_THRESHOLD = 16       # a column wider than this must be allowed to wrap
126
127
128 def convert_table(header: list[str], align: list[str], rows: list[list[str]],
129                  number: str, title: str) -> str:
130     ncol = len(header)
131     widths = [max([_display_width(header[j])
132                   + [_display_width(r[j]) for r in rows if j < len(r)]
133                   for j in range(ncol)])
134              ]
135     natural = sum(widths) + 2 * ncol
136
137     wrappable = [j for j in range(ncol)
138                  if align[j] == "L" and widths[j] > _WRAP_THRESHOLD]
139     use_tabularx = natural > _LINE_BUDGET and bool(wrappable)
140
141     if use_tabularx:
142         # Share the wrapping width between the text columns in proportion to
143         # how much text each of them actually carries.
144         # Proportional-to-length shares starve short-but-not-short-enough
145         # columns, so damp the ratio with a square root and floor it.
146         raw = {j: max(widths[j], 1) ** 0.5 for j in wrappable}
147         total_wrap = sum(raw.values())
148         shares = {j: max(len(wrappable) * raw[j] / total_wrap, 0.55) for j in wrappable}
149         scale = len(wrappable) / sum(shares.values())
150         parts = []
151         for j in range(ncol):
152             if j in wrappable:
153                 parts.append(r">{\hsize=%.3f\hsize\raggedright\arraybackslash}X"
154                             % (shares[j] * scale))
155             else:
156                 parts.append({"r": "r", "c": "c"}.get(align[j], "l"))
157         open_env = r"\begin{tabularx}{\textwidth}{\textwidth}" + "".join(parts) + "}"
158         close_env = r"\end{tabularx}"
159         shrink = False
160     else:
161         spec = "".join({"r": "r", "c": "c"}.get(a, "l") for a in align)
162         open_env = r"\begin{tabular}{\textwidth}" + spec + "}"
163         close_env = r"\end{tabular}"
164         shrink = natural > _LINE_BUDGET      # all-numeric wide table: scale it
165
166     lines = [r"\begin{table}[htbp]", r"\centering",
167             r"\caption*{\normalsize 表" + number + "\u3000" + inline(title) + "}",
168             r"\vspace{-2pt}", r"\zihao{-5}"]
169     if shrink:

```

```

169         lines.append(r"\resizebox{\textwidth}{!}{%")
170     lines += [open_env, r"\toprule",
171              " & ".join(inline(h) for h in header) + r" \\", r"\midrule"]
172     for row in rows:
173         lines.append(" & ".join(inline(c) for c in row) + r" \\")
174     lines += [r"\bottomrule", close_env]
175     if shrink:
176         lines.append("}")
177     lines.append(r"\end{table}")
178     return "\n".join(lines)
179
180
181 def parse_align(sep_row: List[str]) -> List[str]:
182     out = []
183     for cell in sep_row:
184         cell = cell.strip()
185         if cell.startswith(":") and cell.endswith(":"):
186             out.append("c")
187         elif cell.endswith(":"):
188             out.append("r")
189         else:
190             out.append("l")
191     return out
192
193
194 def split_row(line: str) -> List[str]:
195     return [c.strip() for c in line.strip().strip("|").split("|")]
196
197
198 def convert(md: str) -> tuple[str, str, str]:
199     """Return (title, abstract_block, body)."""
200     md = re.sub(r"<!--.*?-->", "", md, flags=re.S)
201     lines = md.split("\n")
202
203     title = ""
204     abstract: List[str] = []
205     body: List[str] = []
206     target = body
207
208     i = 0
209     pending_table: tuple[str, str] | None = None # (number, title)
210     while i < len(lines):
211         line = lines[i].rstrip()
212
213         # ---- headings
214         m = re.match(r"^# (.+)\$", line)
215         if m:
216             title = m.group(1).strip()
217             i += 1
218             continue
219         m = re.match(r"^## (.+)\$", line)
220         if m:
221             head = m.group(1).strip()
222             if head == "摘要":
223                 target = abstract
224                 i += 1
225                 continue
226             target = body
227             hm = re.match(r"^(d+)\$s+(.*)\$", head)
228             if hm:
229                 target.append(r"\section{" + inline(hm.group(2)) + "}")
230             else:

```

```

231         target.append(r"\section*" + inline(head) + ")")
232         target.append(r"\addcontentsline{toc}{section}" + inline(head) + ")")
233         i += 1
234         continue
235     m = re.match(r"^### (.+)$", line)
236     if m:
237         head = m.group(1).strip()
238         hm = re.match(r"^([\d.]+|[A-Z]\.\d+)\s+(.*)$", head)
239         target.append(r"\subsection*" + inline(head if not hm else head) + ")")
240         i += 1
241         continue
242
243     # ---- fenced code
244     if line.startswith("```"):
245         j = i + 1
246         buf = []
247         while j < len(lines) and not lines[j].startswith("```"):
248             buf.append(lines[j])
249             j += 1
250         target.append(r"\begin{lstlisting}[style=pseudo]")
251         target.extend(buf)
252         target.append(r"\end{lstlisting}")
253         i = j + 1
254         continue
255
256     # ---- display math
257     if line.strip() == "$$:
258         j = i + 1
259         buf = []
260         while j < len(lines) and lines[j].strip() != "$$:
261             buf.append(lines[j])
262             j += 1
263         expr = "\n".join(buf).strip().rstrip(",.")
264         target.append(r"\begin{equation}")
265         target.append(expr)
266         target.append(r"\end{equation}")
267         i = j + 1
268         continue
269
270     # ---- image -> figure float
271     m = re.match(r"^!\[图(\d+)\s*([^\]]*)\]\s*(((\^)+)\s*)$", line)
272     if m:
273         num, alt, path = m.group(1), m.group(2), m.group(3)
274         cap = alt
275         k = i + 1
276         while k < len(lines) and not lines[k].strip():
277             k += 1
278         cm = re.match(r"^图%s\u3000(.+)$" % num, lines[k].strip()) if k < len(lines) else None
279         if cm:
280             cap = cm.group(1)
281             i = k
282             width = "0.96" if path.endswith("00_esp_schematic.pdf") else "0.92"
283             target += [r"\begin{figure}[htbp]", r"\centering",
284                       r"\includegraphics[width=%s\textwidth]{%s}" % (width, path),
285                       r"\caption*{\normalsize 图" + num + "\u3000" + inline(cap) + "}",
286                       r"\end{figure}"]
287             i += 1
288             continue
289
290     # ---- standalone figure caption already consumed above; skip strays
291     if re.match(r"^图\d+\u3000", line.strip()):
292         i += 1

```

```

293         continue
294
295     # ---- table title line
296     m = re.match(r"表(\d+|A\d+)\u3000(.+)\$", line.strip())
297     if m:
298         pending_table = (m.group(1), m.group(2))
299         i += 1
300         continue
301
302     # ---- table body
303     if line.strip().startswith("|"):
304         header = split_row(line)
305         align = parse_align(split_row(lines[i + 1]))
306         rows = []
307         j = i + 2
308         while j < len(lines) and lines[j].strip().startswith("|"):
309             rows.append(split_row(lines[j]))
310             j += 1
311         num, ttl = pending_table if pending_table else ("", "")
312         target.append(convert_table(header, align, rows, num, ttl))
313         pending_table = None
314         i = j
315         continue
316
317     # ---- blockquote
318     if line.startswith(">"):
319         buf = [line.lstrip("> ").strip()]
320         j = i + 1
321         while j < len(lines) and lines[j].startswith(">"):
322             buf.append(lines[j].lstrip("> ").strip())
323             j += 1
324         target.append(r"\begin{quote}\small " + inline(" ".join(buf)) + r"\end{quote}")
325         i = j
326         continue
327
328     # ---- ordered / unordered list
329     if re.match(r"^\d+\.\s+", line) or re.match(r"^[*]\s+", line):
330         ordered = bool(re.match(r"^\d+\.\s+", line))
331         env = "enumerate" if ordered else "itemize"
332         items = []
333         j = i
334         while j < len(lines) and (re.match(r"^\d+\.\s+", lines[j]) or re.match(r"^[*]\s+", lines[j])):
335             items.append(re.sub(r"^\d+\.\s+", "", lines[j].rstrip()))
336             j += 1
337         target.append(r"\begin{%s}[itemsep=1pt,topsep=3pt]" % env)
338         target += [r"\item " + inline(it) for it in items]
339         target.append(r"\end{%s}" % env)
340         i = j
341         continue
342
343     # ---- blank
344     if not line.strip():
345         target.append("")
346         i += 1
347         continue
348
349     # ---- plain paragraph
350     target.append(inline(line.strip()))
351     i += 1
352
353     return title, "\n".join(abstract).strip(), "\n".join(body).strip()
354

```

```

355
356 # ----- preamble
357
358 PREAMBLE = r"""
359 \documentclass[UTF8,zihao=5,a4paper]{ctexart}
360 \usepackage[a4paper,top=2.5cm,bottom=2.5cm,left=2.5cm,right=2.5cm]{geometry}
361 \usepackage{amsmath,amssymb,bm}
362 \usepackage{graphicx}
363 \usepackage{booktabs}
364 \usepackage{caption}
365 \usepackage{enumitem}
366 \usepackage{listings}
367 \usepackage{xcolor}
368 \usepackage{setspace}
369 \usepackage{array}
370 \usepackage{tabularx}
371 \usepackage{fancyhdr}
372 \usepackage{xurl}
373
374 %%FONTS%%
375
376 \onehalfspacing
377 \setlength{\parindent}{2em}
378 \captionsetup{skip=4pt}
379
380 % Line breaking. \sloppy (tolerance 9999) let TeX stretch the inter-word and
381 % inter-CJK glue without bound, so a line holding a long unbreakable inline
382 % formula was padded until the Chinese characters visibly drifted apart. Use a
383 % permissive but finite tolerance instead, let long inline maths wrap at its
384 % relations and binary operators, and cap how far the CJK glue may stretch.
385 \tolerance=3000
386 \emergencystretch=2em
387 \relpenalty=200
388 \binoppenalty=400
389 \hfuzz=2pt
390 \hbadness=3000
391 \xeCJKsetup{CJKglue=\hskip 0pt plus 0.04\baselineskip}
392
393 \pagestyle{fancy}
394 \fancyhf{}
395 \renewcommand{\headrulewidth}{0pt}
396 \fancyfoot[C]{\thepage}
397
398 \ctexset{
399   section/format = {\centering\zihao{4}\bfseries},
400   subsection/format = {\raggedright\zihao{5}\bfseries},
401   section/beforeskip = 12pt plus 2pt minus 2pt,
402   section/afterskip = 8pt,
403 }
404
405 \lstdefinestyle{pseudo}{
406   basicstyle=\ttfamily\zihao{-5},
407   breaklines=true, breakatwhitespace=false,
408   columns=fullflexible, keepspaces=true,
409   frame=single, rulecolor=\color{gray!55},
410   framesep=4pt, xleftmargin=6pt, xrightmargin=2pt,
411   showstringspaces=false, extendedchars=true,
412 }
413 \lstdefinestyle{pycode}{
414   language=Python,
415   basicstyle=\ttfamily\zihao{6},
416   keywordstyle=\color{blue!65!black},

```

```

417 commentstyle=\color{green!42!black}\itshape,
418 stringstyle=\color{orange!75!black},
419 breaklines=true, breakatwhitespace=false,
420 breakindent=12pt, postbreak=\mbox{\textcolor{gray}{\hookrightarrow}\space},
421 columns=fullflexible, keepspaces=true,
422 showstringspaces=false, extendedchars=true,
423 numbers=left, numberstyle=\tiny\color{gray}, numbersep=6pt,
424 frame=leftline, rulecolor=\color{gray!45}, framesep=5pt,
425 xleftmargin=17pt, tabsize=4, upquote=true,
426 }
427
428 % 手工编号的图表题注（与正文交叉引用一致）
429 \captionsetup[figure]{labelformat=empty,justification=centering}
430 \captionsetup[table]{labelformat=empty,justification=centering}
431 ""
432
433
434
435 # ----- fonts
436
437 #: Preferred first, portable fallback second. The submission machine normally
438 #: has TeX Gyre Termes and Noto CJK; a bare TeX Live may not.
439 FONT_CHOICES = {
440     "main": ["TeX Gyre Termes", "Liberation Serif", "DejaVu Serif"],
441     "mono": ["Noto Sans Mono CJK SC", "DejaVu Sans Mono"],
442     "cjkmain": ["Noto Serif CJK SC", "Fandol Song", "Noto Sans CJK SC"],
443     "cjksans": ["Noto Sans CJK SC", "Fandol Hei"],
444     "cjkmono": ["Noto Sans Mono CJK SC", "Noto Sans CJK SC"],
445 }
446
447
448 def _installed_families() -> set[str]:
449     try:
450         out = subprocess.run(["fc-list", ":", "family"], capture_output=True,
451                             text=True, check=True).stdout
452     except (OSError, subprocess.CalledProcessError):
453         return set()
454     return {name.strip() for line in out.splitlines() for name in line.split(",")}
455
456
457 def font_block() -> str:
458     available = _installed_families()
459
460     def pick(key: str) -> str:
461         for name in FONT_CHOICES[key]:
462             if not available or name in available:
463                 return name
464         return FONT_CHOICES[key][-1]
465
466     return "\n".join([
467         r"\setCJKmainfont{%s}" % pick("cjkmain"),
468         r"\setCJKsansfont{%s}" % pick("cjksans"),
469         r"\setCJKmonofont{%s}" % pick("cjkmono"),
470         r"\setmainfont{%s}" % pick("main"),
471         r"\setmonofont{%s}[Scale=0.82]" % pick("mono"),
472     ])
473
474
475 def build_tex(title: str, abstract: str, body: str, code: str) -> str:
476     parts = [PREAMBLE.replace("%FONT%", font_block()), r"\begin{document}", ""]
477     # --- page 1: title + abstract only
478     parts += [

```



```

540 ↪ u3002"
        "\u6309\u4f9d\u8d56\u987a\u5e8f\u6392\u5217\u5f1a\u60c5\u666f\u6a21\u578b\u4e0e\u4f18\u5316\u3001\u6392\u653e\u2014\u529f\u
↪ u7387\u524d\u6cbf\u3001\u7ed8\u56fe\u6837\u5f0f\u4e0e\u56fe\u96c6\u751f\u6210\u3001\u786e\u5b9a\u6027\u6838\u9a8c\u3001\u63d0\u4ea4\u
↪ u7a3f\u88c5\u914d\u3002")
541     for n, name in enumerate(CODE_MODULES, start=1):
542         path = SRC / name
543         if not path.exists():
544             continue
545         out += [r"\subsection*{E.%d\quad \texttt{%s}}" % (n, esc(name)),
546               r"\lstinputlisting[style=pycode]{%s}" % (path.as_posix())]
547     return "\n".join(out)
548
549 out.append("\u4e3a\u63a7\u5236\u7bc7\u5e45\u5f0c\u672c\u9644\u5f55\u53ea\u7ed9\u51fa\u51b3\u5b9a\u5168\u6587\u6570\u503c\u7ed3\u679c\u
↪ u7684\u4e24\u6bb5\u6838\u5fc3\u5b9a\u4e49\u5f1b"
550         "\u5b8c\u6574\u53ef\u8fd0\u884c\u6e90\u7801\u968f\u7535\u5b50\u652f\u6491\u6750\u6599\u63d0\u4ea4\u3002")
551 BUILD.mkdir(exist_ok=True)
552 for n, (module, names) in enumerate(CORE_EXTRACTS, start=1):
553     snippet = BUILD / ("code_excerpt_%d.py" % n)
554     snippet.write_text(_extract(module, names), encoding="utf-8")
555     out += [r"\subsection*{E.%d\quad \texttt{%s}\uff1a%s}"
556           % (n, esc(module), "\u3001".join(esc(x) for x in names)),
557           r"\lstinputlisting[style=pycode]{%s}" % snippet.as_posix()]
558 return "\n".join(out)
559
560
561 def main() -> None:
562     include_code = "--no-code" not in sys.argv
563     full_code = "--core-code" not in sys.argv
564     md = MANUSCRIPT.read_text(encoding="utf-8")
565     title, abstract, body = convert(md)
566     tex = build_tex(title, abstract, body, code_appendix(full_code) if include_code else "")
567
568     BUILD.mkdir(exist_ok=True)
569     tex_path = BUILD / "paper.tex"
570     tex_path.write_text(tex, encoding="utf-8")
571
572     for run in range(2):
573         proc = subprocess.run(
574             ["xelatex", "-interaction=nonstopmode", "-halt-on-error",
575             "-output-directory", str(BUILD), str(tex_path)],
576             cwd=ROOT, capture_output=True, text=True)
577         if proc.returncode != 0:
578             log = (BUILD / "paper.log").read_text(encoding="utf-8", errors="replace")
579             errs = [l for l in log.split("\n") if l.startswith("!")][:12]
580             print("XeLaTeX 失败: \n" + "\n".join(errs))
581             sys.exit(1)
582     print("已生成", BUILD / "paper.pdf")
583
584
585 if __name__ == "__main__":
586     main()

```